

## The structure of micelles and microemulsions

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## The structure of micelles and microemulsions

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### Abstract

From simple micelles in water, nearly spherical aggregates of amphiphilic molecules, to bicontinuous microemulsions, oil and water microheterogeneous mixtures stabilised by a surfactant film with both local and large-scale disordered structures, the world of surfactant-containing systems is fascinating. Depending on a subtle balance of attractive and repulsive interactions between molecules at interfaces, an extraordinarily rich polymorphism of aggregated structures can be observed. After summarising the basic general constraints controlling surfactant aggregation and the formation of interfaces, different structures of micelles and microemulsions are reviewed and related to interfacial film properties and the intermolecular interactions inside them. With such systems, both static (time-averaged) and dynamic properties control structures equally well. Structures of very different systems can be described in simple ways in terms of surfactant film average curvatures and flexibility.

This review was received in its present form in April 1989.

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## 1. Introduction: definition of micelles and microemulsions

A surfactant molecule contains at least one polar hydrophilic part and at least one apolar hydrophobic unit, such as a hydrocarbon chain. Typical examples are soaps (sodium alkanoates) or phospholipids which form membranes. These molecules may be also called amphiphilic or tensioactive. The coexistence of two opposite types of behaviour (hydrophilic and hydrophobic) inside the same molecule is the origin of local constraints which lead to spontaneous aggregation into microscopic labile structures observed with surfactants in solution in oil or water.

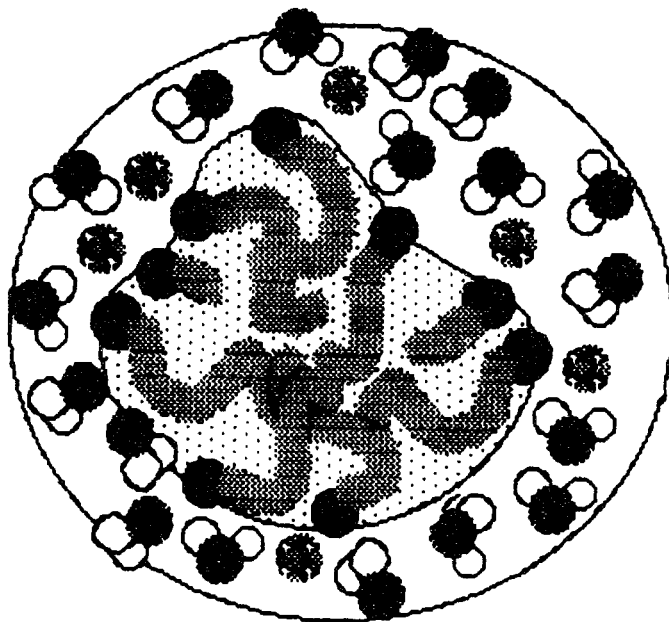
These supermolecular aggregates, at thermodynamic equilibrium, give surprising macroscopic properties to the solution. They could be explained in terms of microscopic structure by McBain (1913) and Hartley (1936). Knowledge of these structures, their equilibrium, formation and growing conditions is important for detergent manufacture, cosmetic industries, mineral flotation and the understanding of chemistry in heterogeneous media. Micellar aggregation is a primitive but easily controllable model for membranes. Therefore, these structures have been intensively studied in the last ten years; nearly two thousand papers indexed with micelle (or microemulsion) together with the keyword 'structure' have been published.

This review attempts to select the main physical aspects relevant to the surfactant aggregate's structure, which can be described in terms of mass, volume, shape, oil-water interface, dynamic equilibrium, shape fluctuations, internal molecular packing, etc. It does not cover the thermodynamic aspects and applications, for which we refer to reviews of Tanford (1973), Wennerström and Lindman (1979a), Lindman and Wennerström (1980), Cazabat and Veyssie (1984), Degiorgio and Corti (1985) and Zana (1987), and special symposium proceedings edited by Mittal and others (1977, 1979, 1982, 1984, 1986).

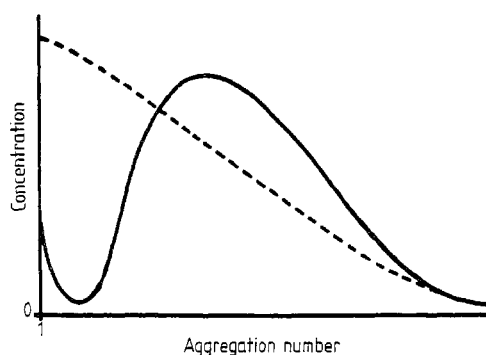
The simplest possible aggregate is the micelle (sometimes called 'direct micelle'), which is a cluster of surfactant molecules in water (figure 1). In such a definition, a micellar solution is a dispersion of surfactant molecules in aqueous or apolar solvent with a characteristic size. A hypothetical measurement of the mass distribution of the aggregates at a given time would display a maximum for a given range of number of molecules included in a given aggregate at a given instant (figure 2). This number  $N_{agg}$  is called the aggregation number.

This micellar behaviour has to be distinguished from ordinary molecular solutions in which molecules are individually dispersed in the solvent, solutions of molecules associated as dimers or oligomers produced by stacking of planar molecules. The progressive aggregation (broken curve in figure 2) leads to a monotonic decrease of concentration with size in the distribution curve: as concentration is increased, aggregate size varies strongly and no characteristic size independent of concentration can be defined in this case. Such aggregation behaviour is observed with porphyrins or bile salts (Zana *et al* 1977), although the origin of aggregation is the same as for micelles ('hydrophobic bonding').

Micellar aggregation can be demonstrated by measurements of physical properties against surfactant concentration. The most significant property is surface (or interfacial) tension (figure 3). As surfactant concentration is increased, surface tension is



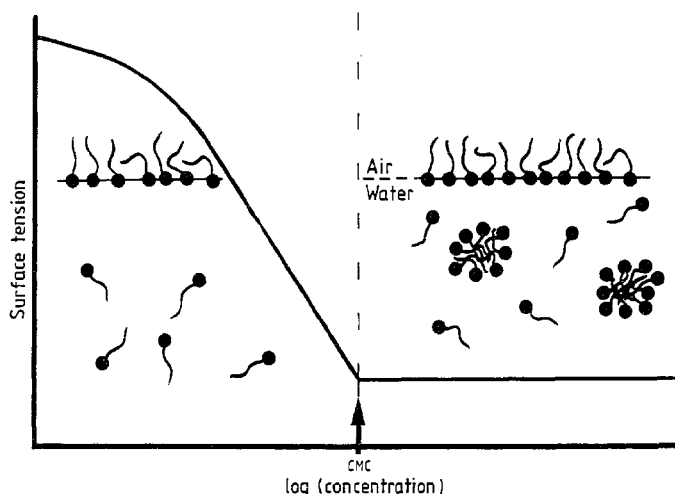
**Figure 1.** Schematic representation of a micelle: an excluded volume including an inner core with disordered hydrophobic chains and an outer shell with headgroups, condensed counter-ions and hydration water.



**Figure 2.** Definition of micellar aggregation: a minimum in the distribution of aggregation number is characteristic of micellar aggregation (full curve), whereas a continuous distribution occurs in non-micellar solutions (broken curve).

lowered because of adsorption of surfactant at the surface. The higher the concentration, the larger the adsorption (adsorbed molecules are in equilibrium with bulk dissolved ones) and the lower the surface tension. This surface tension decrease occurs up to a well defined concentration which is called critical micellar concentration (CMC). Above it, tension remains nearly constant, which expresses a constant concentration of molecularly dissolved molecules ('monomers'). The excess concentration aggregates into micelles, in equilibrium with monomers. The word 'micelle' refers to aqueous solutions (direct micelle), micellar aggregation in apolar solvents occurs in 'reverse' or 'inverted' micelles (Ekwall 1969).

Once a micellar solution is concentrated, micelles become ordered over large distances, resulting in mesophases (lyotropic liquid crystals). The approach of a liquid



**Figure 3.** Surface (air-water) tension as a function of surfactant concentration for an aqueous micellar solution. Schematic structure of the solution is shown below and above the critical micellar concentration (CMC).

crystal can be readily detected by flow birefringence of the micellar solution, which is optically isotropic at rest.

We restrict our discussion to solutions where only short-range interactions exist, the local order does not extend to more than few average sizes of an aggregate. This eliminates the field of lyotropic liquid crystals (mesophases) (Ekwall 1975, Skoulios 1978, Charvolin 1983), where order extends into macroscopic distances, at least in one dimension, leading to Bragg peaks in scattering patterns. Reference to mesophases is, however, frequently made in this paper since it is of interest for comparison between structures occurring in samples having similar compositions.

The most important property of micellar solutions is their ability to solubilise hydrophobic solutes. In the same way, reverse micelles dissolve water. Once water is solubilised in the micellar core, it is common to use again (but incorrectly) the expression 'reverse micelle': this expression should be reserved for binary-binary surfactant-oil mixtures but is widely used in the literature and we will use it too. For direct micelles, solubilisation of an apolar molecule results in 'mixed micelles' or 'swollen micelles'. The ability to dissolve another molecule is called the 'solubilising power' of the micellar solution. The first studies of phase diagrams in micellar systems (Ekwall *et al* 1969a) have shown that short-chain alcohols can greatly enhance the solubilising power of the solution: the added alcohol is therefore called 'cosurfactant'.

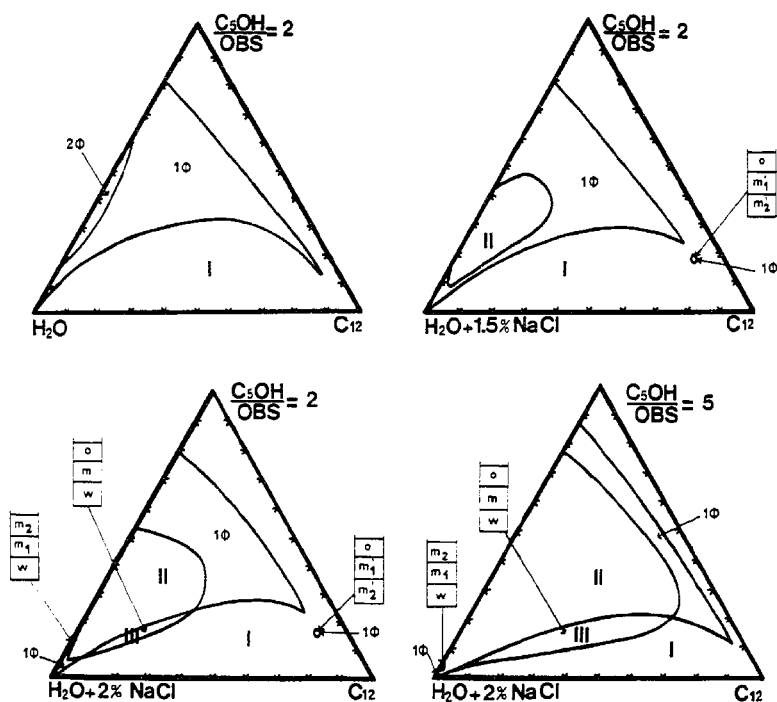
When a large amount of oil is added to a surfactant in solution in a hydrophobic solvent, or when a large amount of water is dissolved in a hydrophobic solution of surfactant, the usual term is o/w (respectively w/o) microemulsion. Since 'large' is not a precise definition, there is no difference in nature between micelles with an added solute or swollen micelle and a microemulsion. So far as physics rather than industrial processes is concerned 'micelles' and 'microemulsion' can be considered as synonyms.

Once the cosurfactant is added, the solubilising power can become infinite: water can be added in any proportion to a solution of reverse micelles (or oil to a solution of direct micelles). Stable and clear quaternary mixtures of water (or brine), oil, surfactant and cosurfactant are called microemulsions (Bellocq *et al* 1984, Bourrel and Schechter 1988). Microemulsions are easily obtained with high surfactant contents

(about 20%). When low surfactant concentrations are used polyphasic systems are obtained, in which excess water or oil separates in a second (or third) phase (Winsor 1948). A typical phase diagram is shown in figure 4: in the Winsor I (WI) and Winsor II (WII) diphasic regions, a microemulsion phase coexists with a phase of excess oil (WI) or water (WII), while the triphasic WIII system consists of a microemulsion in equilibrium with two phases of excess oil and water. According to the hydrophilicity of the surfactant and conditions (ionic strength, cosurfactant content, temperature), all or only some of these polyphasic systems are observed.

Microemulsions with low surfactant content are much more interesting for application purposes and efficient surfactants or mixtures of surface-active compounds are searched for. Successful formulation of a microemulsion requires knowledge of the physics of surfactant association, specially when 'stability' against temperature or salt concentration is needed in industrial applications.

There is no difference in nature between micelles, reverse micelles and microemulsions: the origin of this nomenclature is due to industrial use, referring to rough chemical composition which can include up to five components: the two solvents water and oil, a surfactant, a cosurfactant and salt.



**Figure 4.** Typical pseudo-ternary phase diagrams of microemulsions (Bellocq *et al* 1981). Components are water or brine (left corner), dodecane ( $C_{12}$ , right corner) and sodium octylbenzenesulfonate (OBS) and pentanol ( $C_5OH$ ) put together into a single pseudo-component (upper corner). With pure water, the phase diagram contains essentially a monophasic microemulsion domain and a diphasic one (I) with a microemulsion (m) in equilibrium with excess oil (o). As ionic strength increases, this phase diagram, typical of a hydrophilic surfactant, turns to more intricate ones with appearance of a second diphasic domain (II) with a microemulsion and excess brine (w) as phases, and further, a triphasic domain (III). Increasing the cosurfactant content has the same qualitative influence as adding salt.

The first part of this review will describe the conflicting constraints due to the amphiphilic nature of surfactant molecules. In the second part, results of structural studies on binary systems (surfactant-solvent) are discussed. The last two parts deal with ternary and quaternary (or more) systems. This classification according to the number of components is much simpler than qualitative designations of systems as micelles or microemulsions whose precise definitions are still controversial (Danielsson and Lindman 1981, Friberg 1982).

## 2. Basic rules

Self-association of surfactants results from their hydrophobic character, whereas long-range repulsions between their hydrophilic parts prohibit complete phase separation. Limited aggregation then results from a balance between two opposite forces of different spatial range. These two main contributions to the free energy of micelle formation are large and opposite, small variations are therefore very important and have large structural consequences. A very rich structural polymorphism is then expected and indeed experimentally observed.

Any surfactant aggregate, whether it is a micelle or a microemulsion, is constrained by steric hindrance and the energy cost of mixing oil and water. We first discuss these general constraints. As a consequence of this balance between two large and opposite forces, segregation of hydrophobic and hydrophilic parts is quite complete, hydrophobic regions being free of hydrophilic parts of molecules. This is quantitatively measured by the transfer energy from water to oil. All geometric constraints described in the following are a consequence of that: constant density, limit of sphericity and the molecular packing criterion.

### 2.1. *Transfer energy from water to oil*

From their solubility in water (McAuliffe 1966), the free energy of transfer of linear hydrocarbons from water to oil (hydrocarbon) was estimated. This free energy does not depend much on the oil used. From additivity rules (valid for linear hydrocarbons only), the free energy increments per group are  $-3.7 \text{ kJ mol}^{-1}$  for  $\text{CH}_2$  and  $-8.8 \text{ kJ mol}^{-1}$  for  $\text{CH}_3$  (Tanford 1973, Abraham 1979). They are  $-3.2 \text{ kJ mol}^{-1}$  for  $\text{CH}_2$  and  $-7.3 \text{ kJ mol}^{-1}$  for  $\text{CH}_3$  for the transfer of hydrocarbons from water to micelles, and about  $-3.0 \text{ kJ mol}^{-1}$  for the transfer of  $\text{CH}_2$  of surfactant from water to micelles. All these values compare well, differences being attributable to hydrocarbon-water contact occurring at the micellar surface. They are also very large, and total segregation of hydrocarbon and water is therefore expected. The similarity of transfers from water to oil and to micelles is a strong indication of a water-free hydrocarbon core in micelles. This is also clear when looking at the free energy of transfer of water molecules from hydrocarbon to bulk as estimated from water's minute solubility in hydrocarbons (less than 100 ppm (Schatzberg 1963)).

### 2.2. *Constant density, molecular volumes*

The important consequence of the combination of long-range attractive van der Waals and short-range repulsive excluded volume intermolecular interactions is that the density of liquids is constant, even over a few molecular distances. Local departures



from this behaviour can be considered as defects such as in the solid state and are energetically costly. On this basis, molecular volumes in liquids are the same whatever the structure, unless strong interactions occur (hydrogen bonding). They depend slightly on the solvent: getting an alkyl chain out of water into hydrocarbon results in an increase of its volume of the order of 10% (Davies and Bury 1930, Hoiland 1976, Desnoyers *et al* 1979, 1980, 1982). These volumes are easily obtained from density measurements or found in tables.

Tables should be used with caution, volumes taken from crystallographic data, representative of the solid state, should be considered with a 10% uncertainty, since they are underestimates for solutions (Shinoda and Soda 1963). Volumes calculated from densities of pure liquids do not account for solvent effects. Tables of molecular volumes or volume increments per group are available for aqueous solutions (Millero 1971, Cabani *et al* 1981), and can be used in tandem with an experimental determination of molecular volume, which can be obtained using precise density measurements. Some useful molecular and group volumes are given in table 1. The volumes of the  $\text{CH}_2$  and  $\text{CH}_3$  groups compare well with those of pure hydrocarbons (Huggins 1941, Doolittle 1951).

The constant density of the micellar core is a constraint that is to be introduced in interpretation of experiments. In theoretical calculations about hydrocarbon packing in micelles, the introduction of the rule of constant internal density is a considerable simplification since it reduces the number of possible conformations of packed chains. However, Bothorel *et al* (1974) and Belle and Bothorel (1974) introduced van der Waals attraction and excluded volume effects through a Lennard-Jones potential, constant density was thus not explicitly stated but was a consequence of these potentials.

**Table 1.** Selected molecular volumes of some widely used surfactants in micellar aqueous solution and some useful volume increments per group.

| Molecule or group                                                         | Volume<br>( $\text{cm}^3 \text{mol}^{-1}$ ) | Volume<br>( $\text{\AA}^3/\text{molecule}$ ) | Reference                       |
|---------------------------------------------------------------------------|---------------------------------------------|----------------------------------------------|---------------------------------|
| $\text{C}_{12}\text{H}_{25}\text{OSO}_3^-, \text{Na}^+$                   | 246.4                                       | 409.1                                        | Corkill <i>et al</i> (1967)     |
|                                                                           | 248.8                                       | 413.1                                        | Musbally <i>et al</i> (1974)    |
| $\text{C}_7\text{H}_{15}\text{CO}_2^-, \text{Na}^+$                       | 141.5                                       | 235.0                                        | Ekwall <i>et al</i> (1963)      |
|                                                                           | 142.4                                       | 236.4                                        | Vikingsstad <i>et al</i> (1978) |
| $\text{C}_8\text{H}_{17}(\text{C}_6\text{H}_4)\text{SO}_3^-, \text{Na}^+$ | 234.8                                       | 389.8                                        | Cabos (1983)                    |
| $\text{C}_{12}\text{H}_{25}\text{N}(\text{CH}_3)_3^+, \text{Br}^-$        | 295.5                                       | 490.6                                        | Corkill <i>et al</i> (1967)     |
| $\text{C}_{16}\text{H}_{33}\text{N}(\text{CH}_3)_3^+, \text{Br}^-$        | 365.4                                       | 606.7                                        | Corkill <i>et al</i> (1967)     |
| $\text{C}_{12}\text{H}_{25}\text{N}(\text{CH}_3)_2\text{O}$               | 255.0                                       | 423.4                                        | Benjamin (1966)                 |
| $\text{C}_{12}\text{H}_{25}(\text{OCH}_2\text{CH}_2)_6\text{OH}$          | 452.9                                       | 752.0                                        | Brown <i>et al</i> (1983)       |
| Aerosol OT in xylene                                                      | 390.7                                       | 648.7                                        | Ekwall <i>et al</i> (1970)      |
| Aerosol OT in heptane                                                     | 383.3                                       | 636.3                                        | Cabos and Marignan (1985)       |
| Butanol in water                                                          | 86.6                                        | 143.9                                        | Franks and Smith (1968)         |
| Butanol in octane                                                         | 99.6                                        | 165.4                                        | De Lisi <i>et al</i> (1984)     |
| Pentanol in water                                                         | 101.9                                       | 169.2                                        | Friedman and Scheraga (1965)    |
| Pentanol in octane                                                        | 116.0                                       | 192.6                                        | De Lisi <i>et al</i> (1984)     |
| $\text{CH}_2$ in water                                                    | 15.9                                        | 26.4                                         | Desnoyers and Arel (1967)       |
| $\text{CH}_3$ in water                                                    | 26.4                                        | 43.8                                         | Hoiland (1976)                  |
| $\text{CH}_2$ in hydrocarbon                                              | 16.3                                        | 27.06                                        | Doolittle (1951)                |
| $\text{CH}_3$ in hydrocarbon                                              | 32.2                                        | 53.46                                        | Doolittle (1951)                |
| $\text{OCH}_2\text{CH}_2$                                                 | 37.1                                        | 61.6                                         | Funasaki <i>et al</i> (1984)    |

### 2.3. Tanford's limit of a sphere, surfactant parameter $v/a_0l_c$

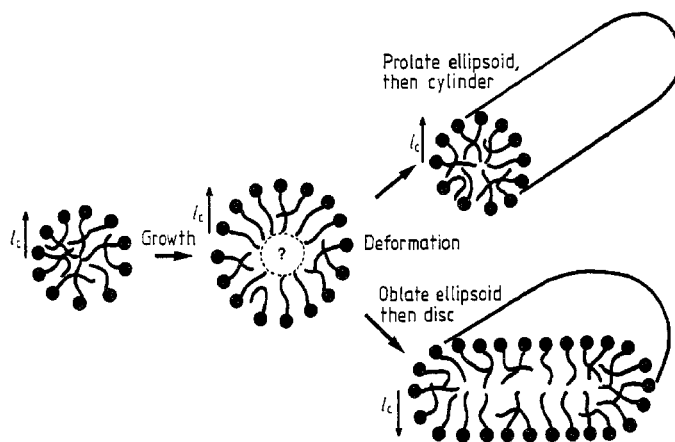
Surfactant headgroups are always located at the water-hydrocarbon interface, but some water-hydrocarbon contact remains and cannot be avoided. A typical case is sodium dodecylsulphate (SDS) as estimated from the comparison of area per headgroup at a micellar interface ( $60 \text{ \AA}^2/\text{molecule}$ ) and for SDS diluted in a cholesterol monolayer at the air-water interface ( $27.5 \text{ \AA}^2/\text{molecule}$  (Pethica 1955)). Complete coverage by headgroups would require a close proximity between them which is incompatible with their mutual repulsion. Because of its minimum area for a given volume, the spherical shape is the best way to minimise this contact area. Micelles are thus expected to be spherical whenever it is sterically possible. To avoid this water-hydrocarbon contact, water is expelled from the micellar core which is composed of dry hydrocarbon of constant density. In order to fill the whole core volume, at least one methyl group has to be present at its centre and, for headgroups to be at the surface, the micellar radius cannot exceed the maximum length of a surfactant molecule. Tartar (1955) and Tanford (1972, 1973) have shown that spherical micelles can grow up to a maximum value of aggregation number  $N_{\text{sph}}$ , any further growth inducing a shape deformation (figure 5). This limit is calculated on purely geometrical grounds since the volume  $v$  and the maximum length  $l_c$  of a linear alkyl chain of  $n$  carbon atoms are:

$$v = 27.4 + 26.9 n \text{ (\AA}^3\text{)}$$

and

$$l_c = 1.5 + 1.265 n \text{ (\AA)}.$$

This limit is strongly dependent on the value of  $l_c$ ; any surface roughness increases  $N_{\text{sph}}$  sharply. Indeed, most of a sphere volume sits close to its surface (a radius increase of 26% doubles the volume). Gruen and De Lacey (1984) have pointed out that a methyl-size hole at the micellar centre is a possible alternative to deformation. The radius increase is  $2.35 \text{ \AA}$ , which results in an increase of  $N_{\text{sph}}$  from 56 to 82 for a hydrophobic chain with 12 carbon atoms. The energy cost is the vaporisation enthalpy of a methyl and is weak ( $\approx 60 \text{ J mol}^{-1}$ ). The same geometrical arguments apply to any defined aggregate shape. The ratio of the area per headgroup ( $a_0$ ) to the average section of the hydrophobic part ( $v/l_c$ ) defines the spontaneous curvature  $R_{\text{sp}}$  of the



**Figure 5.** Limit of sphericity of a micelle as given by Tanford. Large micelles cannot be of a spherical shape as a hole should occur at their centre.

interface.  $v/a_0l_c$  is the fundamental geometric quantity for structural descriptions. Its value has to be kept in mind for any surfactant-solvent couple during structural investigation. A  $v/a_0l_c$  value higher than one indicates a spontaneous bending of the interface towards oil, while a lower value means spontaneous bending towards water (direct micelles). For concentrated solutions, the qualitative estimate of  $R_{sp}$  is no longer sufficient since the oil-water interface has to be characterised by two curvatures on each point. The general relation between  $v/a_0l_c$ , the average curvature  $H$  and gaussian curvature  $K$  is (Barnes *et al* 1988a, b):

$$v/a_0l_c = 1 + \langle H \rangle l_c + \frac{1}{3} \langle K \rangle l_c^2$$

where the averages are macroscopic.

The  $v/a_0l_c$  ratio depends on the surfactant chemical structure ( $l_c$  and  $v$ ) and on surface repulsions between headgroups ( $a_0$ ). The desired curvature (and thus type of aggregate) may be obtained upon a correct choice of the surfactant molecule and solvent conditions (type of solvent, ionic strength, etc), using the  $v/a_0l_c$  ratio as a guide (figure 6) (Israelachvili *et al* 1976, 1980, Mitchell and Ninham 1981). However, this ratio has to be used with caution as it accounts only for geometrical considerations:

(i)  $l_c$  is taken as the length of the fully extended alkyl chain; introducing chain elasticity and entropy together with edge effects (for finite aggregates) into free energy

| $a_0l_c/v$ | Aggregate shape                                                                                                           | Type of surfactant                                 |
|------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| 3          |  Spherical micelles                    | Single chain<br>Ionic or zwitterionic              |
| 2          |  Cylinders (that may be flexible)      | Single chain<br>Non-ionic or ionic with added salt |
| 1          |  Flexible lamella vesicles             | Double chain                                       |
|            |  Lamellar phases                       |                                                    |
|            |  Reverse micelles (in apolar solvents) | Double chain<br>Small area per headgroup           |

Figure 6. Schematic diagram of possible aggregate shapes according to the  $a_0l_c/v$  criterion (Israelachvili *et al* 1980). Note that reverse micelles are formed with an apolar solvent while all other cases refer to binary water-amphiphile solutions.

shows that different aggregate shapes are allowed and can coexist, they are governed by small changes in conditions and are difficult to predict on simple grounds (McMullen *et al* 1984a, b).

(ii) Moreover,  $v/a_0l_c$  is a time-averaged curvature. When the interface is very flexible (with adsorbed cosurfactant or at high temperature), curvature fluctuations give rise to apparent repulsion of entropic origin (Helfrich 1978, Safinya *et al* 1986).

### 3. Principles of methods of investigation

Each structural method works on a characteristic spatial or time domain. The resolution is an important quantity to be known before any interpretation. The spatial (or temporal) resolution is the smallest distance (or time interval) that can be separated with a given method. Smaller details are smoothed out.

The structure, as studied by any physical method, is thus smoothed, all the inhomogeneities of smaller size than the resolution being left out. It is therefore not meaningful to introduce overinformative details in a structural model: given a method and its resolution, a suitable model for the interpretation has to be chosen (figure 7). In the following, methods are discussed, starting from the lowest resolution (macroscopic methods) and ending with NMR which works on a molecular spatial domain. A gap in experimental methods exists between direct visualisation (optical microscope) which works on a  $100\text{ }\mu\text{m}$  scale and light scattering ( $1\text{ }\mu\text{m}$ ). This is currently being filled with video-enhanced microscopy (Evans *et al* 1985); direct images of supra-aggregate structures are observed which opens a new field of structural investigation (Miller *et al* 1987a, b). Since electron microscopy requires freezing and drying of the sample, this technique has not yet been widely used to investigate micelle and microemulsion structure. Transmission electron microscopy, however, can in principle go down to the resolution needed to observe typical scale  $10\text{ nm}$  of the size of surfactant aggregates. Freeze fracture electron microscopy (Dubochet *et al* 1984, Talmon 1986) has been questioned due to artefacts produced during the shadowing and replication procedures. A new technique, known as direct imaging (DI), has recently proven to

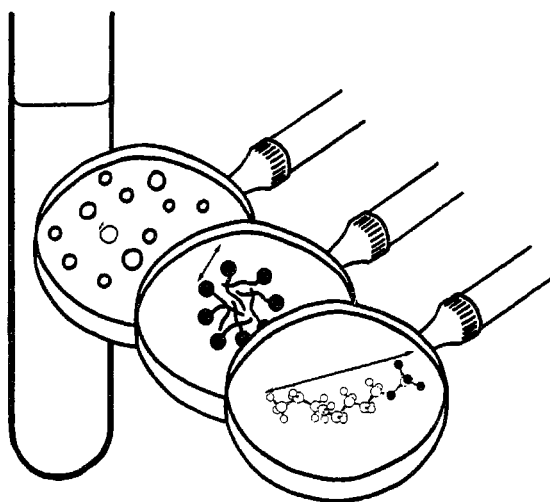


Figure 7. Diagram showing the problem of spatial resolution in experimental methods.

be extremely powerful and, instead of being disturbed by artefacts, the D<sub>1</sub> technique makes use of the artefacts to distinguish between water and oil domains (Jahn and Strey 1988). Since D<sub>1</sub> has, independently of neutron scattering, been able to make a clear distinction between connected and disconnected droplets, it is likely that wide use will be made of this technique in structural studies in the future.

Methods of investigation for dynamical properties are also discussed, taking into account spatial resolution, going from motions of the aggregates, of molecules in the aggregates and of atoms in the molecules. It is restricted to some experimental methods which are the most powerful, general and mainly the most direct, thus reliable ones. A more complete compilation of experimental methods can be found in recent reviews (Wennerström and Lindman 1979a, Zana 1987, Kahlweit *et al* 1987).

### 3.1. Macroscopic techniques

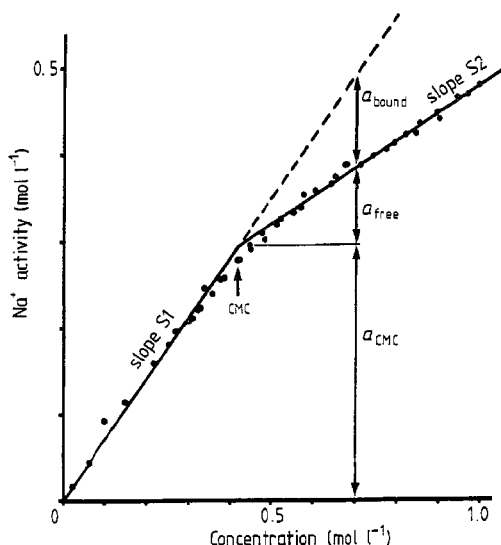
Because of the characteristic size of heterogeneities, all these methods suffer from a lack of spatial resolution. They are very useful for the observation of transitions, as an abrupt variation of macroscopic properties necessarily reveals some change in microscopic structure. For example, critical micelle concentrations are measured by a large number of macroscopic techniques (Mukerjee and Mysels 1971). The transition from small to large micelles is accompanied by a significant increase in viscosity (Ekwall and Holmberg 1965, Ekwall *et al* 1971, Nagarajan *et al* 1982, Porte and Appell 1984) and the appearance of anisotropic susceptibilities (Hoffmann *et al* 1981a, b, c, Porte *et al* 1984). The percolation thresholds are easily observed from macroscopic transport measurements (Laguës *et al* 1978, Laguës 1979, Laguës and Sauterey 1980, Lagourette *et al* 1979, Dvolaitzky *et al* 1980, Zana *et al* 1982, Clausse and Heil 1983, Cazabat *et al* 1984a, b). Some macroscopic methods, as applied to structural studies, are briefly discussed in the following.

One of the earlier methods used in colloid science is viscosity (McBain 1913). This method is still frequently used in polymer science. Intrinsic viscosity  $[\eta]$ , obtained by an extrapolation to infinite dilution, is related to the specific volume  $V_{sp}$  by  $[\eta] = a V_{sp}$ ,  $a$  being a form factor. The particle is defined as the content of a hydrodynamic volume which includes the colloidal particle and all its bound species (condensed counter-ions and hydration water). Because of the large penetration of solvent into the particle, intrinsic viscosity can be used to determine the mass of polymers in solution. This is not the case for micelles. When the size, shape and fraction of bound counter-ions of the particle are already known from other independent determinations, hydration numbers can be obtained. The method seems to work since reasonable values are found, but important problems do exist. The form factor is well established only for spherical particles and its value  $a = 2.5$  is valid under stick boundary conditions at the surface ( $a = 1$  with slip conditions). Accounting for slip, stick or intermediate conditions can be attempted (Altenberger *et al* 1985).

Electrical conductivity and ion activity can be used when ions are present, i.e. with ionic surfactants. Their measurement as a function of concentration provides a measure of the fraction of bound ions. Indeed, the mobility (thus electrical conductivity) of ions is reduced as they bind to an aggregate of low mobility (Evans 1956, Kimizuka and Satake 1962, Satake *et al* 1963, Zana 1980, Lianos and Zana 1983). Activity of ions is less, too, when they are embedded in an aggregate, their chemical potential being less than for their free state (Botré *et al* 1959, Satake *et al* 1969, Sasaki *et al* 1975, Gaboriaud *et al* 1984, Sepulveda and Cortès 1985). For the interpretation of

these measurements, drastic simplifications are usually made, as shown in figure 8. The results are then strongly dependent on the way the experiment is interpreted and cannot be considered as reliable when used alone, only variations along series are significant. Moreover, the fraction of bound counter-ions depends on the cut-off distance (see §4.1.7) which arbitrarily separates bound and free ions. This separation is decided very clearly in theoretical calculations. On the contrary, the way the instrument proceeds on microscopic distances for making such a discrimination is not known. The result is thus an effective quantity: it accounts for the effect through the way in which it is interpreted; it is not a mean value as would be expected for a macroscopic method. These considerations explain why there is such a large scatter in 'experimental' values of bound counter-ion fractions (Sepulveda and Cortès 1985). Of greater significance are measurements of surfactant activity. The two-states model (free or micellar) is clear and the concentration of free surfactant is determined (Sasaki *et al* 1975, Cutler *et al* 1978, Vikingstad 1979). For ionic surfactants above their CMC, the concentration of free surfactant is shown to decrease with total concentration, which is in disagreement with the widely used pseudo-phase model.

Birefringence measurements are useful in studying anisotropic structures. Macroscopically permanent birefringence occurs with lyotropic mesophases which are ordered on large scales. Anisotropy in electric (Rouvière *et al* 1979a, b, Hoffmann *et al* 1981a, b, Schorr and Hoffmann 1981, Neeson *et al* 1983, Guéring and Cazabat 1983) or magnetic susceptibility (Meyer *et al* 1982, Porte *et al* 1984, Rosenblatt and Zolty 1985) or rheological properties (Backus and Scheraga 1951, Scheraga and Backus 1951, Löbl *et al* 1984, Hirsch *et al* 1984a) detects microscopic anisotropies that are averaged out in the absence of an external field. If these effects cannot be ascribed to molecules themselves (Hoffmann and Schorr 1985) or to critical fluctuations (Cantù *et al* 1985), they are due to aggregates. Apart from this easy way to detect anisotropic structures, these effects can be used to align samples on a macroscopic scale, a useful technique being to perform scattering (Thurn *et al* 1984, Hayter and Penfold 1984, Cummins *et*



**Figure 8.** Counter-ion activity as a function of total sodium octanoate surfactant concentration (Stenius and Ekwall 1967). The ratio of slopes above (S2) and below (S1) the CMC gives a measurement of the fraction of free counter-ions.

al 1987a, b), NMR or linear dichroism experiments (Johansson *et al* 1976, Ulmius *et al* 1979). Nevertheless, as orienting fields are generally intense, the possibility of structural modification has to be considered (Clark and Ackerson 1980, Brady *et al* 1984, Rehage *et al* 1986). A promising route for such studies is the use of nematic phases (see § 4.1.3) which are solutions of macroscopically oriented micelles (Samori and Mattivi 1986, Laurent and Samori 1987).

### 3.2. Scattering methods

Scattering measurements give direct access to the Fourier transform of the electron density and are therefore the only direct method of studying surfactant solution structure *in situ*. Any scattering experiment measures the intensity  $I$  of radiation (of wavelength  $\lambda$ ) scattered at an angle  $2\theta$  of the direct beam. The momentum transfer or scattering vector  $q$  is defined by

$$q = 4\pi \sin\theta/\lambda.$$

The scattered intensity is measured between two limits  $q_{\min}$  and  $q_{\max}$ . The information content of the experimental result is limited to a resolution of  $2\pi/q_{\max}$ , usually about 2 nm, in some cases 1 nm when large angles are used. No long-range correlation, object or structural feature larger than  $2\pi/q_{\min}$  can be observed by a scattering technique. Usually, the largest distance measured is about 100 nm for neutron or x-ray scattering and a few  $\mu\text{m}$  in the case of light scattering. Mass and osmotic compressibility can be accessed by absolute measurement of the intensity. Several reviews on the principles, methods and limitations of scattering methods applied to surfactant systems are available (Hayter 1985, 1987, Chen 1986a, b, Magid 1986). We give here only a brief outline of the most widely used expressions.

The scattering of any set of elemental volumes  $v_i$ , each having a scattering length density  $\rho_i$ , is given by

$$I(q) = \frac{d\Sigma}{d\Omega}(q) = \sum_{v_i} \sum_{v_j} \rho_i \rho_j \exp[-q \cdot (r_i - r_j)]$$

where  $r_i$  is the coordinate of the volume  $v_i$  relative to the origin. This formula is valid for any choice of elemental volumes  $v_i$ , large in regard to atomic sizes and smaller than  $2\pi/q_{\max}$ . In the most general case, this formula has to be used to calculate the scattering of the presumed structural model for a surfactant solution and compared with the experiment (Barnes and Zemb 1988).

There are some favourable cases where this general expression reduces to an analytic calculation of the scattering on absolute scale: the most widely used case is the case of monodisperse spheres interacting through steric (hard-sphere) repulsion and an exponential attractive or repulsive potential. In the case of monodisperse spheres of number density  $n$ , the expression of  $I(q)$  reduces to (figure 9)

$$I(q) = nP(q)S(q)$$

where the form factor  $P(q)$ , which expresses the scattering cross section of one particle is (for spheres of radius  $R$ )

$$P(q) = \left( \int_0^R (\rho(r) - \rho_{\text{ext}}) e^{iqr} dr \right)^2$$

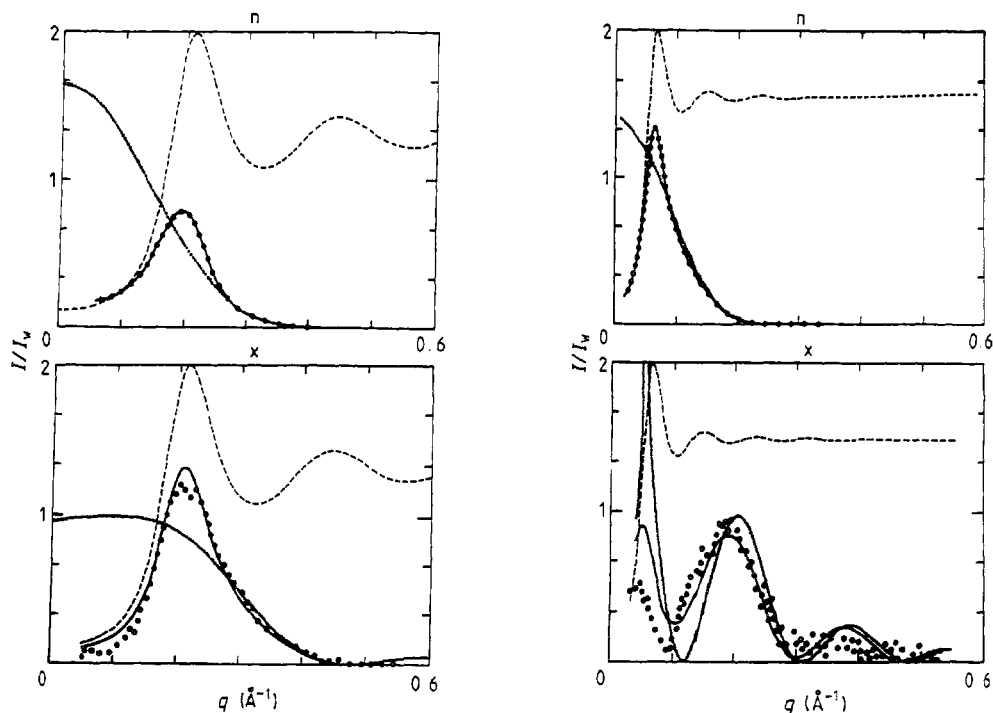


Figure 9. Typical small-angle neutron (n) and x-ray (x) scattering spectra of ionic micellar solutions of sodium octanoate (left) and sodium dodecylsulphate (right). The total scattered intensity  $I(q)$  (experimental: dots; calculated: full curve) is split into intramolecular (form factor  $P(q)$ , dotted curve) and intermolecular (structure factor  $S(q)$ , broken curve) scattering.

where  $\rho(r)$  is the scattering length density of the particle at the radius  $r$  and  $\rho_{\text{ext}}$  the scattering length density of the solvent. Typical values in the case of spherical micelles are given in figure 10. Approximate expressions are available for the case of polydisperse spheres. Fitting procedures including polydispersity are to be regarded with caution since any scattering curve can be reproduced by a set of monodisperse ellipsoids or polydisperse spheres. Scattering experiments can be used to eliminate inappropriate structural models and the reproduction of an absolute scaled scattering spectrum should always be cross-checked with conductivity, specific area and phase boundaries.

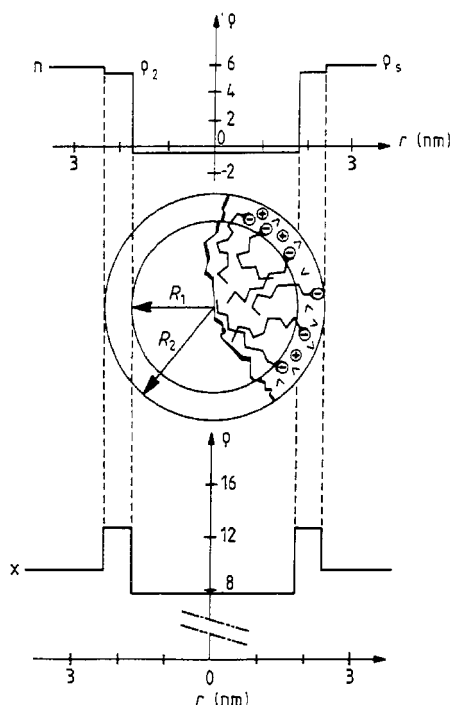
The structure factor  $S(q)$  is defined by

$$S(q) = 1 + \frac{1}{N} \int_0^\infty (g(r) - 1) e^{iqr} dr$$

where  $g(r)$  is the correlation function between particles.  $S(q)$  is the interparticle interference scattering and is related to the interactions. In favourable cases, the structure factor can be calculated analytically—for hard-sphere interaction (Percus and Yevick 1958) and for hard spheres and a screened electrostatic potential (Hayter and Penfold 1981). In more general cases, the HNC integral equation is needed to determine  $S(q)$  (Hansen and McDonald 1976).

For the case of monodisperse spheres, a self-consistent fitting procedure can be used to elucidate micellar structure: an estimate of aggregation number, hydration and





**Figure 10.** Standard model of radial scattering length densities of a micellar solution, for neutrons ( $n$ ) and x-rays ( $x$ ), according to resolution constraints.

effective charge is made. These three parameters allow complete calculation of  $P(q)$  and  $S(q)$ . This result is compared with the scattering results and parameters are changed until agreement with experiment is obtained. If radii are used as a fitting parameter instead of aggregation numbers, unphysical results can be obtained. Since this procedure nearly always converges, cross-checking of the absolute scaling and surface area is needed.

When micelles are polydisperse, the factorisation  $I(q) = P(q)S(q)$  is no longer strictly valid: in the so-called 'decoupling approximation', approximate expressions are available (Hayter 1985, Chen 1986a, b). When the microstructure is locally cylindrical or lamellar, approximate expressions are available (Vonk *et al* 1988, Teubner and Strey 1987, Warr *et al* 1988a), but the calculation of the full scattered intensity requires numerical integration of the fundamental expression.

Without requiring any structural models, some general parameters can be extracted from a scattering experiment. These are:

(i) the invariant  $Q^*$ , is the second moment of the scattered intensity (Luzzati *et al* 1976):

$$Q^* = \int_0^\infty q^2 I(q) dq = 2\pi^2(\bar{\rho} - \rho_{ext})^2 \varphi(1 - \varphi)$$

where  $\varphi$  is the volume fraction of the dispersed medium and  $\bar{\rho}$  is the average scattering length density of the particle.

(ii) the total interfacial area  $\Sigma$ , the macroscopic manifestation of the area per headgroup  $\sigma$  ( $\approx 1 \text{ nm}^2$  per molecule), can be derived by

$$\lim_{q \rightarrow \infty} I(q)q^4 = 2\pi\Delta\rho^2\Sigma$$

thus allowing measurement of the interfacial area  $\Sigma$  independently of any structural model if the scattering has been measured at large enough angles (typically  $qR \geq 10$ , if  $R$  is the radius of droplets).

(iii) Other characteristics of the structure can be obtained from the experimental curve by more sophisticated experiments: average curvature of the interface by labelling (Auvray *et al* 1985, Huang and Kotlarchyk 1986), balance between Gaussian and average curvature and average chord delimited by the interface on random segments taken in the structure (Méring and Tchoubar 1968).

In any case, scattering studies are very powerful techniques to eliminate some structural models. However, a scattering experiment alone cannot be a proof of the microstructure. X-ray and neutron scattering can be used with a bandwidth  $q_{\min}/q_{\max}$  of the order of 100 typically. Classical light scattering gives a low  $q$ -range but a much smaller bandwidth of the order of 10.

For interpretation of quasi-elastic scattering, structure has to be inferred via a dynamical model of time correlations of fluctuations of objects. When the structure is a dilute dispersion of monodisperse spheres, one may use the expression of the effective diffusion coefficient  $D_{\text{eff}}$

$$D_{\text{eff}}(q) = \frac{D_0}{S(q)} (1 + H(q)) = \left( \frac{kT}{6\pi\eta R} \right) (S(q))^{-1} (1 + H(q))$$

approximate expressions of the hydrodynamic correction term  $H(q)$  are available (Ackerson 1978). If the solution is not made of monodisperse spheres, interpretation can only be qualitative.

### 3.3. NMR relaxation methods

Nuclear magnetic resonance (NMR) is a basic tool for investigating geometric and dynamic properties at a molecular level. Any non-zero nuclear spin can be observed (Brévard and Granger 1981). When spectral lines do not overlap, each atom can be observed individually. This technique thus has the highest spatial resolution. This obvious advantage has some drawbacks: lower-resolution information such as particle size, although present, is difficult to extract from such abundant information. NMR methods should then be used for structure refinement, when some lower-resolution data are already known. The latter should then be introduced as a constraint for the interpretation. Lastly, obvious constraints should not be forgotten: atoms are tied to one another by chemical bonds of known length and angles, and only few dihedral angles are allowed defining the trans and gauche conformations. Because of spectral resolution problems, studies on  $^1\text{H}$  NMR are limited,  $^{13}\text{C}$  or  $^2\text{H}$  (from specific deuteration) are preferred for hydrocarbon chains studies, and headgroups or counter-ions can be observed by their relevant nucleus.

For quantitative structural studies, two main methods are available: vicinal spin-spin coupling constants which are conformation dependent, and relaxation which allows both spatial and dynamic studies. Thus, in this latter case, structural studies depend on an implicit dynamical model. The former has been rarely used because of the want of good spectral resolution. Conformational analysis has been only performed near headgroups by means of  $^1\text{H}$ - $^1\text{H}$  (Menger and Jerkunica 1977, Maitra and Eicke 1981, Maitra *et al* 1983, Ueno and Kisimoto 1983) or other vicinal coupling constants (Belaïd and Chachaty 1984, Chevalier and Chachaty 1984, Faure *et al* 1985, Menger and Carnahan 1986, Menger *et al* 1987).

NMR relaxation is due to the modulation of magnetic interactions by rotational Brownian motion, providing geometrical and dynamic (correlation times  $\tau_c$ ) information. The modulation of the magnetic energy is characterised by an autocorrelation function  $G(t)$  whose Fourier transformation is the spectral density  $J(\omega)$  of motion (figure 11). NMR relaxation times are combinations of spectral densities at relevant Larmor frequencies (Abragam 1961). Spin-lattice relaxation ( $T_1$ ) is thus important when molecular motions occur in the Larmor frequency range. Slow motions having correlation times larger than the inverse Larmor frequency are inefficient in inducing relaxation. Experiments can be carried out at different Larmor frequencies to measure motions on different time scales and are complementary. Very low frequencies (kHz) are accessible by rotating frame relaxation ( $T_{1\rho}$ ). Spin-spin relaxations ( $T_2$ ) probe very slow motions through  $J(0)$ . The highest available magnetic field with commercial NMR spectrometers (14.1 T) sets the upper limit of the NMR frequency domain at 5 GHz. Detailed studies of fast molecular motions would require a factor of the order of ten more.

Motions in heterogeneous media can be separated into two parts: slow motions of the structure itself and internal fast motions of molecules. Each part corresponds to a different time scale and these motions are uncorrelated; the autocorrelation function is thus a sum of two independent autocorrelation functions pertaining to these motions. This idea, introduced by Charvolin and Rigny (1973) and developed for micellar systems by Wennerström *et al* (1974, 1979), is a two-step application of the general spectral density for a time hierarchy of  $N$  motions which reduce order parameters by successive geometrical increments  $S_i$  (Davis *et al* 1978, Paddy *et al* 1981, Dufourc and Smith 1986) (figure 11):

$$J(\omega) = \sum_1^N \left( \prod_{k=1}^i s_{k-1} - \prod_{k=1}^i S_k \right) \frac{\tau_{ci}}{1 + \omega^2 \tau_{ci}^2}.$$

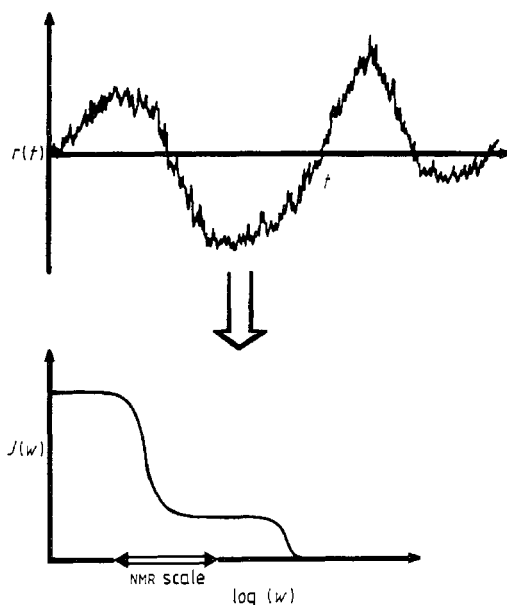
For a fast internal axially anisotropic motion coupled to a slow overall isotropic one, the spectral density is

$$J(\omega, S, \tau_{cs}, \tau_{cf}) = S^2 J(\omega, \tau_{cs}) + (1 - S^2) J(\omega, \tau_{cf})$$

where  $\tau_{cf}$  and  $\tau_{cs}$  are the correlation times for the fast and slow motions, and  $S$  is an order parameter defined as the time average due to fast motions

$$S = \langle 3 \cos^2 \theta - 1 \rangle / 2$$

$\theta$  being the angle between the C—H bond and the symmetry axis of the fast motion locally perpendicular to the interface for  $^{13}\text{C}$ - $\{^1\text{H}\}$  relaxation. Any relaxation time and the nuclear Overhauser enhancement can be calculated from this spectral density (Wennerström *et al* 1974, 1979, Ahlnäs *et al* 1983, Lindman *et al* 1983).



**Figure 11.** Rotational Brownian diffusion  $r(t)$  of a surfactant  $^{13}\text{C}$  nucleus in a micelle with two different rates of motion (top), and the deduced two-step spectral density  $J(\omega)$  (bottom).

Quadrupolar relaxation can also be interpreted in the same way (Henriksson *et al* 1977a). This model is simple, and experiments at three Larmor frequencies allow the determination of the three parameters  $\tau_{\text{cs}}$ ,  $\tau_{\text{cf}}$  and  $S$ . The slow motion includes the rotation of the micelle and the diffusion of the surfactant molecule at its spherical surface (Alexandre *et al* 1973). Studies of anisotropic slow motions are possible (Söderman *et al* 1985b). The anisotropy of fast motions is accounted for by  $S$  only, their rate ( $\tau_{\text{cf}}$ ) is assumed to be the same for any orientation (Halle and Wennerström 1981). This simplification may seem unrealistic since methylene motions in alkyl chains occur by discrete jumps between trans and gauche positions (van Gunsteren *et al* 1981, Jorgensen *et al* 1984). It takes advantage of the fact that fast motions do not enter the frequency dependence of  $T_1$  to model them as simply as possible, but to what extent the model for fast motions influences the other parameters in a fitting procedure is still a question.

Another problem is the possibility for cross-correlation effects to be important with such anisotropic motions (Hubbard 1969, 1970, Werbelow and Grant 1977). Although equilibrium recovery curves were always observed as monoexponential, their systematic inspection seems judicious. The development of more refined spectroscopic methods using these cross-correlation effects (Vold and Vold 1978) is promising for the detailed study of fast local motions (Prestegard and Grant 1978, Fuson and Prestegard 1983, Brown *et al* 1985).

### 3.4. Raman and infrared spectroscopies

Vibration frequencies of molecular bonds and angles are measured by these techniques. The spatial and time resolutions are very high, to such an extent that these methods are not very sensitive to the environment of the molecule. Raman and IR spectra are intrinsic characteristics of molecules. Given a molecular conformation, they are identical in gas, liquid, solution and solid states unless a specific interaction occurs

(solvation, hydrogen bonding, etc). Spectral differences reveal changes in molecular conformation and Raman and IR spectroscopies have mainly been used for conformational analysis of hydrocarbon chains of surfactants. The vibration frequencies being much larger than the rate of conformational isomerisation, the spectra are sums of subspectra of all the conformers present in the sample, the complete information is thus contained in the spectra. Nevertheless, few and essentially qualitative conclusions have been drawn from both these techniques. The reason for this is the large number of conformations and the extremely small frequency differences between the conformer subspectra (Mizushima and Simanouti 1949, Snyder 1967, Snyder *et al* 1982). Broad composite bands are then observed which can only be characterised by the frequency of their maximum and their band width. Raman C-C and C-H stretching bands contain lines corresponding to the all-trans conformation (easily assigned with spectra of dry crystalline samples), and additional bands which are assigned as gauche-containing conformations (Lippert and Petitcolas 1972, Okabayashi *et al* 1974, 1975, 1982, Kalyanasundaram and Thomas 1976, Rosenholm *et al* 1976, 1977, Burns *et al* 1982, Amorim da Costa and Santos 1983, Brooker 1984).

IR spectroscopy became possible with the important sensitivity enhancement brought about by the introduction of Fourier transform IR spectroscopy, allowing the study of aqueous solutions. Only shifts of band maxima could be observed (Umemura *et al* 1980, 1981, Kawai *et al* 1984) and, like Raman spectroscopy, only relative variations over different samples can be studied. Another limitation is the sensitivity; the solutions must be fairly concentrated and, if the influence of micellisation is to be studied, spectra should be recorded below the CMC. Most studies then deal with short chain surfactants which have large CMC and pre-aggregates are observed below the CMC, making the observation of unassociated molecules difficult. The main conclusions of these studies are the expected results: the micellar core is liquid-like (gauche conformations are present) and the probability of all-trans conformation is larger in micelles than for monomeric molecules and increases with concentration (as the area per headgroup decreases). The combined use of these two spectroscopic techniques, together with the use of deuterated compounds (Rosenholm *et al* 1976), could lead to more quantitative results, if improvements in sensitivity (Kamogawa *et al* 1984) are achieved. When surfactant molecules contain a function which has some isolated band (a double bond, the polar head, etc) (Okabayashi *et al* 1982, Okabayashi and Abe 1980, Okabayashi and Matsushita 1984, Umemura *et al* 1980, 1981, Cameron *et al* 1982, Brooker 1984, Kawai *et al* 1984) clearer conclusions can be drawn.

### 3.5. Methods involving the use of probes

The methods previously described make use of intrinsic probes. X-ray or neutron scattering use only intrinsic electron or scattering length, NMR uses non-zero nuclear spins. Some techniques such as electron spin resonance (ESR), fluorescence, UV or visible absorption are not so general. Most organic molecules (especially surfactants) do not contain unpaired electrons nor absorb light. When the medium has not itself the necessary spectroscopic property to enable such experiments to be performed, it can be introduced into the medium by means of added molecules called probes. The addition of the foreign molecule is assumed not to perturb the structure under study, but it is generally not possible to ensure that this is actually the case. These methods are then only useful to back up known results, or conclusions which need to be confirmed by more reliable methods. Nevertheless, they should not be completely

rejected as they may allow useful complementary experiments to be carried out on known structures: as an example, dynamic fluorescence or ESR can work on time domains below the time resolution of NMR.

Generally, the interpretation of such experiments is dubious because, for unique experimental data, two important factors are unknown: the structure under study and the localisation of the probe in this structure. The incorrect use of such probe methods had led to an incorrect model of micelles (Menger 1979, 1984). Thus, because added hydrophobic probes have the same spectroscopic properties as in polar solvents, it was concluded that water should penetrate the micellar core (where probes were assumed to sit) (Kalyanasundaram and Thomas 1977, Menger *et al* 1978, 1981, Menger 1979, Menger and Boyer 1980, Menger and Bonicamp 1981, Menger and Chow 1983, Menger and Doll 1984, Martens and Verhoeven 1981, Turro and Okubo 1981, Russell and Whitten 1982). An alternative (and correct) interpretation is that these probes solubilise in the interface (Wennerström and Lindman 1979b, Zachariasse *et al* 1981, 1984, Ganesh *et al* 1982, 1984). Probes are also used to monitor the 'local' or 'effective' polarity of their solubilising site (Mukerjee *et al* 1977, Donchi *et al* 1980, Zachariasse *et al* 1981, 1984). Such quantities have never been clearly defined and reference to macroscopic dielectric constants is generally made. Since heterogeneities have molecular dimensions, descriptions by means of macroscopic quantities defined from statistical physics are forbidden. The meaning of such experiments is thus obscure and their practical use questionable.

A classical example is the determination of the so-called microviscosity or fluidity by studying the motions of probes. Once the structure is known, probe motions reflect internal motions of the host, provided that dimensions of the solvent compartment are large enough for edge effects to be neglected and for reference to bulk solvent to be significant. The problem of localisation of the probe remains and requires an independent determination, although it is frequently neglected. When solutes solubilise at the interface, experiments have to be interpreted at a molecular level (Aoudia *et al* 1984a, b). Hydrophobic cores were found to be as fluid as bulk alkanes (Singer 1982, Shinitzky *et al* 1971, Pownall and Smith 1973, Kalyanasundaram *et al* 1975, Zachariasse 1978, Emert *et al* 1979, Turro *et al* 1979, Turro and Okubo 1981, Stilbs *et al* 1983a). Cores of water-in-oil microemulsions could also be studied with water-soluble probes but their adsorption at the polar interface cannot be excluded (Wong *et al* 1976, 1977, Valeur and Keh 1979, Keh and Valeur 1981).

## 4. Binary micelles

### 4.1. Micelles in water

Sometimes called 'direct micelles', binary micelles in water are obtained by spontaneous dissolution of surfactant. They have to be distinguished from vesicle solutions which are usually obtained as clear solutions upon dispersion with high-power ultrasound (sonication), i.e. addition of mechanical energy to produce a state which is metastable for most surfactant molecules tested so far.

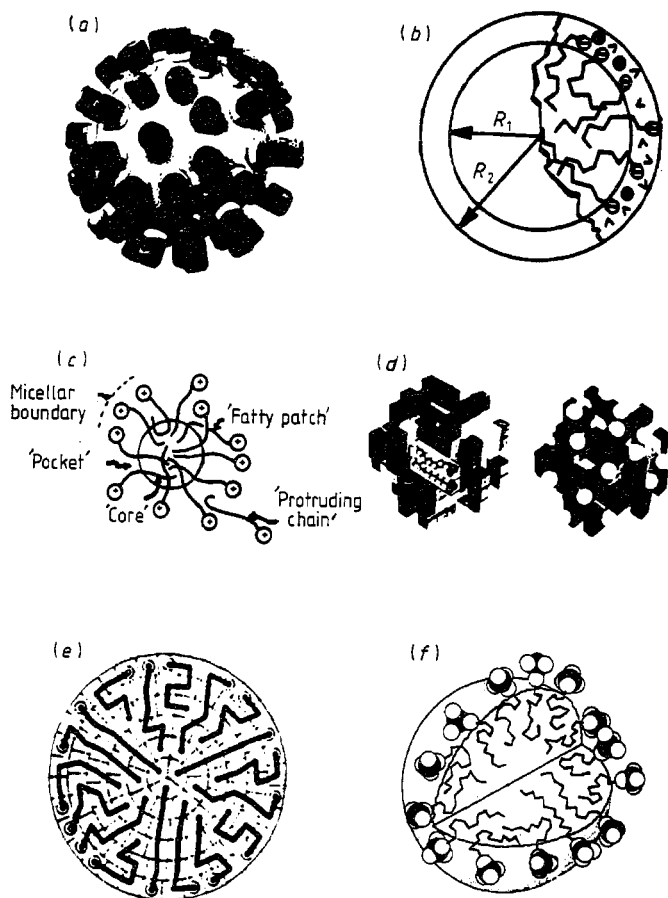
**4.1.1. Current theoretical models of micellar structure.** It is now accepted that dilute micellar aggregates are spherical droplets of aggregated surfactant molecules: the first direct proof has been obtained by Reiss-Husson and Luzzati (1964) by comparing the x-ray scattering spectra on an absolute scale with lamellar and spherical geometries.

Before this crucial work, it was accepted that small-angle scattering had 'proved' that micellar solutions were a dilute dispersion of surfactant bilayers of the same nature as surfactant vesicles. This 'proof' was only supported by the similarity of shape between scattering spectra of bilayers and micelles.

Micellar aggregation occurs because two opposite contributions to the free energy of micellisation are present. The task of thermodynamic models is to calculate the standard chemical potential  $\mu_0(N)$  of a surfactant molecule in a micelle of aggregation number  $N$ . The size distribution is directly obtained and properties are deduced from the relevant partial derivatives of the free energy. These are to be compared with experimental structural data.  $\mu_0(N)$  may include many terms according to the degree of sophistication of the model. The main difficulty is to describe the mutual repulsions of headgroups in the interface which include electrostatic interactions at a molecular level (a dielectric constant cannot be defined), steric hindrance, solvation, etc. A complete thermodynamic theory of micellar aggregation would be of enormous intricacy and consequently rather obscure, the few main contributions have been examined by several authors. Some known structural information has necessarily been introduced into models and a single term in  $\mu_0(N)$  expressions often includes several physically distinct phenomena. Tanford (1974a, b, 1977) has considered hydrophobic and interfacial free energies and was able to calculate with some success the free energy for different sizes and shapes; the area per headgroup was related to the micellar size and shape by simple geometrical considerations since the micellar core is water free. The hydrophobic contribution was obtained from the known free energy of transfer of a surfactant molecule from water to hydrocarbon corrected for water-hydrocarbon contact at the interface (which is equivalent to adding an interfacial tension term). The interfacial free energy proportional to the inverse area per headgroup was taken from experimental pressure-area isotherms at water-hydrocarbon interfaces. The problem is that the high curvature of micelles is not accounted for by pressure-area data which concern flat interfaces. Israelachvili *et al* (1976, 1980) and Mitchell and Ninham (1981) calculated curvature contribution by modelling the interface as a curved capacitor. This model, which retains the same area per headgroup dependence as Tanford's, is especially valid for zwitterionic or ionic surfactants at high ionic strength for a complete electrical neutralisation of the interface to occur at small distances.

An alternative is to calculate the electrostatic free energy by a numerical resolution of the Poisson Boltzmann equation in a cell (see § 4.1.7) (Gunnarsson *et al* 1980). Upon addition of a rough estimate of the free energy of chain packing inside the micellar core as a third term and rotational and translational entropy, McMullen *et al* (1984a, b) have shown that many aggregate shapes are allowed and may be in equilibrium because of their small energy differences, so that a defined shape cannot be *a priori* decided for a model. Extreme refinement of chemical potentials is insignificant compared with the basic assumptions of such theories. Indeed, all these calculations work with chemical potentials varying as the logarithm of mole fractions, thus under ideal solution conditions. This is obviously a formally incorrect assumption as interactions must be present for aggregation to occur, a problem which has been addressed by Mitchell and Ninham (1981). The important feature in thermodynamic theories is the cancellation of large and opposite contributions which makes the system very sensitive to slight changes in external conditions.

Very different and contradictory models for micellar structure have been proposed, and the discrimination between them has occurred only recently (figure 12). The original oil droplet model is due to Hartley (1936): micelles are spherical with a liquid



**Figure 12.** Different structural models of micelles as they have appeared in the literature. (a) Hartley (1936) spherical micelle with a disordered, water-free, hydrophobic core; (b) Hartley micelle as used for the interpretation of scattering experiments; (c) Menger (1979) micelle with extensive water penetration in the core; (d) Fromherz (1981a, b) surfactant block model with maximum order inside the core; (e) Dill (1984a, b) and Cantor (1984) model for filling the micellar core with hydrocarbon chains, no chain folding is allowed; (f) Gruen (1985a, b) model of chain packing inside the micellar core is the result of simple constraints (constant inner density, headgroups sitting at a spherical interface).

water-free core made of hydrocarbon chains; headgroups are at the interface. This model is still used for the resolution of scattering experiments, an outside 'interfacial' layer contains the headgroups, bound counter-ions and hydration water. Two contrasting models of internal structure have appeared: Menger (1979) proposed a loose micelle with a large penetration of water inside its core. The extensive water-hydrocarbon contact allows the interpretation of spectroscopic and reactivity experiments using probes (see § 3.5). In contrast, Gruen (1985a, b) proposed a model similar to Hartley's in that the micellar core is liquid and water free, and has calculated hydrocarbon chain packing at constant density inside it. Dill and Flory (1980, 1981), Dill and Cantor (1984), Dill (1984) and Cantor and Dill (1984) have presented similar calculations, but the surfactant alkyl chains were not allowed to fold. From the idea that the internal state of micelles is intermediary between that of a liquid droplet of uncorrelated alkane molecules and that of a crystalline lamellar phase, Fromherz (1981a, b, 1984) builds up a micelle with preformed trimers of parallel correlated surfactant molecules.



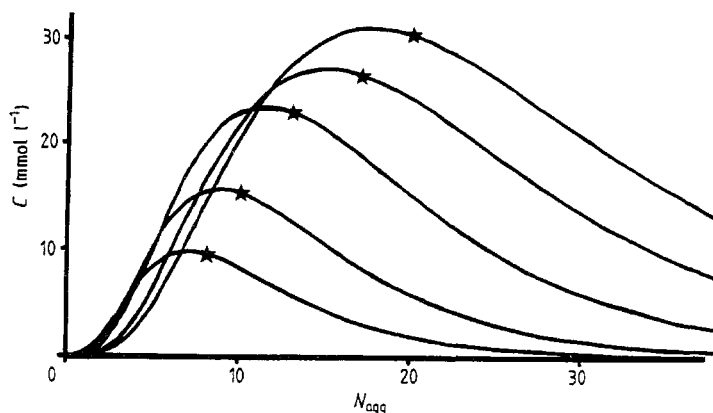
Dimensions of a trimer block are expected to correspond to the correlation length of molecules within the block. Micelles are built with perpendicular blocks up to the aggregation number and a well defined cubic shape results from this model.

**4.1.2. Aggregation number, volume and average shape.** Aggregation numbers are measured by means of scattering methods (see § 3.2). This way is more direct than many other methods described in the literature and is free of any choice of thermodynamic model. The main feature of experimental results is that micelles are either small, their size being 1 to 1.5 times the maximum possible size of a perfect smooth sphere, or very large. Small micelles are obtained with ionic and zwitterionic surfactants, while large ones are made up with non-ionics or ionics under conditions of high ionic strength (Chen 1986a, b). This is consistent with the idea that size is controlled by repulsive interactions between headgroups at the interface. When repulsions increase, the surfactant parameter  $a_0 l_c / v$  increases and micelles get smaller.

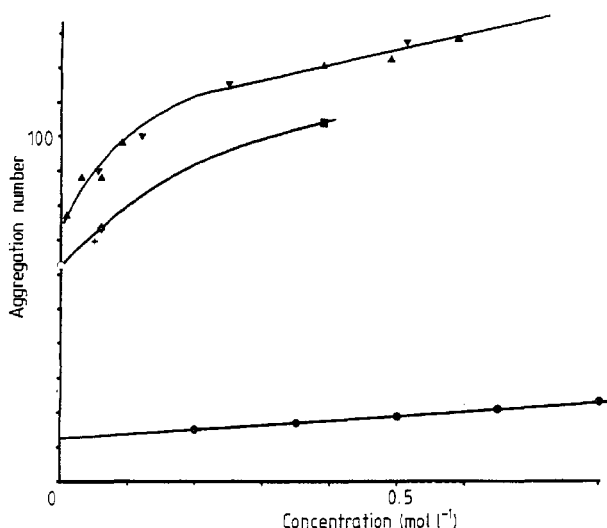
As a consequence, size and CMC are related: surfactants with low CMC aggregate into large micelles, the higher the CMC, the smaller the micelles. Experimental evidence which supports this picture is size variations upon changing interfacial electrical charge. Thus, a continuous size increase (and CMC decrease) is observed as salt is added in ionic micellar solution (Emerson and Holzer 1967, Hayashi and Ikeda 1980, Bendedouch *et al* 1983a, b, Hayter and Penfold 1983, Cabos 1983, Imae and Ikeda 1985). The same effect is obtained when the surfactant solution is concentrated, increasing the ionic strength (Hayter and Zemb 1982, Hayter and Penfold 1983, Zemb *et al* 1983, Bendedouch *et al* 1983a, b, Magid *et al* 1983, Drifford *et al* 1984, Berr *et al* 1986b, Sheu *et al* 1987). Salt does not actually change the electrical charge per headgroup but screens interheadgroup repulsions upon condensation effects. A true headgroup charge variation can be obtained with derivatives of weak acids (carboxylates, phosphates, etc): lowering the pH causes protonation of some anionic headgroups with a subsequent charge variation. A similar effect can be obtained by complexation with charged species such as calcium cations (Trauble and Eibl 1974, Tocanne *et al* 1974, Verkleij *et al* 1974). Alkylcarboxylic acids are water insoluble (infinite aggregation) while the corresponding carboxylates form small micelles. Corkill *et al* (1970) and Ikeda *et al* (1979) have shown a size decrease as a zwitterionic surfactant is converted into a cationic one.

Chevalier *et al* (1985) allowed the charge of alkylphosphates to vary from one as in conventional surfactants to two. Highly charged micelles are found to be small and polydisperse, their size showing a strong concentration dependence. The size of monocharged surfactant micelles exceeds slightly the limit of sphericity and depends only slightly on concentration. Some general trends can be outlined from available experimental data. Short-chain or highly charged surfactants form spherical polydisperse micelles whose size is lower than the limit of sphericity (figure 13). Long-chain (above about eight carbon atoms in a linear chain) monocharged surfactant micelles reach quickly a plateau in their size dependence with concentration (figure 14). Micelles are slightly aspherical (Cabane *et al* 1984, 1985, Berr *et al* 1986a) and further growth is hindered. Size increase implies a strong deformation and occurs abruptly as a sharp transition (improperly called second CMC), giving rise to the very large micelles discussed in the next section.

Non-spherical micelles should be expected whenever the radius of a sphere, including the measured number of surfactant molecules, exceeds the length of the fully extended surfactant molecule.



**Figure 13.** Distribution of aggregation numbers of disodium octylphosphate micelles for several surfactant concentrations  $C$  (Chevalier *et al* 1985). The Schulz distribution is assumed, only widths (variance) are from experiments.  $C$ —CMC are 0.12 M, 0.24 M, 0.47 M, 0.71 M and 0.95 M, giving polydispersity curves with increasing mean aggregation numbers (stars).



**Figure 14.** Variation of aggregation numbers with concentration ( $C$ —CMC) for sodium octanoate (●, Hayter and Zemb 1982) and sodium dodecylsulfate (SDS) with molecular volumes from crystallographic data: ▲, Hayter and Penfold (1983); ▼, Sheu *et al* (1985); with experimental molecular volumes: ◆, Cabane *et al* (1984, 1985); +, Triolo *et al* (1985); ■, Berr *et al* 1986b. The aggregation number at the CMC (○) is from light scattering (Mysels and Princen 1959).

The calculation of size distribution requires knowledge of standard chemical potentials of molecules in micelles as a function of the aggregation number  $N$ . Several attempts have been made with different analytical forms of chemical potentials (Tanford 1973, 1974a, b, 1977, Israelachvili *et al* 1976, 1980, Eriksson *et al* 1979, 1981, McMullen *et al* 1984a, b). The results depend strongly on starting assumptions, the most complete potential being that of McMullen *et al* (1984a, b). Nevertheless, without any assumption other than ideal solution behaviour, the width  $\sigma$  of the size distribution ( $\sigma^2 = \langle N^2 - \bar{N}^2 \rangle$ ) is related to the concentration (mole fraction of micellised molecules  $x_{\text{mic}}$ )

dependence of the average aggregation number  $\tilde{N}$  as (Hall and Pethica 1966, Corkill *et al* 1969, Israelachvili *et al* 1976)

$$\sigma = \tilde{N} \frac{\partial N}{\partial \ln(x_{\text{mic}})}.$$

This relation expresses that monodisperse micelles have a size independent of concentration. On the other hand, the concentration dependence of aggregation number provides the only direct method of measuring polydispersity. This is useful in interpretation of scattering experiments (Chevalier *et al* 1985). When ideal solution conditions are not fulfilled, intermicellar interactions influence  $\tilde{N}$  (Ben-Shaul and Gelbart 1982, Gelbart *et al* 1984) and this relation is no longer valid.

It is indeed hard to distinguish between size distribution and non-spherical shape in small-angle scattering spectra. Theoretical considerations or examination of adjacent anisotropic phases (see § 4.1.3) may be of considerable help for that purpose.

**4.1.3. Large micelles.** The area per headgroup in an ellipsoid being less than that of a sphere, the transition to large micelles occurs upon a reduction of interheadgroup repulsions. It may then be caused by salt or surfactant addition, or solute solubilisation (such as alcohols (Ekwall *et al* 1969a)). Difficulties in experimental studies of such either concentrated or ternary solutions have been overcome in early work by a crude modelling as rigid cylinders without mutual interactions (Mazer *et al* 1976, 1977, Missel *et al* 1980, 1982, 1983). If most conclusions cannot resist a careful reinterpretation, the small enthalpy difference between spheres and cylinders has been recognised and measured (Mazer and Olofsson 1982). Micellar growth is then controlled by entropy and a large polydispersity is expected.

Because of the limited length of a surfactant molecule, at least one dimension of large aggregates is equal to twice this length, leading to cylinders with spherical end caps or lamellae with toroidal rims. Such aggregates may be flexible. Energetically unfavourable edges can be avoided upon formation of toroids from cylinders or vesicles from lamellae, but the energy reduction may be compensated by a decrease of deformation entropy. Porte (1983) has shown that toroids are unlikely because of their bending entropy, but that vesicles should occur in those conditions.

Anisotropic aggregates are easily detected by induced birefringence or depolarised light scattering measurements. Most structural investigations of these systems are quasielastic light scattering studies. Intermicellar structure factors as well as hydrodynamic interactions (Ackerson 1978) are generally ignored.

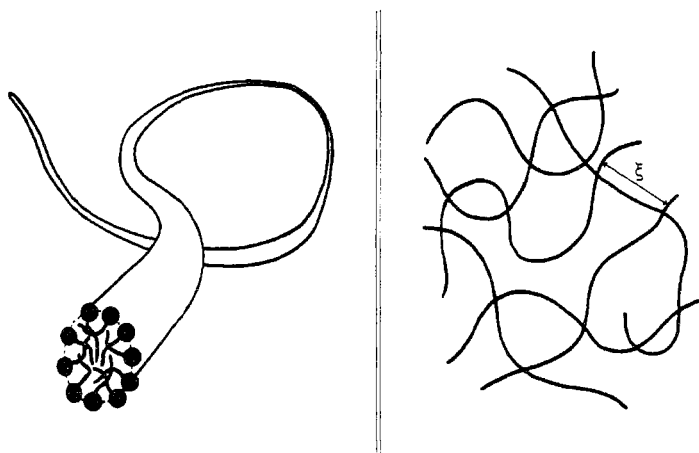
In this case, attractive interactions could be misinterpreted as micellar growth. Such interactions occur with non-ionic surfactant systems but may also occur in ionic ones (see § 4.1.7) (Appell and Porte 1983, Bendedouch and Chen 1984, Chao *et al* 1985). On the other hand, the experimental autocorrelation functions obtained from light scattering are not single exponentials. Biexponential decays corresponding to overall translational motion and internal structure fluctuations have been observed as varying the scattering angle (Flamberg and Pecora 1984, Pecora 1984). Most experimental evidence is then only of qualitative significance, if shear does not induce structural perturbations, shear alignment should allow more reliable conclusions to be made, aggregate dimensions and flexibility being observed (Cummins *et al* 1987a). As salt is added, rod-like micelles are formed which become flexible as they grow further (Porte *et al* 1980, Porte and Appell 1981, Appell and Porte 1981, Appell *et al* 1982,

Hoffmann *et al* 1983, Imae and Ikeda 1984a,b Imae *et al* 1985, van de Sande and Persoons 1985).

The transition from rigid to flexible micelles as well as the persistence length of coiled micelles depends on the nature of the surfactant molecules.

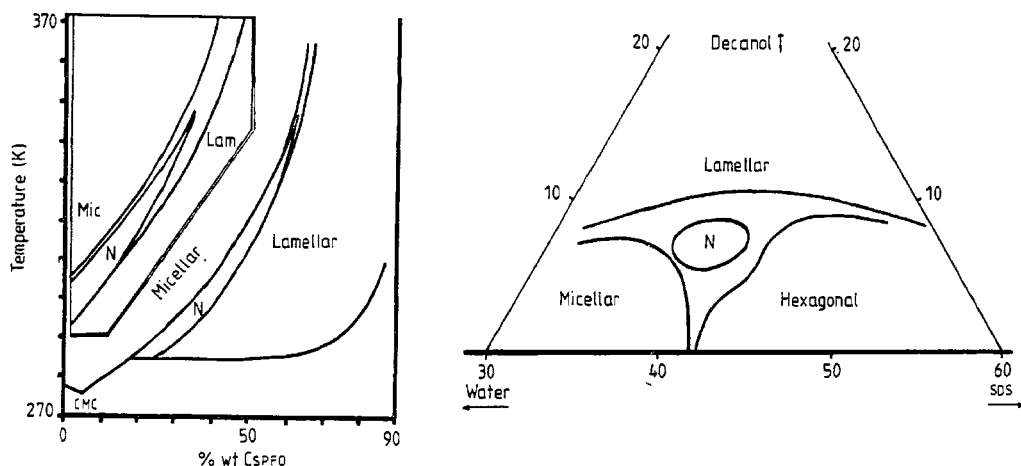
Indeed, the strength of interfacial interactions governs the variation of area per headgroup and thus the flexibility. When flexibility is strongly hindered as in hexadecylpyridinium salicylate micelles, interacting rigid rods are observed, resulting in a viscoelastic behaviour of the solution (Gravsholt 1976, Ulmius *et al* 1979, Hoffmann *et al* 1981a, Kalus *et al* 1982, Löbl *et al* 1984). Micellar growth is limited and a size decrease is observed at high concentrations; flexibility is nevertheless achieved with high enough electrolyte concentrations (Hoffmann *et al* 1981b, 1983 Angel *et al* 1984).

This behaviour is reminiscent of that of polymers: a rod-to-coil transition is observed while adding salt into a polyelectrolyte solution (Drifford *et al* 1985), a coiled polymer in a good solvent changes its conformation suddenly into a rod as a non-solvent is added (Lim *et al* 1983). Candau *et al* (1984, 1985) have noticed that the behaviour of large micelles as monitored by light scattering follows the same scaling laws as semidilute polymer solutions (Imae and Ikeda 1986, Kato *et al* 1987). With such an analogy, the description of micellar solutions as a dispersion of interacting particles is abandoned for a network of flexible rods correlated over a characteristic length  $\xi$  (figure 15).



**Figure 15.** Schematic diagrams of large flexible micelles (left) and their semidilute solution with a rough definition of the correlation length (right).

As a micellar solution is concentrated, micelles grow in size and ordered mesophases made up with infinite aggregates occur. In certain cases, the micellar size may remain finite and anisotropic micelles (large to small axis ratio  $< 3$  (Charvolin *et al* 1979, 1984, Hendrikx *et al* 1983)) get ordered in concentrated solutions, giving rise to nematic mesophases (Lawson and Flautt 1967, Rosevear 1968, Radley *et al* 1976, Fujiwara *et al* 1979, Fujiwara and Reeves 1980, Forrest *et al* 1980, Forrest and Reeves 1981, Gelbart *et al* 1985). With nematic phases, some continuity exists between micellar solutions and infinite aggregates in mesophases (figure 16). The incomplete powder average due to macroscopic ordering allows more structural information to be obtained out of experiments. Internal segregation in multicomponent micelles could be demonstrated in this way, the small headgroup compound being concentrated in the low-curvature



**Figure 16.** Phase diagrams showing the location of nematic phases (N): Left: temperature-concentration phase diagram of caesium perfluorooctanoate (CsPFO) and water (Boden *et al* 1979, 1987b). Right: part of ternary phase diagram of sodium decylsulphate (SDS), decanol and water (Hendriks and Charvolin 1981).

regions (Hendriks *et al* 1984). This important observation invalidates the use of elastic moduli in thermodynamic treatments of micellar aggregation: indeed the assumption that a linear relation exists between bending radius and energy is no longer correct if internal segregation occurs.

A relation between structures of isotropic and neighbouring anisotropic phases has often been stated, at least as an aid to discussion (Scriven 1976, 1977, Wennerström and Lindman 1979a, Danielsson *et al* 1982, Hoffmann 1984). Indeed, structural studies of concentrated micellar solutions made up of anisotropic aggregates are difficult because of loss of information due to orientational averaging, while this is not the case for adjacent ordered mesophases which coexist with isotropic solutions. Assuming a realistic standard chemical potential per molecule in a cylindrical micelle of aggregation number  $N$  as (Israelachvili *et al* 1976)

$$\mu(N) = \mu(\infty) + \frac{\alpha RT}{Np}$$

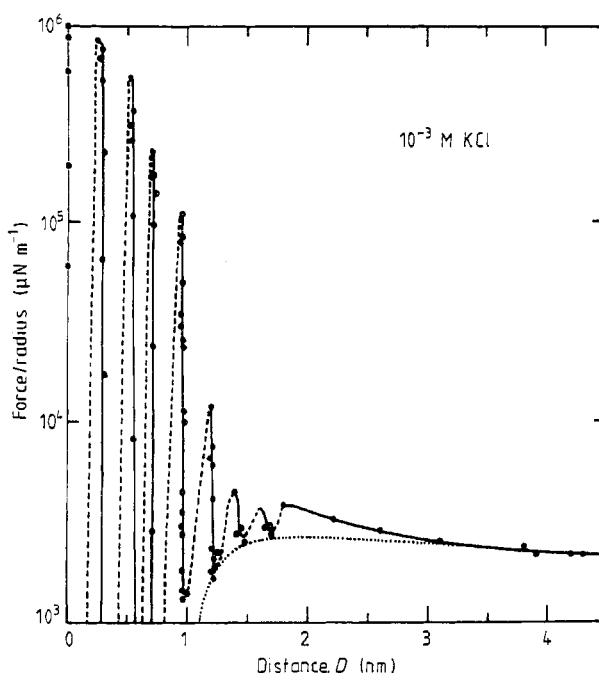
where  $\alpha > 0$  and  $p$  is the surfactant parameter  $p = v/a_0 l_c$ , a hexagonal mesophase is found to be able to coexist only with a concentrated solution of cylindrical aggregates ( $p = 1/2$ ), while a lamellar mesophase ( $p = 1$ ) can coexist with a dilute micellar solution (Wennerström 1979).

**4.1.4. Hydration, water penetration in the micellar core interface thickness.** Water is the surrounding medium of direct micelles and some water molecules are bound to the hydrophilic groups at the surface. The dehydration of the interface is necessary for two micellar cores to merge and requires the dehydration energy. Hydration water is thus defined as belonging to the hard volume of the micelles and is of importance for their stability.

The definition of hydration numbers requires the splitting of water into two physically distinct states as bound and free, which may seem arbitrary. Hydration of the interface first occurs through hydrogen bonds of high energy. If two micelles come into contact and coalesce, hydration water has to be removed. Hydration water

participates in the stabilisation of aggregates. The definition of hydration is thus the number of water molecules which are not expelled when micelles come into contact. This is also the definition currently used for protein structure studies (Tardieu *et al* 1987). The case of micellar colloids is simpler, since, in contrast to proteins, only one type of hydration water exists (water cannot enter hydrophobic regions).

Different definitions have been previously used, mainly for technical reasons: early studies of hydration made use of volume differences between dry particles (from static scattering methods) and hydrated ones (from dynamic measurements), hydration water is then that which moves with the particle in a two-state model. It was shown that these two different hydration numbers are frequently identical, but it should be recognised that all surfactant molecules studied so far were quite similar, differences could occur for chemically different headgroups. Several other hydration processes may occur: especially water structuring near hydrophobic regions (Badiali 1985, Tarazona and Vicente 1985). This phenomenon has been demonstrated by measurement of direct forces between mica plates (Pashley and Israelachvili 1984, Israelachvili 1985) (figure 17). It should not be inferred that unbound water is completely free. In ionic micellar solutions, electric fields are present everywhere in the solution and, because of the high dielectric constant of water, orientation of water molecules occurs everywhere.



**Figure 17.** Force between two mica plates immersed into water, as a function of their separation (Pashley and Israelachvili 1984). Oscillations are present at short separations, corresponding to removal of discrete hydration sheets.

Between water and hydrocarbon, the region of mixing is called the 'interface'. The interface is therefore defined as the region where neither water nor hydrocarbon may be considered as free of the other component. The interface thickness is related to the interfacial tension.

It should be noted that the interfacial area depends strongly on its localisation for systems with high curvature (with  $v/a_0l_c$  very different from one). This problem is difficult with direct micellar systems of high curvature and the choice of a clear definition for the interface should be provided every time an area per headgroup value is put forward.

According to this definition of the interface, each type of hydrophilic group has a well defined hydration number. These numbers are usually obtained by comparing the micelle hydrodynamic volume with its dry volume (Kushner and Hubbard 1954, Mukerjee 1964, Courchene 1964, Ekwall and Holmberg 1965, Schott 1967) or by self-diffusion measurements of water (Lindman and Brun 1973, Lindman *et al* 1976, 1980a, b). It is then assumed that only strongly bound water moves with micelles (two-site model). A more direct way, according to the above definition, is to evaluate the hard volume through interaction measurements (Hayter and Zemb 1982, Chevalier *et al* 1985). In cases of ionic micelles with strong condensation of counter-ions, the hard volume may be as large as twice the dry volume.

It was inferred that water could enter very deeply into the micellar core, forming 'fjords' in it (Menger 1979). This erroneous conclusion was based on incorrect interpretation of experiments involving the use of added probes (see § 3.5). The absence of water inside the micellar core has been clearly demonstrated by NMR relaxation (Clemett 1970, Podo *et al* 1973) and small-angle neutron scattering (Cabane *et al* 1984, 1985) experiments. Some water-hydrocarbon contact occurs, however, as headgroups are not bulky enough to allow a complete coverage of the hydrocarbon. The first methylene of hydrocarbon chains is also exposed to water because of its fast protrusive motion out of the micelles.

For this contact area to be a minimum, the interface has to be smooth, as observed by small-angle neutron scattering: the interface thickness is less than the experimental resolution ( $5\text{\AA}$ ) (Tabony 1984, Cabane *et al* 1985). This picture agrees with the small compressibility of micellar interfaces (Bloor *et al* 1984).

**4.1.5. Fluctuations of the interface.** Small micelles have an average spherical shape. This picture comes from experimental measurements which provide averages. Monte Carlo calculations allow simulations of the structure of an isolated micelle and have shown considerable shape fluctuations (Haan and Pratt 1981, Owenson and Pratt 1984, Pratt *et al* 1986). The time (or number) average is spherical. The width of the distribution of radii (distances between headgroups and centre of mass) is as large as the average radius. Headgroups remain, however, fully exposed to water. This wide distribution is entirely due to shape fluctuations and is neither an average interface thickness nor a water penetration into the micellar core. On the contrary, Ljunggren and Eriksson (1984) have shown that these large fluctuations occur only when the interfacial tension is low.

Fluctuations from a spherical shape require an area increase which is limited by interfacial tension. When the interfacial energy is less than  $kT$ , large fluctuations are allowed, which is not the case for micelles ( $\sigma = 15\text{ mN m}^{-1}$ ). Monte Carlo calculations provide larger shape fluctuations but the model is rough. The resolution of these calculations is only one third the length of a hydrocarbon chain. The chain flexibility is restricted by the lattice spacing, so that possibilities for a spherical micelle to occur are few. Conclusions should be confirmed through a more refined model and experiments. Shape fluctuations have been proposed for the interpretation of small-angle neutron scattering experiments with  $^2\text{H}$  labelled surfactants (Cabane *et al* 1985) (see

§ 4.1.6 and figure 19). The best theoretical simulations available are compatible with these experimental results (Haan and Pratt 1981, Halle *et al* 1988).

Some neutron spin-echo inelastic scattering experiments (Hayter and Penfold 1983, Huang *et al* 1987) allowed a direct measurement of the contribution of the shape fluctuations of the interface.

**4.1.6. Packing of hydrocarbon chains.** Hydrocarbon chains are confined in a medium of defined average shape and have to fill its whole volume with a constant density. The micellar core being liquid-like (Roberts and Chachaty 1973) and free of water (see §§ 2.1 and 4.1.4) its density is that of a liquid alkane. Headgroups have, moreover, to sit at the interface. *Gauche* conformations have an energy 2.3 kJ higher than *trans* ones (Flory 1969). The alkyl chains must obey these three constraints.

Both experimental and theoretical approaches have been used in studying this problem. Theoretical ones consist in calculations of the free energy of all the conformations which satisfy these three constraints (Bothorel *et al* 1974, Belle and Bothorel 1974, Gruen 1981, 1985a, b, Gruen and De Lacey 1984, Ben-Shaul *et al* 1984, 1985, Ben-Shaul and Gelbart 1985, Szleifer *et al* 1985). A different approach from Dill and Flory (1980, 1981), Dill and Cantor (1984) and Cantor and Dill (1984) suffers from the inability of chains to fold; further, it disagrees with experimental observations of the occurrence of terminal methyl groups at the interface.

Current theoretical studies assume that micelles are only spherical on the time average, significant instantaneous deviation being possible and headgroups being located at the surface, and that the internal core is 'liquid'. They show a great disorder in the micellar core, in agreement with its liquid-like character. The orientational order is larger for the first methylenes (close to the headgroups) than at the end of the alkyl chains. Chains were also found uncorrelated (Gruen 1985a), which seems surprising allowing for the constraints and since correlations do exist in bulk liquid alkanes (Bothorel 1968, 1974, Bothorel *et al* 1972, Fourche and Bothorel 1969). This orientational order can be estimated through an order parameter  $S = \langle 3 \cos^2 \theta - 1 \rangle / 2$ ,  $\theta$  being the angle between the radial vector and the normal to the H-C-H plane of a methylene (Edholm 1982). This order parameter is twice that found by  $^{13}\text{C}$  NMR relaxation (see § 3.3) and allows successful comparisons with experimental data (Gruen 1985a, b) (figure 18).

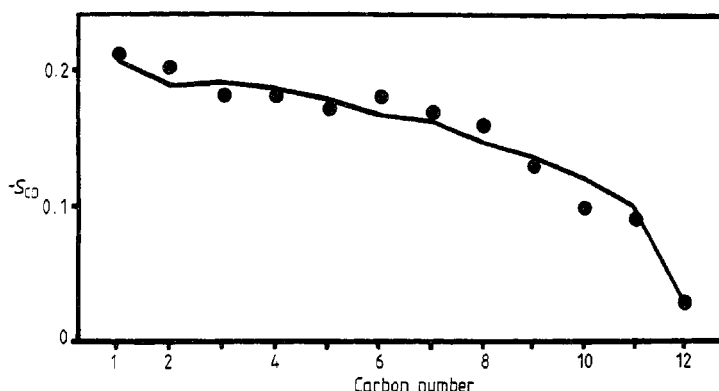


Figure 18. Alkyl chain order parameter profile as obtained from  $^{13}\text{C}$  NMR relaxation experiments (dots) of Walderhaug *et al* (1984) and Gruen's (1985a, b) packing calculations (full curve) for dodecyltrimethylammonium chloride micelles in water.



Experimental determination of the order parameter along the chain shows a very slight dependence of the order parameter profile on area per headgroup (Mély *et al* 1975, Charvolin and Hendriks 1985).

The radial distribution of methylenes has been studied by small-angle neutron scattering and does not fully agree with chain packing calculations. Bendedouch *et al* (1983a) have deduced a distribution of terminal methyl groups which peaks towards the micelle centre. This is a result of the poor experimental resolution. A more disordered picture emerges from high-resolution experiments (Cabane *et al* 1985) which show broader distributions than theory (figure 19). Packing calculations agree with molecular dynamics (van der Ploeg and Berendsen 1983) and Monte Carlo (Haile and O'Connell 1984) simulations carried out under similar hypotheses, namely a well defined aggregate shape. This discrepancy could be ascribed to shape fluctuations which have not been introduced into theoretical models. When some fluctuations in the headgroup positions are allowed, the group's distributions indeed broaden in the correct way (Woods *et al* 1986). The internal structure models are thus essentially correct but the choice of one particular (average) shape for the micelle introduces too much order into the structure. Further progress in this field requires more structural data to be introduced in the model, namely shape anisotropy and fluctuations as discussed in § 4.1.5.

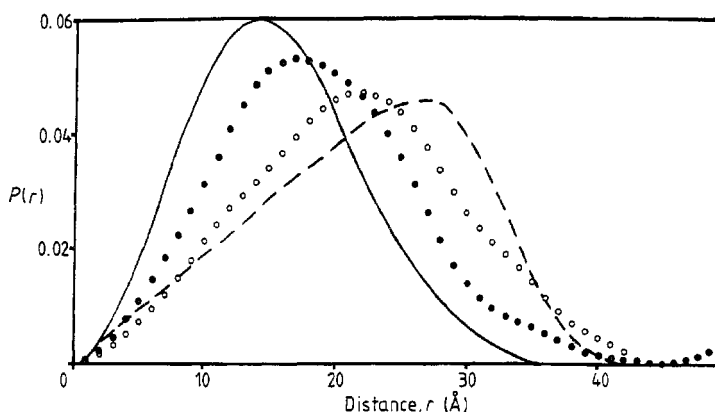


Figure 19. Distribution of distances (Patterson function  $P(r)$ ) for methyl and  $\gamma$  methylene groups in SDS micelles as obtained from experiments (Cabane *et al* 1985):  $\circ$ , methyl;  $\bullet$ ,  $\gamma$  methylene; and theoretical calculations (Gruen 1985a): ---, methyl; —,  $\gamma$  methylene.

**4.1.7. Effective charge, bound counter-ions, interactions between ionic micelles.** The structural charge of a micelle (molecular charge times aggregation number) is considerably reduced by counter-ion condensation, especially when large concentrations of salt are added. This leads to a smaller charge, as experienced far from the micelle (from another micelle). This net charge, reduced by counter-ions 'near' the interface, is referred to as 'effective' charge.

It is therefore usual to classify counter-ions as bound and free (see § 2.6), which is a good approximation when compared with the resolution of most experimental techniques (Clifford and Pethica 1964, Lindman and Brun 1973, Lindman *et al* 1976, 1982, 1984a, b Kamenka *et al* 1974, 1977, Smith 1979, Stilbs and Lindman 1981, Almgren *et al* 1985, Lindström *et al* 1985, Jansson and Stilbs 1985).

'Bound' counter-ions are only electrically attracted toward the micellar surface, there is no real chemical binding and counter-ions remain fully hydrated (Gustavsson and Lindman 1975, Lindman *et al* 1976).

With ionic surfactants, a highly charged polyion should result from aggregation of hydrophobic ions. Counter-ions of opposite charge are then attracted toward the micelle as in any polyelectrolyte solution. This counter-ion 'condensation' contributes to micelle formation, their presence close to the charged interface reduces its charge density and makes the aggregate stable by weakening electrostatic repulsions between headgroups. It is usual to separate counter-ions into two states as condensed (or bound) and free, and a fraction of bound counter-ions  $\alpha$  is defined. For scattering studies, a counter-ion is considered to be 'bound' if it is included in the excluded 'hard sphere' volume of the aggregate. It turns out to be nearly equivalent to the definition of the  $\zeta$  potential, where the ions are bound when they move with the Brownian motion of the particle. The 'bound' state is therefore much more general than a chemical binding.

The radial concentration profile of heavy counter-ions can be studied in the wide-angle region of small-angle x-ray scattering patterns. The very high resolution of NMR is of no help because of the fast exchange rates of ions between different environments. Bound counter-ions allow charged headgroups to get close to each other, providing the micellar aggregate with its stability. They thus belong to the micelle. Removing counter-ions from the surface causes an increase of area per headgroup. Increasing the charge of headgroups causes more counter-ions to condense.

Given an ion, compensation occurs when one attempts to modify the effective interface charge, for example by pH variation around the pK of the headgroups, so that the effective charge per unit area is kept constant around a reasonable value of some  $\mu\text{C cm}^{-2}$  for a given  $a_0 l_c / v$  (Chevalier *et al* 1985, Drifford *et al* 1985). Intermicellar interactions are repulsive with monovalent counter-ions and the micelles are stable. The potential between ionic micelles is, in all cases studied so far, well approximated as the sum of a hard sphere and screened Coulomb potentials, even if the real potential is obviously more complicated (Cabos *et al* 1979, Corti and Degiorgio 1981, Hayter and Penfold 1981a, b, 1983, Hayter and Zemb 1982, Magid *et al* 1983, Triolo *et al* 1983, Bendedouch and Chen 1983, 1984, Cabane *et al* 1984, 1985).

As salt is added, or with multivalent counter-ions, the strong intermicellar correlations become vanishingly small. For high ionic strengths, the electrostatic repulsive potential is weakened to such an extent that an attractive van der Waals potential may predominate (Bendedouch and Chen 1984, Chao *et al* 1985) and demixing phenomena are observed (Appell and Porte 1983). This DLVO-type description should be modified: because of counter-ion size and correlations, electrostatic interactions can be attractive.

Looking at counter-ions as point-like electrical charges only is an oversimplification. Counter-ions have their own specificity. Although it may not be the only important factor, the size of hydrated ions strongly influences the micellar behaviour. Given a charge, hydrophobic (less hydrated) counter-ions show larger condensation effects (Marra 1986) and give rise to a smaller area per headgroup. Thus, hexadecyltrimethylammonium or hexadecylpyridinium bromides aggregate into large micelles which turn into hexagonal mesophases when concentrated, while chlorides form spherical micelles which get ordered as cubic mesophases (Anacker and Ghose 1968, Ulmuis *et al* 1978, Almgren *et al* 1979, Porte *et al* 1980, Porte and Appell 1984). With cations, the differences are less (Mukerjee *et al* 1967, Cabos 1983, Berr *et al* 1986a), however, complexation of cations with a cryptand results in a considerable growth of counter-ion

size and a concomitant reduction of micellar size (Evans *et al* 1986b, Miller *et al* 1987a, b). Highly hydrated hydroxide counter-ions are adsorbed far from the interface on cationic micelles, which are consequently small (Lianos and Lang 1983, Hashimoto *et al* 1983). The same occurs with organic counter-ions, but their behaviour is more difficult to rationalise (Jansson and Stilbs 1985, 1987). The viscoelastic behaviour of hexadecyltrimethylammonium salicylate micellar solutions is also due to the specific high condensation of the counter-ion (Ulmius *et al* 1979, Hoffmann *et al* 1981a, b, 1983, Kalus *et al* 1982, Löbl *et al* 1984, Angel *et al* 1984, Olsson *et al* 1986). Upon a photochemical transformation of the counter-ion, Wolff and von Büнау (1984a, b, 1986), Wolff *et al* (1987) and Müller *et al* (1984) drastically change the macroscopic properties of micellar solutions. Studies with hydroxide or organic counter-ions have shown the limits of conventional theories of micellar aggregation (primitive models of electrostatics): spontaneous vesicles are formed (Ninham *et al* 1983b, c, Talmon *et al* 1983, Hashimoto *et al* 1983, Brady *et al* 1984, 1986) and vesicle-to-micelle transitions can be observed. This suggests that some unknown additional interactions which are usually neglected become important (Evans *et al* 1986a, Ninham and Evans 1986).

The electrical potential of the aggregate, although screened by counter-ion 'condensation', is efficient over large distances. The larger the ionic strength (surfactant and added electrolyte concentrations), the smaller its range. In direct micelles, the screened electrostatic potential between two colloidal droplets is therefore a valid approximation (Hayter and Penfold 1981a, b):

$$u(r) = \frac{Z_0^2 e^2}{\epsilon (1 + \kappa \sigma_0 / 2)^2} \frac{\exp[-\kappa(r - \sigma_0)]}{r}.$$

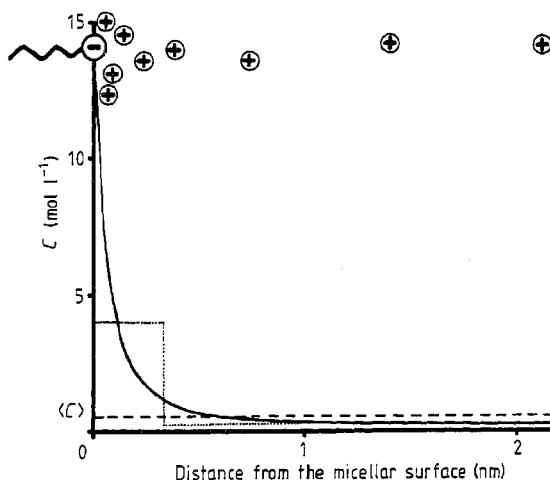
This electric potential applies for two rigid spheres of charge  $Z_0$  and diameter  $\sigma_0$  separated by a continuous medium where counter-ions are considered as point-like and the ionic strength fixes the inverse Debye length  $\kappa$ . As a rule of thumb, the inverse Debye length (in angströms) in pure water at 25 °C is given by  $\kappa = 0.33\sqrt{c}$ ,  $c$  being the ionic strength expressed in moles per litre.

This potential applies only outside of the hard sphere 'core', which includes the 'bound' counter-ions and the hydration water. It can be shown that this one-component macrofluid (OCM) approach is formally equivalent to the more rigorous calculation of pair distribution functions between micelles using HNC (the hypernetted chain, see below) to solve the 'primitive model' where no 'bound' counter-ions exist since all counter-ions are taken into account with their own diameter in this two-component model (Belloni 1986a).

The counter-ions are distributed in water according to their electrostatic energy. The counter-ion distribution is given by the Poisson-Boltzmann equation. This equation has no analytical solution, and different types of approximations are used in order to solve it analytically or numerically. The most reasonable compromises between approximations and computational heaviness are Poisson-Boltzmann cell numerical methods (Marcus 1955, Alexandrowicz 1962, Alexandrowicz and Katchalsky 1963, Katchalsky *et al* 1966, Bell and Dunning 1970, Westra and Leyte 1979, Delville 1980, 1984, Delville *et al* 1984, Gunnarsson *et al* 1980, 1982, Bratko and Vlachy 1982, Belloni *et al* 1984). In these procedures, the solution is modelled as an electroneutral cell containing one aggregate of uniform charge density immersed in a continuous solvent where point-like ions sit. The cell volume is the inverse number density of monodisperse aggregates and has the same symmetry as the aggregates. As ions are assumed point-like and influenced by the polyion screened potential, ion-ion correlations are neglected.

This method compares well with numerical simulations (Monte Carlo, Brownian dynamics (Linse *et al* 1982, Åkesson and Jönsson 1985)) for monovalent counter-ions and low charge densities. It may be unsuccessful, however, in accounting for intermicellar properties (osmotic pressure) because micelle-micelle correlations are not considered in a cell model (Wennerström *et al* 1982, Linse and Jönsson 1983, Jönsson *et al* 1984).

In the case of failure, some more accurate models are available such as the hypernetted chain (HNC) (Elkoubi *et al* 1977, Rogers 1980, Beresford-Smith and Chan 1982, Linse and Jönsson 1983, Kjellander and Marcelja 1984, Belloni 1985). Theoretical calculations usually show different kinds of counter-ions as going away from the charged surface which can be separated into two classes, 'condensed' and 'free' (figure 20), once a clear definition of the boundary has been adopted, giving significance to the two-state model for ions.



**Figure 20.** Radial concentration profile of counter-ions around a spherical micelle (CMC = 0.14 M,  $C = 0.5$  M,  $N_{\text{agg}} = 45$ ) as calculated from Poisson-Boltzmann cell method.  $\langle C \rangle$  is the average concentration, the dotted line shows the two-step (bound and free) approximation.

However, this separation remains arbitrary. Inside the separation surface, bound ions belong to the micelle while, outside of it, ions are free. The inner volume may be chosen as one which is already known, such as the excluded volume in scattering experiments, or the hydrodynamic one. The particle is usually defined in classical colloidal science as the volume inside the surface of zero shear where the zeta potential is measured by electrophoretic experiments. Electrostatic calculations of absorption excesses at the micellar surface provide some theoretical grounds for this so-called 'dressed micelle' model (Evans *et al* 1984).

The free energy of surfactant assembly is composed of four terms

$$\mu_N^0 = \mu_1^0 + \Delta g_{\text{HC}} + g_{\text{surface}} + q_{\text{packing}} + g_{\text{int}}$$

where  $g_{\text{int}}$  is the interparticle interaction term which is usually neglected, whereas the transfer to the hydrocarbon environment, the packing and the surface balance between electrostatic repulsion and water/hydrocarbon contact are important (Mitchell and

Ninham 1981, Evans and Ninham 1983). In fact  $g_{\text{surface}}$  is a balance between electrostatic repulsion and an interfacial tension term

$$g_{\text{surface}} = \gamma A_0 + \int_0^{\sigma_0} \psi'_0 d\sigma$$

where  $\psi'_0$  is the electric potential at the surface when the surface charge per area is  $\sigma$ . Then, the geometry of the aggregate imposes a relation between the surface potential and charge: in the case of a spherical micelle, one obtains a quantitative prediction of the surface charge by the so-called 'dressed micelle' model (Evans *et al* 1984). The optimal micelle has aggregation number and charge such that surface free energy is minimum

$$\frac{\partial g_{\text{surface}}}{\partial A} = 0$$

and thus one obtains a constraint for the electrostatic free energy

$$\gamma = \frac{\partial g_{\text{el}}}{\partial A}.$$

This theory, known as the 'dressed micelle' model, allows reasonable prediction of surface charge when the aggregation number and interfacial tension are known.

Because of ion-ion correlations (counter-ion excluded volume effect) and interface electrical polarisation, deviations from Poisson-Boltzmann behaviour may be considerable (Nägele *et al* 1985). Indeed, because of the mobility of amphiphilic molecules at a dielectric discontinuity, the hypothesis of uniform charge density at the interface no longer applies when charged counter-ions (especially multivalent ones) approach it; a surface polarisation occurs which can be easily accounted for by the introduction of fictitious image charges in electrostatic calculations (Kjellander and Marcelja 1985a). Depending on the location of surface charges about the dielectric discontinuity, attraction or depletion of counter-ions results from the image charge phenomenon (Kjellander and Marcelja 1984, 1985b, Linse 1986, Bratko *et al* 1986). On the other hand, because of ion-ion correlations with bulky, multivalent counter-ions, the electrostatic attraction of micelles by counter-ions may exceed the intermicellar repulsion when it is weakened enough by counter-ion screening (far enough from the charged micellar surface). An intermicellar attraction is then found which is electrostatic in origin and can be considered as a condensation of micelles around counter-ions (Muller and Derjaguin 1983, Guldbrand *et al* 1984, Svensson and Jönsson 1984, Kjellander and Marcelja 1985a, b, 1986a, b, Belloni 1986b).

**4.1.8. Attraction between micelles: cloud points and phase equilibria.** Dispersion forces between identical particles are always present and attractive, so that attractive interactions between micelles always occur. Such interactions may also have different origins. Electrostatics are so important and long range with ionic systems, that they allow attractive interactions to be neglected, except when high salinity ( $0.1 \text{ mol l}^{-1}$  screens electrostatics at distances shorter than the average intermicellar distance, leaving the van der Waals attractions as the main interaction). Also, this is not the case for non-ionic surfactant systems. These surfactants are of wide industrial use and the most studied molecules are polyoxyethylene alkylethers of general chemical formula  $\text{H}(\text{CH}_2)_x(\text{OCH}_2\text{CH}_2)_n\text{OH}$  often abbreviated as  $\text{C}_x\text{EO}_n$ . As attractive interactions become important, the solution becomes cloudy (turbid) and demixing occurs: the

monophasic solution splits into a concentrated solution and a very dilute one (figure 21). Coexistence curves may be upper or lower consolute boundaries according to whether the diphasic domain is at the low- or high-temperature side of the boundary (Chakhovskoy 1956, Mitchell *et al* 1983). Closed diphasic domains showing both behaviours are sometimes observed (Herrmann *et al* 1966, Lang and Morgan 1980, Lang 1984) (figure 22). Critical points are always present. Note that an upper consolute boundary in water and simultaneous lower consolute boundary in oil explains the full ternary phase diagram of non-ionic microemulsions (Kahlweit *et al* 1988).

Phase diagrams have been put into a unified picture and explained through simple geometrical arguments by Mitchell *et al* (1983). An interesting feature of these binary systems is that, depending on compounds, the direct ( $L_1$ ) and reverse ( $L_2$ ) micellar domains are either separated by a lamellar mesophase domain or form a unique monophasic domain in the phase diagram. In this latter case (figure 23), surfactant and water can be mixed in any proportion. The structural inversion from direct to reverse micelles upon water removal, which is reminiscent of that of microemulsions (see § 6), is currently a subject of intensive study. Mitchell *et al* (1983) have predicted generic phase diagrams from geometrical considerations (figure 24). However, it does not account for the possible 'melting' of lamellar phases into reverse micelles, the  $L_2$  phase is thus not predicted. The structure of this  $L_2$  phase is not known.

In the dilute region, where structure is the simplest to study, the non-ionic micelles are generally small at the CMC ( $N_{\text{agg}} \approx 80\text{--}200$ ).

Considerable growth may occur, depending on compounds, even for fairly dilute solutions. The reason for the rapid growth is the small size of headgroups together with the absence of long-range repulsive interactions between them ( $a_0 l_c/v$  is low). Growth is induced by dehydration of headgroups which allows a smaller area per headgroup. Such a dehydration occurs on concentrating and on heating, so that non-ionic micelles grow on heating contrary to what is expected on the basis of entropy. As the demixing boundaries are approached, according to different interpretations, micelles either remain small (Staples and Tiddy 1978, Hayter and Zulauf 1982, Triolo *et al* 1982, Zulauf and Rosenbuch 1983, Zulauf *et al* 1985, Magid *et al* 1984a, Zana and Weill 1985, Nash *et al* 1987) or grow in size and anisotropy (Nilsson *et al* 1983,

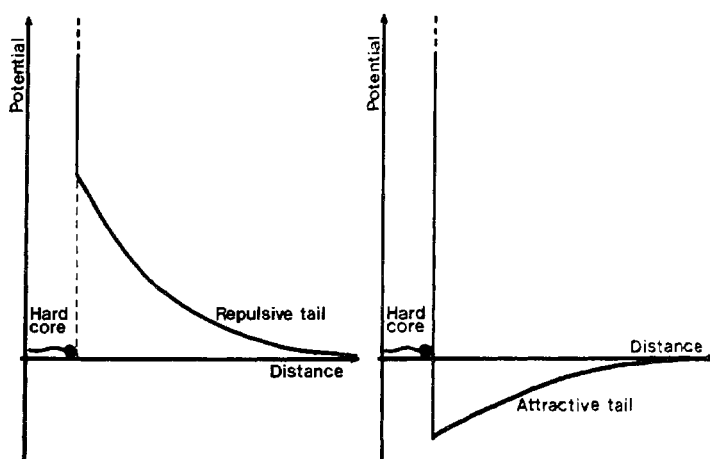
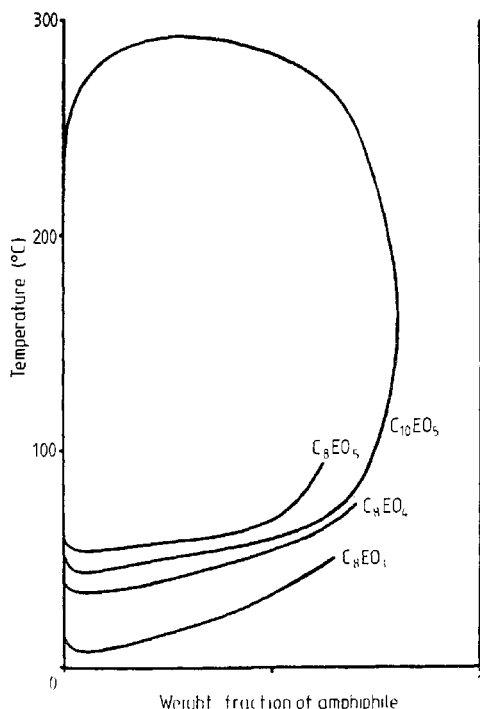


Figure 21. Two types of potentials about a micelle. Left: hard core and long-range repulsive (electrostatic). Right: hard core and long-range attractive (van der Waals).



**Figure 22.** Typical phase diagrams of non-ionic amphiphilic polyoxyethylene alkylethers ( $C_nEO_n$ ). The low-temperature region is shown for  $C_8EO_3$ ,  $C_8EO_4$  and  $C_8EO_5$  (Chakhovskoy 1956), diphasic domains are at high temperature. The complete phase diagram of  $C_{10}EO_5$  shows both lower and upper consolute boundaries, resulting in a closed demixing domain.

1985, Nilsson and Lindman 1983, 1984, Ravey 1983, Brown *et al* 1983, Brown and Rymden 1987, Löfroth and Almgren 1984, Corti *et al* 1984, Kato and Seimiya 1986).

New insights into this problem may come from studies of shear-aligned anisotropic micelles (Cummins *et al* 1987b). Their experimental studies are contradictory since the discrimination between solutions of strongly attracting small micelles and large ones (still attracting) is difficult. Scattering methods alone seem to be unsuccessful in discriminating between these two behaviours: the high- $q$  tail of scattering patterns can be identical with spheres or cylinders of about the same radius; the large critical scattering at low  $q$  hides the discriminating part of the particle scattering (Lum Wan *et al* 1987). Interparticle correlations for non-spherical particles or large attractive potentials (the MSA approximation is not valid (Tau and Reatto 1985)) cannot be calculated in a simple way.

The range of the attractive potential which may include several distinct physical phenomena is not known (a single Yukawa tail may not be appropriate (Reatto and Tau 1984, Ravey *et al* 1986)). It seems that multitechnique approaches are more reliable as results have to be consistent with one another. They have, moreover, to be consistent with thermodynamics. A growth into large flexible micelles was invoked as the only way to explain the low-concentration domain of occurrence of critical points (Kjellander 1982, Goldstein 1986). Polydispersity seems to be able to induce phase separation (Dickinson 1979, Smith and Rowlinson 1980). For the upper consolute boundary of dioctyllecithin-water phases diagram to be predicted under temperature-independent potential conditions, micellar growth and polydispersity were also invoked

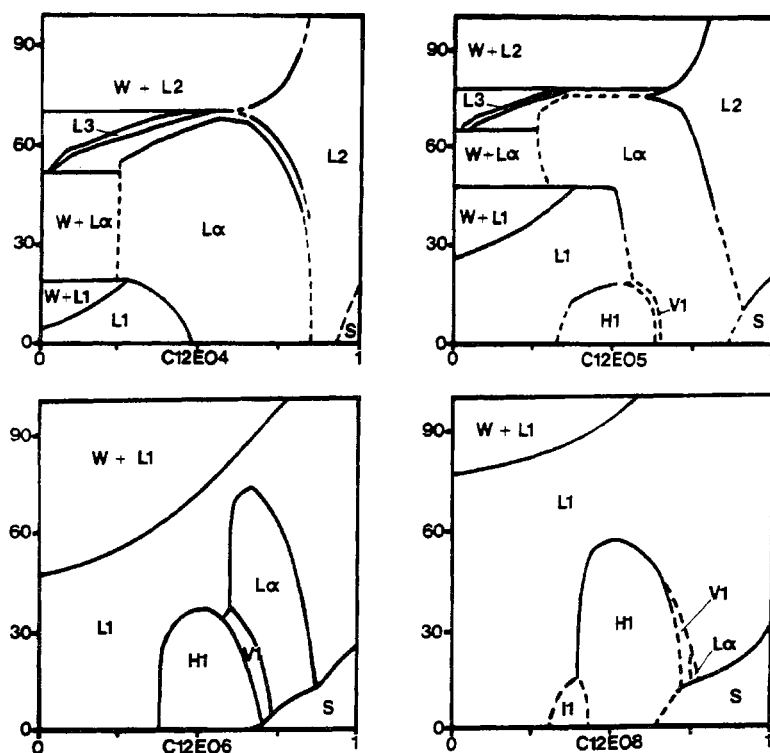


Figure 23. Phase diagrams of some water–non-ionic amphiphile binary mixtures. The  $x$  axis is weight fraction of amphiphile and the  $y$  axis is temperature ( $^{\circ}\text{C}$ ). Different domains are abbreviated as: W = water or dilute aqueous solution of surfactant (below the CMC), S = solid, L1 = direct micellar solution, L2 = water-swollen liquid surfactant, L3 = isotropic solution, I1 = cubic mesophase made of direct spherical micelles, H1 = hexagonal mesophase, V1 = bicontinuous cubic mesophase,  $L\alpha$  = lamellar mesophase.

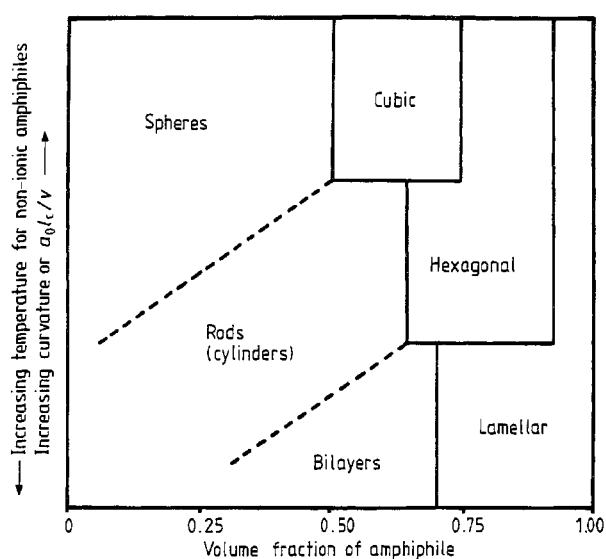


Figure 24. Schematic phase diagram of water–surfactant binary mixtures against interface curvature (Mitchell *et al* 1983).

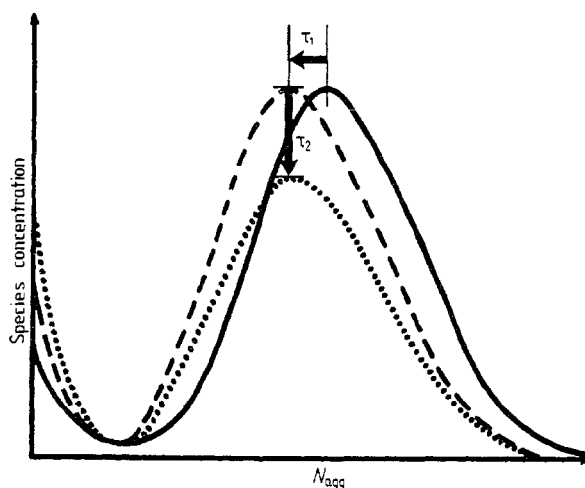


(Blankschtein *et al* 1985). Alternative explanations which keep moderate micellar size may be possible (Reatto and Tau 1987).

*4.1.9. Dynamic aspects: Brownian motion and molecular exchange.* Because of their small size and internal liquid-like character, very fast dynamic processes occur in micellar solutions. They are numerous according to the time or spatial scale considered. They are split into several different types, mainly because experimental methods work for a given time and spatial resolution and are only sensitive to some of these motions. At low resolution, translational and rotational motions of the micellar entities occur. On a smaller scale, motions of surfactant molecules in micelles are observed: exchange of molecules between solvent and micelles, surface translational diffusion, molecular rotational motion. Next, rotational intramolecular motions (*gauche-trans* isomerisations) are considered. Time scales are related to the spatial extent of motions as they are for all Brownian motions: energy is the same ( $kT$ ), so that small-amplitude motions are fast. Translational diffusion of micelles is studied by means of quasielastic light scattering (QELS), pulsed field gradient spin-echo NMR (PFGSE) (Stilbs 1987) or diffusion of radioactive tracers (Kamenka *et al* 1984). QELS provides mutual diffusion constants which are equal to self-diffusion constants at the CMC when intermicellar interactions vanish (Pusey 1978), i.e. at infinite dilution of aggregates. The latter are directly obtained from PFGSE or tracer diffusion (Nilsson *et al* 1983, Nilsson and Lindman 1983, 1984). A method based on polarography has been also used for self-diffusion measurements (Zana and Mackay 1986). Self-diffusion constants are related to the hydrodynamic radius of the micelles through Stokes-Einstein relations, and to obstruction by other micelles (Jönsson *et al* 1986). This is a powerful means of measuring micellar size as a function of concentration. The size is that of the hydrated micelle. Mutual diffusion constants depend, moreover, on intermicellar static and hydrodynamic interactions. QELS as a function of scattering vector and self-diffusion measurements provides a set of complementary data (Pusey 1978) which allows a complete description of dynamic structure when a correct model for static interactions is used.

Exchange of surfactant molecules between water and micelles used to be studied with chemical relaxation methods (temperature or pressure jumps, fast mixing by stopped flow) or ultrasonic relaxation. The current theoretical description of these processes is due to Aniansson *et al* (1976) with some further refinements (Almgren *et al* 1977a, b, Lessner *et al* 1981a, b, Wall and Aniansson 1980, Elvingsson and Wall 1986). Two relaxation processes are observed (Lang *et al* 1975). The fast one is molecular exchange at a constant fraction of micellised molecules and the second one is due to variations of this fraction through micelle formation or breakdown (figure 25). The rate constants of entrance of surfactant molecules into micelles is always the same ( $k_+ \approx 10^9 \text{ mol}^{-1} \text{ s}^{-1}$ ) (Aniansson *et al* 1976, Hoffmann *et al* 1981d), which suggests a diffusion-controlled process. The exit rate ( $k_-$ ) is strongly dependent on the alkyl chain length and only weakly on the nature of headgroups (anionic, cationic or non-ionic). The variation of  $k_-/N_{\text{agg}}$  with chain length gives an energy barrier per methylene equal to the free energy of micellisation (the equilibrium constant  $K = \exp(\Delta G/RT)$  is equal to  $N_{\text{agg}} k_+/k_-$ ); it follows the CMC variations (Aniansson *et al* 1976, Hoffmann 1978).

The diffusion of surfactant molecules at the interface is a two-dimensional process (headgroups remain at the interface). The diffusion constants as measured in lamellar mesophases by PFGSE are of the order of  $10^{-11}$  to  $10^{-10} \text{ m}^2 \text{ s}^{-1}$  (Charvolin and Rigny



**Figure 25.** Effect of the two dynamic processes of micelle formation and breakdown on the aggregation number distribution as inferred by Zana and Aniansson. The fast process ( $\tau_1$ ) shifts the distribution curve (full curve to broken curve) keeping the number of micelles constant, while the slow one ( $\tau_2$ ) corresponds to disappearance or formation of micelles from monomers (broken curve to dotted curve).

1971a, b, 1973, Roberts 1973, Lindblom and Wennerström 1977, Lindblom *et al* 1979). As the interface is curved, this is equivalent to a rotation. The typical values of micellar radii (1–2 nm) give rotational correlation times of the order of  $10^{-9}$  s, which is the frequency domain of NMR relaxation experiments. In such experiments, however, the diffusion at the curved interface is obscured by the rotation of the micelle itself which occurs on the same time scale. Faster diffusion at a micellar interface ( $10^{-9}$  m<sup>2</sup> s<sup>-1</sup>) is measured through quasielastic small-angle neutron scattering experiments (Tabony *et al* 1983, Tabony 1985). This is mainly due to the different time scales of the methods rather than to structural differences between micelles and mesophases. Quasielastic neutron scattering probes smaller distances and diffusion is thus free of the influence of defects, which is especially true for mesophases.

Molecular and intramolecular rotations are difficult to separate experimentally. They occur on time scales of comparable magnitude and are certainly correlated. Most NMR theories assume uncorrelated motions, which is their main limitation. These motions are restricted in amplitude and anisotropic because of packing constraints in the micellar core. The approach of Wennerström *et al* (1979) is the most realistic one in spite of some approximations (anisotropic motions are assumed to be isotropic in rate and uncorrelated).

Motions of the same type and time scale are combined together, so that the full spectral density is described with only three parameters: a slow motion correlation time  $\tau_{cs}$ , a fast one  $\tau_{cf}$  and an order parameter  $S$  which characterises the anisotropy of fast internal motions (see § 3.3) (Ahlnäs *et al* 1983, 1984, Lindman *et al* 1983, Walderhaug *et al* 1984, Söderman *et al* 1985a, b, Stébé *et al* 1985, Néry *et al* 1986, Eriksson *et al* 1987). Slow-motion correlation times are in the nanosecond range as discussed above, their decrease with micellar radius fully agrees with this interpretation (Söderman *et al* 1987). Fast movements are in the picosecond range, similar but a little bit slower than in liquid alkanes, and they do not depend on aggregate size. The weak dependence of  $\tau_{cf}$  on the position of the carbon atom in the alkyl chain is a strong indication for occurrence of correlations between internal motions whatever

the calculations of Gruen (1985a) have shown. Order parameters ( $S$ ) are the same as those measured by  $^2\text{H}$  NMR in mesophases and those from packing calculations (see § 4.1.6). This excellent set of consistent results should suggest confidence in multiple-movement models, but these three order parameters are averages performed over different time scales: in  $^{13}\text{C}$  NMR relaxation on micelles, motions faster than the slow one (1 GHz) enter the averaging, in  $^2\text{H}$  NMR on mesophases averaging is carried out down to the MHz range and packing calculations are full-time averages. The consequence of such agreement is that no motion slower than 1 GHz influences molecular orientation in either micelles or mesophases.

#### 4.2. True reverse micelles: micelles in organic apolar solvents

True reverse (or inverted) micellar solutions are binary solutions made up with surfactant and an organic apolar solvent, i.e. without any water. The micellar core is then made of polar headgroups. Hydrophobic chains are directed towards the solvent.

There is some controversy about the possibility of surfactants aggregating without any water or polar impurity (Eicke and Christen 1978). It seems nevertheless that such cases may occur, since the amount of impurities should be important for one molecule to be present in each micellar core. Some surfactants do not aggregate without added water (Becher 1960, Kitahara 1965, Güveli 1984), but the great majority do (Elworthy 1959, Becher 1960, Kitahara *et al* 1962, Fowkes 1962, Eicke and Christen 1974, Eicke and Rehak 1976, Zulauf and Eicke 1979, Tavernier *et al* 1981, Sudhölter and Engberts 1982, Ravey *et al* 1984, Ravey and Buzier 1984, Güveli 1984).

In spite of its chemical instability and difficulties of purification (Fletcher *et al* 1984, Delord and Larché 1984), the most popular surfactant is sodium di2-ethylhexyl-sulfosuccinate or Aerosol OT (AOT) for its low cost, its ability to solubilise large amounts of water at room temperature and the important available experimental studies. Structural data dealing with true reverse micelles are few, attention being focused on ternary systems containing water. Many non-ionic surfactants are of poor purity, and techniques which provide number average aggregation numbers (vapour pressure lowering, membrane osmometry, vapour pressure osmometry) have often been misinterpreted. Measurements by these techniques have to be extrapolated down to infinite dilution (to the CMC) in order to obtain interaction-free structural data. On this basis, the general belief that true reverse micelles are only oligomers (dimers, trimers, etc) is unsound. Claims of concentration-independent aggregation numbers and of large differences between number average and mass average aggregation numbers from light scattering indicating polydispersity (Sirianni *et al* 1962) seem dubious. These two statements are, moreover, contradictory since polydisperse micelles necessarily have concentration-dependent aggregation numbers (see § 4.1.2). In this field, general reviews should then be read with caution (Eicke 1980).

Some general trends can, however, be outlined. Although it is not always the case, a CMC above which micellar aggregation occurs can be defined and measured. The CMC can be rather high (Ravey *et al* 1984, Ravey and Buzier 1984). Below the CMC, surfactant molecules associate into oligomers, and rather large micelles can exist above it (Elworthy 1959, Tavernier *et al* 1981, Kumar *et al* 1984, Kotlarchyk *et al* 1985) and are in equilibrium with sub-CMC small aggregates (Eicke *et al* 1975). Aggregates should be called micelles only when aggregation numbers are enough (at least 10) for a deep minimum in the size distribution to be possible. The absence of size variation with

concentration was observed by Huang *et al* (1984), as well as the occurrence of an attractive intermicellar potential.

In the absence of water, self-association of water molecules cannot be the origin of aggregation, and true reverse micelles are in that way different from those containing water in their core. Headgroups are dipoles, especially ionic ones that exist as ion pairs in apolar solvents. Electrostatic dipole-dipole attraction between headgroups should then be important and long range because of the low dielectric constant of the surrounding solvent, and considered as the driving force for aggregation. Association of two dipoles into a dimer results in a quadrupole of lower energy and so on. The situation is similar to ion cluster formation in polymer ionomers (Eisenberg 1970, Yarusso and Cooper 1983, Dreyfus 1985), the unfavourable elastic energy of the polymer chain being replaced by entropy of mixing. A larger aggregation is possible but limited by entropy and interfacial interactions which determine the curvature. Hydrocarbon chain-solvent interactions are of importance in these latter interactions (Eicke and Christen 1974, Eicke 1979). Thus, Little and Singleterry (1964) and Little (1970) related the micellar size of dinonylnaphthalene sulfonates to the solvent solubility parameter  $\delta$  of Hildebrand. Such interactions are short range as concentration does not affect the structure. Dynamics of surfactant molecules are similar to those of direct micelles (Faure *et al* 1987).

The main interest of reverse micelles is their ability to dissolve water in their core, forming water-in-oil microemulsions as discussed below.

## 5. Ternary systems

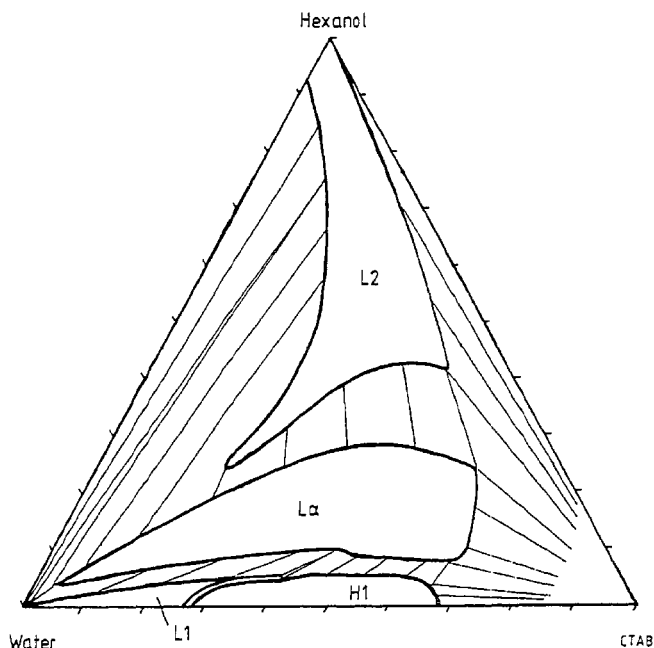
Phase diagrams of ternary systems are more complex than those of binary systems. Figure 26 is a typical example, illustrated by the hexadecylammonium bromide-water-hexanol system (Ekwall *et al* 1969a). At high water content, the direct micellar phase  $L_1$  and in the oil corner, the 'reverse micelle' or 'water-in-oil' microemulsion are the two isotropic phases. Between them, the usual liquid crystalline mesophases occur, in this case the lamellar and hexagonal phases.

The picture which emerges from the micellar structure of binary solutions of surfactants and water ( $L_1$ ) is the following: a fluctuating oil droplet with a smooth interface is built by the hydrocarbon tails. The interface contains charged headgroups with a huge amount of water (typically ten molecules per headgroup) included in the micellar 'hard sphere' excluded volume: when two micelles come into contact, this water volume is not expelled. The residence time scale of a molecule in a given aggregate is in the nanosecond range, probably the same as the shape fluctuations. How does this picture change when a third component is added?

Additional components can be classified into several classes:

(i) With short-chain alcohols as third component, the solubility of oils in micellar solutions is greatly enhanced: alcohols are called 'cosurfactants'. The extreme case where all the surfactant is replaced by alcohol, leaving a clear and stable ternary mixture of oil, water and alcohol without any surfactant ('detergentless microemulsions') is beyond the scope of our survey of surfactant systems (Smith *et al* 1977). A complete analysis of phase diagrams and related surface tensions has been made by Puig *et al* (1987).

(ii) Of practical interest in detergency applications is the case where the third component is insoluble in the pure main solvent (solubilisation). Apolar molecules



**Figure 26.** Hexanol/water/hexadecyltrimethylammonium bromide (CTAB) ternary phase diagram. L1 and L2 are direct and reverse micellar domains, H1 and  $L\alpha$  are hexagonal and lamellar mesophases (Ekwall *et al* 1969a).

are easily solubilised in direct micelles. On the other hand, water can be solubilised in a binary mixture of oil and surfactants. In most cases, the solubilising power towards a third component is low: the solubilisation limit of hydrocarbons is of the order of one to five solute molecules per surfactant.

(iii) Some exceptions to this low-solubilisation limit are known: the cases of AOT for anionic amphiphiles, of nonylphenol polyethyleneoxide for non-ionic surfactants (Marsden and McBain 1948) and of double-chain cationic ammonium surfactants such as didodecyldimethylammonium bromide (DDAB) (Fontell *et al* 1986).

(iv) Another ternary system is the 'mixed micelle', where a mixture of surfactants is used in water. This is a very active field because of industrial applications: a synergic effect in solubilising power is expected, since the surfactant shape parameter  $a_0 l_c / v$  can be adjusted at will with the chemical composition of the surfactant mixture.

Although studies on solubilisation behaviour in surfactant systems are numerous, reliable results in a system under good control are few, mainly because of the lack of phase diagram determination. Two classical reference works giving complete phase diagrams in ternary surfactant systems are due to the Swedish group (Ekwall *et al* 1969a, b). Solubilisation by a surfactant solution occurs below the CMC: a hydrophobic molecule can induce micellar nucleation. The effect of a foreign substance on a binary solution is shown to increase the thermal stability and the domains of existence of the micellar phases. The work of Ekwall's group is also a critical review of most of the ternary phase diagrams known ten years ago.

The case of added salt is somewhat different: we consider a mixture of salt, water and surfactant as a binary system, since the main effect of the salt is to screen out the electrostatic interactions between ionic headgroups; surfactant, salt and water mixtures are therefore described in § 4.

### 5.1. Models of structure

The situation for ternary systems is more complicated than for binary systems: when a model of spherical interacting droplets breaks down, i.e. when an autocohereant iterative fitting procedure does not work, one has to invoke locally lamellar, rod-like or disc-like structures. This is geometrically possible because of local segregation of the components in a mixed aggregate: alcohols, which have a lower  $a_0 l_c / v$  parameter than surfactants, tend to accumulate in the cylindrical regions rather than at the highly curved parts of the ellipsoid like swollen micelles: this effect has been directly observed by neutron scattering using deuterium-labelled molecules (Hendrikx *et al* 1984).

Unfortunately, the scattered intensities for these structures cannot be calculated analytically. One has either to neglect interactions between particles, an assumption which cannot be right in real systems (even at very low concentration of the order of 1% volume fraction, interactions between aggregates are important). On the other hand, one could consider only the power law behaviour of the scattered intensities, but this procedure can be very misleading. The only path is to calculate numerically the scattering of the proposed structure on an absolute scale (Barnes and Zemb 1988). This is the main reason why most published results in ternary systems use some kind of spherical droplet model with an interaction potential. On the other hand, this quite simple assumption has been claimed to hold surprisingly well for some surfactants such as AOT (Kotlarchyk 1984) (figure 32).

If the microstructure is a collection of spherical droplets, the basic rule is simple: since the area per surfactant headgroup is known, the radius and the concentration of the droplets are known at any composition. Assuming hard sphere interaction, the scattered intensity is calculated *ab initio* for any composition. It turns out that the predicted scattering is very close to the observed one for the ternary system most studied so far (AOT-water-oil), but this could be a very peculiar property of the AOT molecule, whose surfactant parameter  $a_0 l_c / v$  can vary without much energy cost. A linear relation (Oakenful 1980) between water content and observed radius is then obtained (Cabos and Delord 1979, 1980, Kotlarchyk *et al* 1982). The radius is fixed by the available interfacial area  $\Sigma$  and the total polar volume fraction  $\varphi$  as

$$R_s = \frac{3\varphi}{\Sigma}.$$

If the microstructure is locally cylindrical, only one  $R_c$  radius of cylinders is possible

$$R_c = \frac{2\varphi}{\Sigma}.$$

If the structure is locally lamellar, only one thickness  $T$  of the bilayer satisfies packing constraints

$$T = \frac{2\varphi}{\Sigma}.$$

In case of asphericity, the Talmon and Prager (1978) model or the cubic random cell model (see § 6.2) (De Gennes and Taupin 1982) could be used. These models have been designed to describe Winsor III microemulsions with equal amounts of water and oil and vanishing surfactant concentration. It is therefore not surprising that these models do not compare to the scattering of ternary systems: Talmon-Prager cannot account for any peak in the scattering and the predicted position of the scattering

peak predicted by the cubic random cell ( $q = 2\pi/\xi$ ) corresponds to about twice the observed position in real systems ( $q = \pi/\xi$ ).

Teubner and Strey (1987) have proposed recently a phenomenological fit to the peak position. Two phenomenological parameters are needed; one length scale  $\xi$  is the correlation length where  $D^*$  is the periodicity (peak position). The scattered intensity then expresses as

$$I(q) = \frac{8\pi/\xi(\rho_{\text{int}} - \rho_{\text{ext}})C_2V}{a_2 + c_1q^2 + c_2q^4}$$

$a_2$ ,  $c_1$  and  $c_2$  can be fitted to the scattering curve and the two distances  $D^*$  and  $\xi$  are obtained from

$$\xi = \left[ \frac{1}{2} \left( \frac{a_2}{c_2} \right)^{1/2} + \frac{1}{4} \left( \frac{c_1}{c_2} \right)^{1/2} \right]^{-1/2}$$

$$D^* = \left[ \frac{1}{2} \left( \frac{a_2}{c_2} \right)^{1/2} - \frac{1}{4} \left( \frac{c_1}{c_2} \right)^{1/2} \right]^{-1/2}.$$

These expressions fit surprisingly well the scattering of a large number of microemulsions: AOT, SDS or non-ionics were tested (Teubner and Strey 1987). The result is that  $D^* \approx 2\xi$ , which means that the set of random Voronoi polyhedra, which is the underlying lattice in the DOC model is a good approximation. It should be mentioned that these expressions of the scattering do not use any modelling of the microstructure.

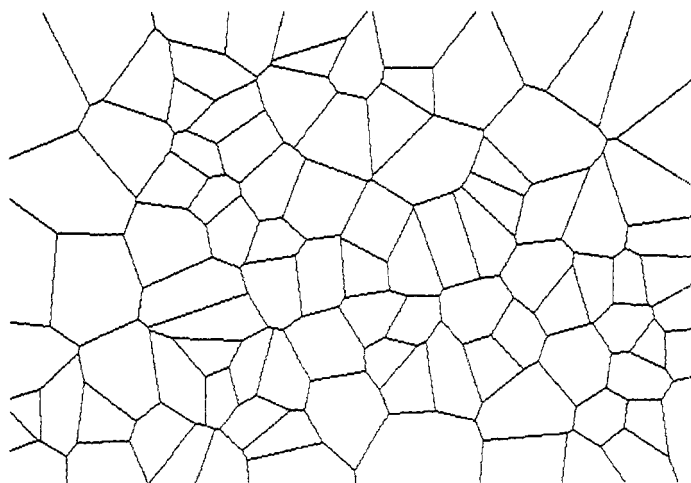
The scaling law towards the polar volume fraction for water-in-oil microemulsions is nevertheless approximately correct for the CRC model

$$q_{\text{max}} = 2\pi/D^* \sim \varphi(1-\varphi).$$

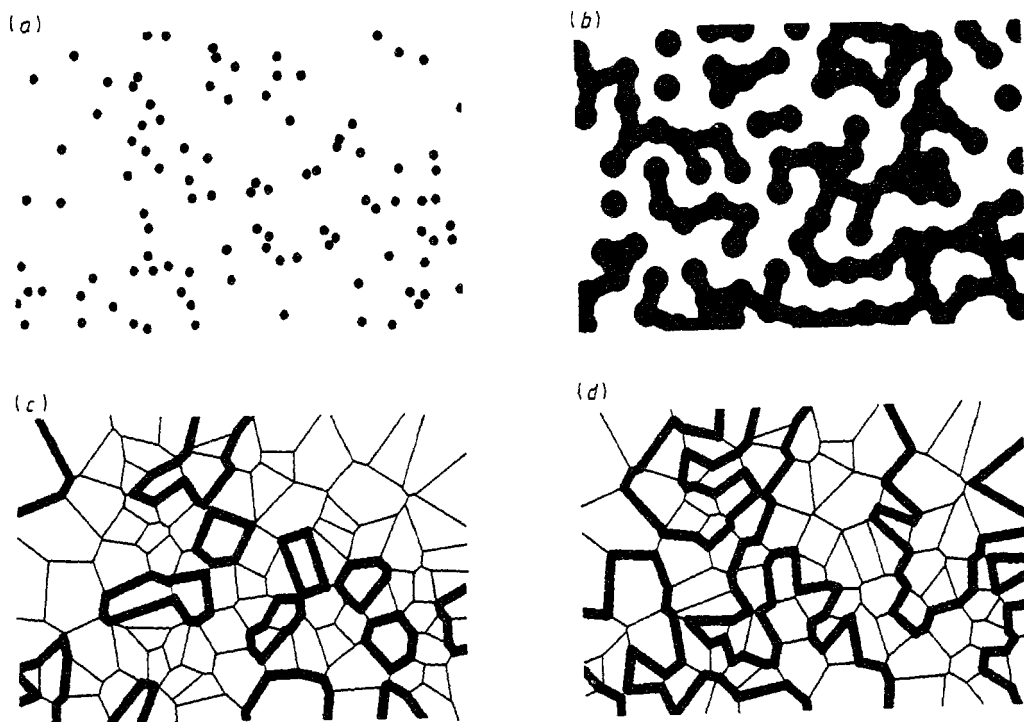
We have proposed recently an alternative model (DOC model), which is a low-resolution picture of any possible type of dispersion of water in oil or oil in water (Zemb *et al* 1987). The DOC model is a low-resolution general model of the possible structure of micelles and microemulsions. The basic assumption of this model is that a surfactant system can be described as two 'phases' (oil and water) separated by an interface of constant non-zero curvature. The structure is macroscopically disordered and has local constraints given by the available volume fraction and specific area (Ninham *et al* 1987).

Since microemulsion scattering does not show sharp Bragg peaks, we start with a set of Poisson points, with a distance of closest approach  $D_{\text{min}}$  between them. These points are the centres of a virtual set of hard spheres of diameter  $D_{\text{min}}$ . This set of Poisson points defines a complete tessellation of space in cells belonging to a random lattice (figure 27). Once these random cells are constructed, *each* cell is filled with water *and* oil separated by a surfactant film, according to the macroscopic volume fractions. If oil is medium 1, the water (medium 2) can be distributed via three possible distributions (figure 28):

(i) Water is located between planar bilayers along the faces of the Voronoi polyhedra in the DOC lamellar model, which looks like a foam. The prediction for this structure is that the scattering peak should move very slowly with water volume fraction, as observed with the tetradecane-DDAB-water system. Covering all the faces of the Voronoi cells induces a structure which is made of an infinity of disconnected cells. If only some of the possible faces are covered, the resulting structure is bicontinuous (Cates *et al* 1988) and a 'sponge'-like structure results (Warr *et al* 1988a). This



**Figure 27.** Two-dimensional partitioning of space with Voronoi polygons: centre points are chosen at random and Voronoi cells are defined as the set of points closer to a given centre than any other centre point. In two dimensions, the average number of cell edges is about 6; it is about 13 in three dimensions.



**Figure 28.** Four examples of DOC structures built from the same Voronoi lattice: (a) connectivity  $Z=0$  reduces to a set of hard spheres. (b) DOC cylinder with connectivity  $Z=2$ : each centre is bound to the two nearest neighbours on average. (c) DOC lamellar with pseudo-volume 2.5% is a dispersion of vesicles. (d) DOC lamellar with pseudo-volume 50% is a random dispersion of flexible bilayers.



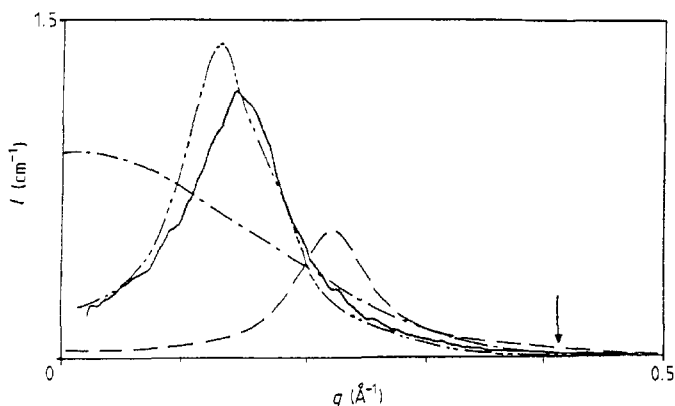
disordered locally lamellar structure has also been observed in the cetylpyridinium chloride-hexanol-water system by Porte *et al* (1988).

(ii) Water is distributed in spheres around the Poisson points. This is very close to hard spheres, except that some coalescence is allowed when two spheres are in contact. The connectivity of this model is  $Z = 0$ . Scattering peak positions for this model can be derived (Barnes *et al* 1988a, b).

(iii) Water is distributed in cylinders joining two adjacent Poisson points. Since each polyhedron has 13.4 nearest neighbours on average, a maximum number of 13 cylinders can leave a point. The average number of cylinders leaving one spherical droplet is called connectivity  $Z$ . Usually observed values of the connectivity are between 1 and 4. This assembly of connected cylinders and spheres is a first-order approximation of the more general random interface of constant curvature. This structure occurs whenever the spontaneous curvature is smaller than the periodicity  $D^*$ .

The generality of these low-resolution structural models of microemulsions is that each of these structures can be readily distinguished from the other ones by scattering experiments alone when the average chord is evaluated together with peak position as the volume fraction is varied. The DOC lamellar, cylinder and spherical models are also a quantitative generalisation of the intuitive notion of droplets coalescing into a bicontinuous structure. The occurrence of three basic local geometries has been predicted for different values of the interfacial film rigidity and spontaneous curvature (Safran *et al* 1984). Similar thermodynamic arguments lead to the same conclusions (Porte *et al* 1986): any structure without long-range order can be approximated locally as cylindrical, spherical or lamellar. Subtle surface chemistry effects would have to be precisely known to predict a whole phase diagram: change of counter-ion from bromide to chloride changes completely the phase diagram and the occurring structures (Gomati *et al* 1987).

Figure 29 shows the predictions of the CRC, Talmon-Prager and DOC models for a ternary microemulsion (Ninham *et al* 1987). The peak position and the shape of the scattering are very close to the prediction of a model of connected random cylinders,



**Figure 29.** Small-angle x-ray scattering of didodecylammonium bromide ternary microemulsion (full curve) compared with the scattering of monodisperse hard spheres (broken curve), the scattering of randomly filled Voronoi cells (chain curve) and the scattering predicted by the DOC cylinder model with a connectivity of 2 (double dot chain curve) (Ninham *et al* 1987). The arrow shows the peak position in the CRC model.

while random Voronoi polyhedra and a dispersion of spheres is incompatible with the observed scattering.

## 5.2. Solubilisation

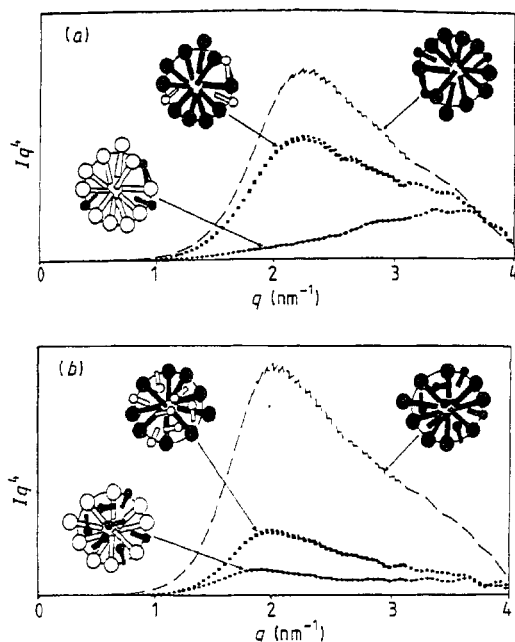
Solubilisation of a compound in any organised assembly (micelles, vesicles, etc) is measured by the solubilising power, which is the maximum additive-to-surfactant mole ratio in the aggregate. Exact measurement of the solubilising power is difficult, since the amount of surfactant and solute present in bulk should be taken into account, so that, for qualitative approaches, the macroscopic overall mole ratio is used instead.

**5.2.1. Surfactant-brine-short-chain alcohol ternary systems.** These systems are precursors of microemulsions. A general coherent frame of the transition of micelles to microemulsions (structural and dynamic changes induced by alcohol) has been given by Hoffmann *et al* (1986).

The most important question is the effect of short-chain alcohols on micellar structure: alcohols are used as cosurfactants to increase the solubility of oils; microemulsions with equal amounts of water and oil can only be stable if an alcohol is present. Only a few surfactants with a shape parameter  $a_0l_c/v$  close to 1 are able to form stable three-component microemulsions. It is therefore crucial to understand the role of alcohols in ternary systems: is the high oil solubility mainly due to the release of steric constraints due to film flexibility or is it due to a modification of the effective  $a_0l_c/v$  which allows very low curvature of the interface (Ben-Shaul *et al* 1986)?

The measurement of the partitioning of alcohol between water and micellar aggregates is the first step in any structural study, since the result allows determination of volume fractions (Stilbs 1982a, 1983, Abuin and Lissi 1983, Quirion and Magid 1986, Treiner and Mannebach 1987). The determination of the concentration of the monomeric surfactant in equilibrium with the micellar aggregate is also a problem. It is known that addition of alcohol decreases the CMC (Zana *et al* 1981, Abu-Hamdiyyah and Rahman 1987), this would mean from a pseudo-phase model that the monomeric surfactant concentration is lowered. However, in ternary systems, the addition of alcohol increases the solubility of the surfactant in water, thus increasing the concentration of monomers (Stilbs 1982b, Hayter *et al* 1984, Lang and Zana 1986, De Lisi *et al* 1987). If butanol is used as a cosurfactant, the micellar structure can be fully destroyed, the solution being a molecular dispersion (Eriksson and Lindblom 1980).

As has been shown by contrast variation neutron scattering experiments (Hayter *et al* 1984) in the case of pentanol solubilisation in octanoate micelles, alcohol molecules are simply adsorbed onto the micellar interface at low alcohol-to-surfactant ratio, without inducing mass modification (figure 30). As the surface gets saturated, excess pentanol penetrates into the micellar core, allowing growth of the micellar aggregate. The size distribution has been calculated from free energy evaluations and this type of mass variation is indeed expected (Ben-Shaul *et al* 1986). The same behaviour is already known for benzene (Eriksson and Gillberg 1966, Candau and Zana 1981). A direct force measurement between surfactant layers with added pentanol suggests also a two-step mechanism: at low concentration, pentanol adsorbs at the interface, while it enters the surfactant layer at higher concentration (Pashley *et al* 1986). This 'two-sites' model of alcohol solubilisation provides an explanation for the solubilisation data of heptanol, in SDS (Russell *et al* 1986, Blokhuis *et al* 1986). Adsorption and



**Figure 30.** Variable contrast small-angle neutron scattering spectra ( $Iq^4$  versus  $q$ ) for the pentanol (CoS)/sodium octanoate (S)/water system. (a) Low pentanol content ( $[\text{CoS}]/[\text{S}] = 0.25$ ). (b) high pentanol content ( $[\text{CoS}]/[\text{S}] = 1$ ). The solvent is deuterated water and protonated materials are in black. In (a) spectra from different contrast conditions are different, especially that with protonated pentanol in deuterated octanoate micelles and water, showing its surface adsorption. In (b) all spectra are identical in shape, intensity differences are due to different proton content in micelles only, showing uniform distribution of pentanol in the micellar volume.

solubilisation can be readily recognised through calorimetric measurements (Kertes *et al* 1985). The concept of interfacial area saturation is suggested by very different experiments. Only one indirect Raman spectroscopic study seems to disagree with this picture of interfacial adsorption followed by penetration into the micellar core (Shih and Williams 1986). This general concept has been extended and seems also to be valid in mesophases (Boden *et al* 1987b). At low alcohol-to-surfactant ratio, the order parameter of the hydrocarbon chains is not modified when octanol is added to the micelle (Jansson and Stilbs 1987).

It has been noticed that the scattering of the micelles is incompatible with a model of spherical droplets at high pentanol content (Zana *et al* 1983, Marignan *et al* 1986). A locally lamellar structure is predicted by the DOC lamellar (foam-like) structural model in the octylbenzenesulphonate/water/pentanol  $L_2$  phase (Marignan *et al* 1983) as the peak position in scattering experiments does not vary with water content.

Few precise results are available about the counter-ion condensation and therefore the effective surface charge variation induced by alcohol adsorption on the interface; since the conductivity increases sharply when alcohol is added to a micelle, the surface charge is supposed to increase, due to the desorption of counter-ions. This very crude picture has recently been questioned by Hartland *et al* (1987).

On the other hand, addition of alcohol has a very strong influence on micellar dynamics, as can be seen by the exchange rate of pyrene probes between micelles

(Malliaris *et al* 1987). Three different relaxation times in chemical relaxation experiments have been observed in such a ternary system (a supplementary exchange process is observed as compared with binary systems). The exchange of pentanol is only diffusion controlled. Surfactant exchange between neighbouring micelles is greatly enhanced when cosurfactant is present. The micellar structures become very labile (Yiv *et al* 1981) and the shape can be modified: micellar anisotropy is induced by pentanol adsorption (Hirsch *et al* 1984a).

One of the most striking features in ternary systems with alcohol is the existence of an oil continuous phase, connected to the oil corner in the phase diagram, with 90% water content (Ekwall 1969, Ekwall *et al* 1986). The structure and origin of this phase remains one of the key problems in microemulsion structure. It is expected that this phase presents locally planar interfaces from the value of the packing parameter. In our opinion these  $L_2$  reverse micelles with very high water content should have the foam-like structure predicted by the DOC or the random surface models (see § 6.1), but this remains to be proven. Lamellar phases occur generally near the water corner in phase diagrams ('swelled' lamellar phases) (Benton and Miller 1983, Kachar *et al* 1984b, Bassereau *et al* 1987). In a different system, a strongly streaming birefringent phase has been observed nearby the usual direct micellar phase  $L_1$ . A foam-like structure identical to the DOC lamellar model has been proposed on the basis of  $q^{-2}$  dependence of the scattering of this phase. But this is not a firm confirmation of the structure, the flat behaviour of the scattering obtained at distances up to about 500 Bragg spacings is still unexplained: alternatives exist in terms of local lamellar or cylindrical geometries, which are both shown to be allowed in thermodynamical models (Porte *et al* 1986).

Although the detailed structure is not known in detail, it has been shown very recently that a statistical thermodynamic treatment can account for the occurrence of direct and reversed micelles in a ternary system. This success in reproducing a complete phase diagram together with ordered structures (lamellar and hexagonal mesophases) has been obtained by combining packing restrictions, electrostatics of counter-ions, an interfacial free energy, entropy of mixing and a hydration force at low water contents (Jönsson and Wennerström 1987).

By mixing some soap and its corresponding acid other types of mixed micelles are obtained which are direct ones in water or reversed micelles in oil. The case of the sodium octanoate/octanoic acid/water system has been studied in detail (Ekwall *et al* 1986). This type of mixed micelle is interesting because a mixture of salt and acid produces an aggregate of variable structural charge per unit area, since the soap has its own structural charge while the acid has none. This type of system is reminiscent of alcohol-containing ones, the alcohol being replaced by an acid. There is, however, a major difference: the surface charge density has to be constant in the systems with an acid as a cosurfactant because proton exchange between soap and acid molecules is much faster than structure reorganisation rates, any component segregation inside an aggregate is prohibited. The effective surface charge density, i.e. the charge of the surfactant headgroups less that of the counter-ion present in the hard sphere volume, was found to be independent of the chemical composition in such a system because of counter-ion condensation (Chevalier *et al* 1985), it adjusts itself to about  $8 \mu C \text{ cm}^{-2}$ .

**5.2.2. Other solutes in direct micelles.** Solubilisation of hydrocarbons by micellar solutions is of crucial industrial importance; not only the maximum amount of solubilised oil, but also the dynamic ratio of oil uptake are the key parameters for

efficient detergency. Due to the practical importance of this field, a large number of review papers are available (McBain and Hutchinson 1955, Ekwall *et al* 1969a, Mukerjee 1980, Nagarajan 1986, Bayer *et al* 1986, Hoffmann and Ulbricht 1987).

The picture emerging from studies on micellar solubilisation is the concept of 'two-site localisation': added molecules are either dissolved in the hydrocarbon core or adsorbed at the interface. This explains why polar molecules such as esters or ketones are much more soluble than hydrocarbons, as solubilising in the interfacial region (Mitra *et al* 1984). The 'adsorbed' and the 'dissolved' state of the host molecules have very different effects on the shape parameter  $a_0 l_c / v$ . The effect on the packing parameter is so strong that transition to and from liquid crystals can be induced (Sagitani and Friberg 1983). Values of adsorbed-to-dissolved mole ratio have been given by Mukerjee (1971).

In the case of in core solubilisation, the hydrophobic volume is increased, whereas in the adsorbed state the main effect of the solute is to modify the area per surfactant headgroup  $a_0$ , which can be either increased by the polar part of the solute or decreased by screening the ionic headgroup repulsion. This picture explains the variation of aggregation numbers with the presence of solubilisates and the sphere-to-rod transition induced by the host molecule (Mukerjee 1980, Hoiland *et al* 1985). Once the sphere-to-rod transition has occurred, the solubilising power increases by a factor of two (Ozeki and Ikeda 1985, Imae *et al* 1986). Geometrical considerations also allow interpretation of the solubilisation of lipids by micelles (Lichtenberg 1985, Dennis 1986).

The concept of two-site solubilisation should allow qualitative prediction of the solubilisation power, expressed in numbers of host molecules per surfactant. Qualitative trends are known, but there is now a lack of any quantitative predictive theory: a much better knowledge of molecular forces is needed in order to explain simple observations. For instance, the solubility of hydrocarbons in the core of direct micelles is much lower than in the same volume of pure hydrocarbon: the physical origin of this basic observation is not known, the 'Laplace pressure effect' being a phenomenological explanation only. This point has been discussed by Funasaki *et al* (1984) for non-ionic micelles. Nevertheless, some tentative quantitative theory of solubilising power for one given micelle (SDS) (Biedermann and Datyner 1981) and a purely phenomenological but useful method of prediction of the solubility (Smith *et al* 1987) have been proposed.

### 5.3. Ternary reverse micelles using AOT

**5.3.1. Pure reverse micelles of AOT, water and oil.** By far the most widely used molecule in ternary water-in-oil microemulsions ('reverse micelle',  $L_2$ ) is AOT (sodium 1,4-bis(2-ethylhexyl) sulphosuccinate)/water/oil system. The branched geometry of the tail allows a very important water uptake in oil. In other words, the hydrophile-lipophile balance is well adjusted or the shape parameter  $a_0 l_c / v$  can be close to 1 (Oakenfull 1980). A monograph dedicated to structure and reactivity in AOT micelles has been edited by Luisi and Straub (1984). For instance, a slight modification in the polar head structure is enough to reduce the maximum solubilised water from  $w = 60$  to less than  $w = 6$  (Faure *et al* 1985). AOT is unfortunately hydrolysed by solubilised water in a few days (Delord and Larché 1984), producing its own short-chain cosurfactant. This is the reason why detailed phase diagrams including liquid crystalline phases are not available for AOT. Only approximate phase diagrams, i.e. determination of

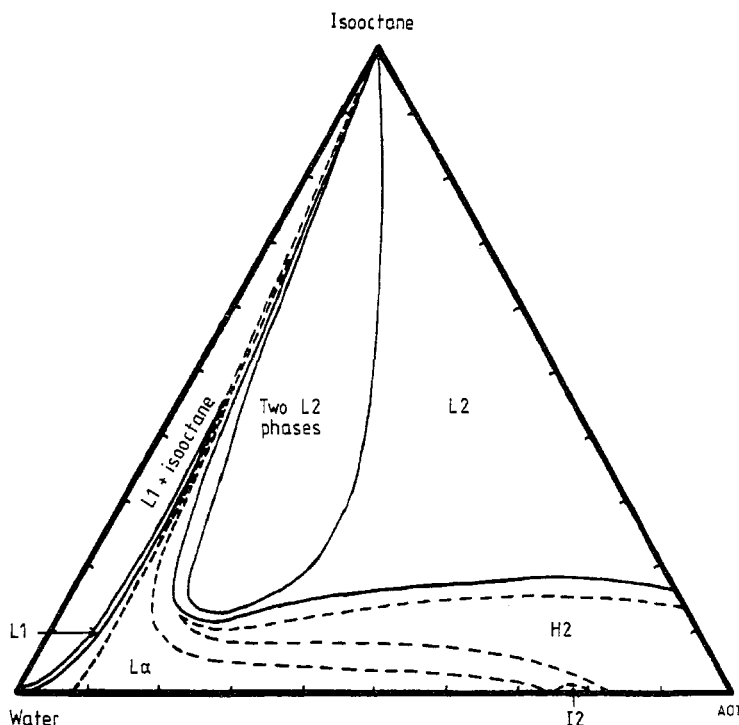


Figure 31. Phase diagram of the AOT/water/isooctane ternary system at 15 °C (Kunieda and Shinoda 1979) showing the large water solubilisation in reverse micelles (L2 domain). Compositions are weight fractions, the small L1 domain is an aqueous micellar solution which turns to lamellar  $L\alpha$ , cubic I2 and reverse hexagonal H2 mesophases upon AOT concentration. Similar phase diagrams with different solvents may be found from Ekwall *et al* (1970).

fluid phase extension, are available (figure 31) (Ekwall *et al* 1970, Rouvière *et al* 1979a, b, Kunieda and Shinoda 1979, Larché and Delord 1985).

At low water content ( $w \leq 30$ ) and low AOT concentration, the microstructure is a dispersion of spherical droplets (Cabos and Delord 1979, 1980). The area per polar head is a constant  $0.6 \text{ nm}^2$ . Therefore, the radius of the reverse micelle varies linearly with the water content (figure 32). The question of shape or shape fluctuations at high  $w$  (about 50) is still under debate since opposite conclusions have been drawn (Kotlarchyk *et al* 1985, Ganz and Boegfer 1986, Peyrelasse *et al* 1986). Direct measurement by neutron spin-echo has allowed determination of interfacial bending constant (Huang *et al* 1987). The value of  $K_0 = 5 \text{ kT}$  is surprisingly high, since a unique feature of the AOT ternary microemulsions, together with the high solubility of water without cosurfactant, is the flexibility of the interfacial film, its spontaneous curvature varies easily to adapt against steric constraints: the size of water cores in micelles varies from 1 to 5 nm without demixing, without cosurfactant. This is due to the branched hydrocarbon chain: the AOT molecule has its cosurfactant linked in the apolar part of the molecule.

A structure of closed unconnected spherical droplets has been observed in the AOT/water/decane system although some coexistence between water droplets and dry

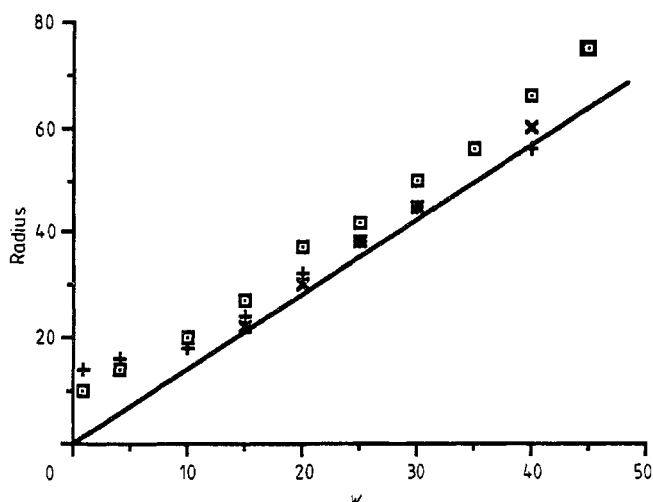


Figure 32. Linear dependence of the radius of a reverse micelle versus the mole ratio  $w = [\text{H}_2\text{O}]/[\text{AOT}]$ , as measured by different techniques: +, small-angle x-ray scattering (Pileni *et al* 1985a);  $\square$ , small-angle neutron scattering (Robinson *et al* 1984 and unpublished results of Zemb);  $\times$ , pulsed radiolysis (Pileni *et al* 1984). Linearity is imposed by sterical constraints, deviation from it at low  $w$  is due to the volume of AOT polar headgroups.

surfactant aggregates has been proposed (Grimson 1987). This behaviour seems, however, very unlikely, since it would mean identical values of the chemical potential  $\mu_N$  for very different aggregation and hydration conditions. In this latter case, it seems that the energy cost for dehydrating some AOT molecules 'in excess' would be large. The structure of the concentrated  $L_2$  phase has been claimed to remain tight stacking of spherical droplets: the AOT microemulsion does not turn into open connected structures till the AOT concentration has reached 30% (Kotlarchyk *et al* 1984a, b, Huang and Kotlarchyk 1986). The reason for the high electrical conductivity observed above 20% water content is therefore a balance between droplet coalescence and core exchanges. A detailed study of the electrical percolation threshold and the relation with droplet concentration fluctuations near the critical point has shown that a dynamical percolation between spherical droplets occurs (Grest *et al* 1985, Peyrelasse *et al* 1988).

Reviews about structural results in AOT reverse micelles are available (Eicke and Kvita 1984, Chen and Kotlarchyk 1985).

A rough comparison of the scattering of hard spheres shows that some attractive potential between the droplets has to be considered (Safran and Turkevich 1983, Kotlarchyk *et al* 1984a, Robinson *et al* 1984, Toprakcioglu *et al* 1984). In the case of AOT, the existence of a critical point has been demonstrated (Huang and Kim 1981) in non-penetrating solvents such as decane, but has not been observed in octane (Kotlarchyk *et al* 1983).

However, the physical origin of the attractive intermicellar potential, probably related to hydrophobic chain interpenetration, remains unclear: for instance, no theory accounts for the differences in critical compositions and temperatures for different oils. The only general rule is that the attractive interaction is proportional to the interpenetration of the two surfactant layers when reverse micelles come into contact (Lemaire *et al* 1983): this very simple interaction potential gives the right variation in water/oil reverse micelles for different penetrating oils. By taking into account the interpenetration of the chains and van der Waals interactions between the water droplets, the virial

coefficients could be fitted for different water droplet radii (and also added alcohol supposed to reduce chain interdigitation in quaternary microemulsions) (Brunetti *et al* 1983, Roux *et al* 1984a, b). Another interpretation attributes the observed attraction to solvent-induced interactions (Pincus and Safran 1987).

Some authors refuse to admit the necessity of this attractive interaction potential to explain the presence of critical points in the phase diagrams, explaining the high scattering at low  $q$  by a high polydispersity of the droplet size (Kotlarchyk *et al* 1988). An interpretation of the scattering due to a polydisperse set of hard spheres could even be reconciled with thermodynamical arguments developed by Safran (1983) and Stecker and Benedek (1984). Indeed, the origin of the attractive potential is not well understood since the measurement by any scattering technique cannot separate polydispersity from attractive interaction effects. However, the attractive interactions are important, since polydispersity cannot induce liquid-gas equilibria in the phase diagram. In the absence of explanation of the origin of these attractive interactions, Huang *et al* (1984) could explain the whole scattering curve by using a square well attractive potential of  $1.5kT$  with a range of 1.1 times the micellar radius.

These attractive interactions induce the formation of micelle clusters; when micelles coalesce a continuous water path can be formed which explains the electrical percolation threshold observed with increasing water content (Eicke *et al* 1986).

The measurement of the counter-ion distribution inside the micelles' water core has been attempted, but no clear conclusion could be drawn because of the low resolution of small-angle x-ray scattering (Cabos and Marignan 1985).

Reactivity studies in the core of reverse micelles show two very different behaviours according to the water-to-surfactant ratio  $w$ : below 10 water molecules per AOT molecule, water is bound to surfactant headgroups and counter-ions. When  $w$  exceeds 10, 'bulk' water appears and reactivity constants (such as hydrated electron life time) characteristic of bulk water are observed (Pileni *et al* 1982). The state of the water molecules has been investigated by NMR: six water molecules per dissociated counter-ion and three per ionic headgroup are immobilised (Wong and Thomas 1977, Wong *et al* 1977), a correct evaluation of water dynamics requires, however, a more complete model than simple isotropic rotation and multiple-frequency experiments carried out with different observable nuclei ( $^1H$ ,  $^2H$ ,  $^{17}O$ ) (Llor and Rigny 1986). Rotational isomerism on the headgroups allows a slight variation of the interfacial area per surfactant headgroup. Isomeric rotation is dependent on the water content and the nature (penetrating or non-penetrating) of the oil (Maitra and Eicke 1981, Martin and Magid 1981). This observation is very important and may be the key to understanding the unique ability of AOT to dissolve water in oil: the molecular conformation adjusts for providing an  $a_0 l_c / v$  value imposed by external conditions, the origin of AOT's solubilising properties being due to its flexibility at a molecular level.

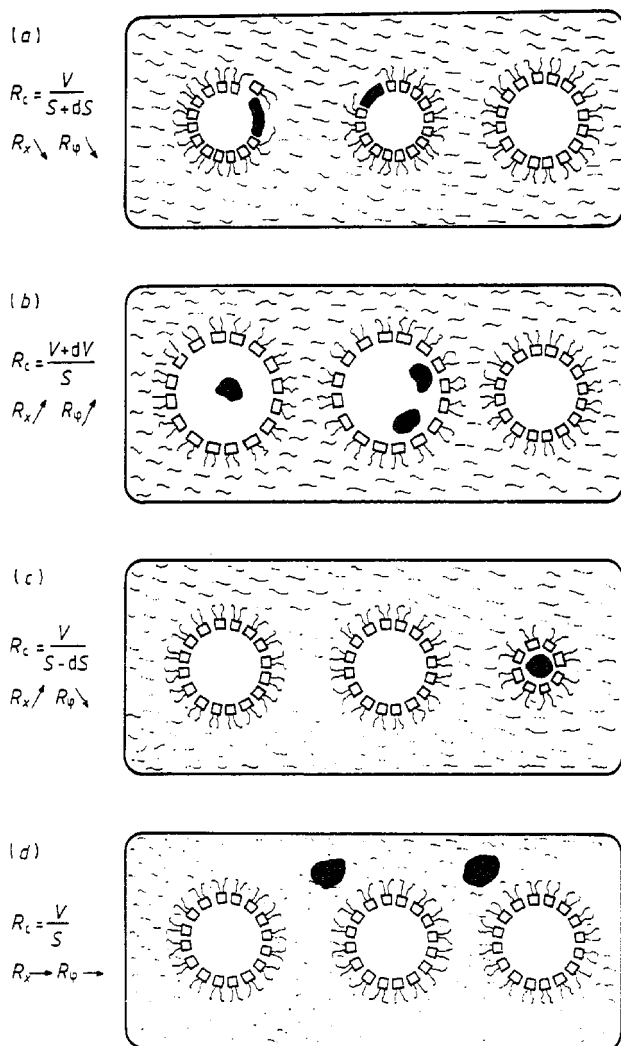
Recently, a molecular dynamics simulation of an aggregate of 72 model surfactant molecules and 1072 molecules of water has been shown to have the following properties: the water pool is a spherical cavity on average with significant roughening of the interface (Brown and Clarke 1988). The outermost water molecules are not hydrogen bonded but strongly coordinated to the counter-ion adsorbed at the interface. The significant solvent penetration into the interface is probably an artefact due to the modelling of the whole surfactant tail as a single spherical bead.

**5.3.2. Solubilisation of additives in reverse micelles.** Addition of solutes to reverse micelles results in quaternary systems, but the results are given here, because the basic structure of the water droplets remains after small additions of solute. A solute can



be located in the water core, in the interfacial region or in the bulk solvent. Solute distribution within these three locations has been indirectly obtained by means of fluorescence experiments (Encinas and Lissi 1986).

However, at low additive concentration, the basic structure of spheres interacting through a hard sphere and a slight attractive potential holds and one is interested in the perturbation of the structure by the additive. Since the radius of the water droplet is fixed by the polar volume and the available interface, four different types of solubilisation can induce four different types of structure perturbation (Pileni *et al* 1985a). From simple geometrical arguments, it is possible to deduce the solute location from the perturbation (figure 33).



**Figure 33.** Solute localisation in reverse micelles: different structure perturbations due to different solubilisation sites. (a) Adsorption at the interface provides a larger total interfacial area at constant volume, and a concomitant radius decrease. (b) Solubilisation into the water cores increases inner volume at constant interfacial area, thus a growth of radius. (c) Solute needs surfactant for its solubilisation, promoting formation of new micelles, interfacial area decreases at constant inner volume and radius increases. (d) Solute dissolves in the solvent, micelles are unaffected.

(i) If solubilisation occurs in the polar volume, its effect is mainly an increase of the average size of the micelle. This case has been observed with chymotrypsin:

(ii) If the solute is located at the interface, adsorption is equivalent to addition of surfactant, thus decreasing the micellar radius. This has been observed with cytochrome c.

(iii) If the solute dissolves in the solvent, no structure modification is expected: this has been observed with diethylether.

(iv) If the micelle is too small to receive a solute molecule without deformation, i.e. at low water content, a segregation occurs between small empty micelles and large objects which are mainly hydrated proteins covered with surfactant (Zampieri *et al* 1986, Chatenay *et al* 1987).

Chatenay has also shown that intermicellar interactions are strongly modified by the presence of a solute. Sheu *et al* (1986) confirmed that at high water content ( $w > 20$ ) a model with two populations of micelles is not needed for data interpretation and that solubilisation of cytochrome in reverse micelles induces an increase of droplet average radius.

The attractive interaction potential between reverse micelles is also strongly modified by the presence of the solute. The presence of alcohols as additives in the interfacial layer modifies the demixing temperature ('cloud point') and the low- $q$  region of the scattering (Howe *et al* 1986). As the 'cloud point' is reached by heating, a demixing into a dilute and a concentrated  $L_2$  phase occurs. As micelles have an identical size in both coexisting phases, the behaviour is similar to a liquid-gas equilibrium. When initial concentration and temperature are set so that the upper and lower concentrations are equal, a critical point is reached. In that context, the physics of reverse micelles is similar to that of pure molecular fluids.

Reverse micelles are very useful systems in biotechnology: it has been shown that whole bacteria (*E. coli*) can be solubilised and remain active in reverse micelles (Haering *et al* 1985). Reverse micelles as a bioseparation tool have been reviewed by Kadam (1986). A general review about hosts in reverse micelles has recently been published (Luisi *et al* 1988).

#### 5.4. Mixed micelles

When two different surfactants are used together, the resulting structure is called a mixed micelle.

Mixtures of surfactants behave very differently from what would be predicted from the properties of their pure components: mixtures are not ideal. This can be simply shown by the measurement of the CMC of a mixture of two surfactants: the CMC of a mixture does not vary linearly with the composition (Bourrel *et al* 1984, Hoiland *et al* 1985). Theories giving the composition of aggregates from the CMC variation with overall chemical composition of the mixture are available (Rubingh 1979, Hua and Rosen 1982, Nagarajan 1985, Nguyen *et al* 1986).

Indeed, the properties of a mixed micelle system depend mainly on the interaction of polar headgroups, which can be detected by volumetric studies (Funasaki *et al* 1986). Mixtures of surfactants with identical headgroups and different chain lengths are close to ideality. Strong deviations are observed with mixtures of ionic and non-ionic surfactants or cationic-anionic mixtures. Mixing cationic and anionic surfactants often results in an insoluble precipitate, but this is not always the case (Hoffmann and

Ulbricht 1987). The structure of these micelles differs from classical micelles. Interacting rodlike micelles are formed at low concentration. When salt is added, the stiff rods turn progressively into flexible ones, giving a viscoelastic character to the solution (Hoffmann *et al* 1983). These rodlike micelles are charged since the micelle chemical composition is not a 1-1 mixture. The presence of connected rods has been deduced from the observation of a scattering peak at very small angles. This system follows the prediction of the DOC model for a connectivity  $Z = 2$ . The size of the flexible rods first increases, and then decreases with concentration because of the very strong intermicellar repulsions (Hoffmann and Ulbricht 1987). This situation is analogous to the vesicle-to-micelle transition, which was surprisingly observed with increasing concentration (Ninham *et al* 1983b, Brady *et al* 1984). At higher concentration, above a transition composition detected as a break in the conductivity, the structure turns to uncharged disc-like micelles being 1-1 anionic-cationic surfactant mixtures in the aggregates.

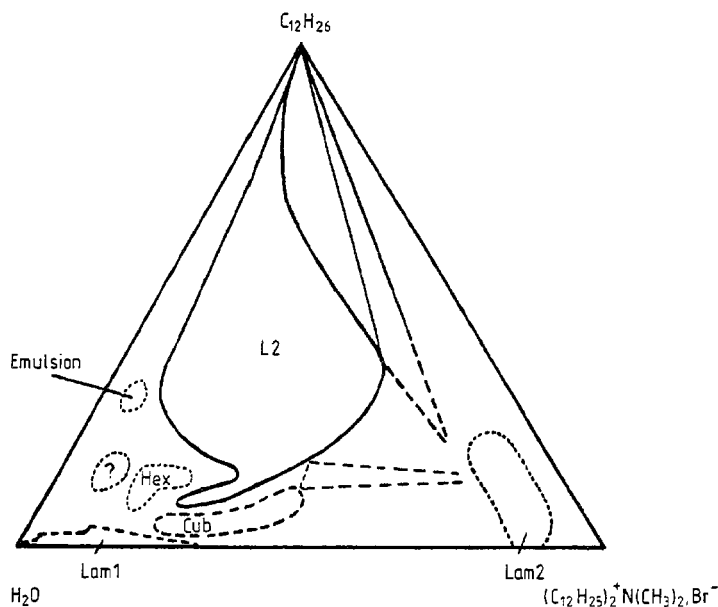
The field of mixed micelles made of non-ionic and ionic surfactants together is still an open field in terms of structural studies, even if these types of mixture are very widely used in industrial applications. A simple picture of interacting spherical droplets cannot account for light scattering experiments (Guéring *et al* 1985). Theoretical expressions of the counter-ion binding to such mixed aggregates has been proposed by Rathman and Scamehorn (1987). The concentration of monomers in equilibrium with mixed micelles and interaction parameters between molecules can be evaluated from CMC values of mixed systems (Asakawa *et al* 1985, Hey *et al* 1985, Wall and Elvingston 1985).

### 5.5. Double-chain cationic surfactants

Double-chain cationic surfactants are convenient for studying three-component microemulsions (without cosurfactant). The prototype of this type of surfactant is didodecyl-dimethylammonium bromide (DDAB), more than 50% of oil can solubilise in water with DDAB, the same amount as with AOT (Angel *et al* 1983). The phase diagrams show a very large  $L_2$  domain, which is very stable since DDAB is not beset with hydrolysis problems (Fontell *et al* 1986). DDAB is nearly insoluble in water and oil, ternary mixtures with DDAB are therefore an ideal microemulsion model system (figure 34).

The phase diagrams have been explored, together with the ordered anisotropic phases appearing around the large microemulsion phase (Fontell *et al* 1986). The differences between phase diagrams obtained in ternary systems with different oils are explained by the different oil uptake by the hydrocarbon tails (Ninham *et al* 1984, Evans *et al* 1986b). A large number of structures can be built up with these types of surfactant since the packing parameter is close to one. Phase diagrams have been studied by shortening one or two of the surfactant tails. For the system didecyl-dimethylammonium bromide/water/hexane, a new type of demixing between two  $L_2$  phases has been demonstrated: two microemulsions are in equilibrium, far from any critical point since temperature has no effect on that demixing (Warr *et al* 1988b).

The electrical conductivity shows inverse percolation behaviour at low water content: as water is added, the conductivity decreases by orders of magnitude at a volume fraction which depends on the type of oil (Chen *et al* 1984, Blum *et al* 1985). A structure as thin cylinders has therefore been proposed (Chen *et al* 1986). It is the only one compatible with the interfacial tension behaviour, due to strong correlations between parallel conducting thin cylinders of oil (Allen *et al* 1987). These disordered



**Figure 34.** Phase diagram of the water/dodecane/didodecyldimethylammonium bromide (DDAB) ternary system. L2 is the three-component microemulsion domain, Lam1, Lam2, Hex and Cub are mesophases (Fontell *et al* 1986).

open connected cylinders have been the first example of a DOC cylinder model: the structure can be described as a network of connected cylinders laid on a virtual Voronoi lattice (Zemb *et al* 1987). This network of connected cylinders, of connectivity close to 2 (figure 35), is the first model compatible with the x-ray scattering peak position (Ninham *et al* 1987). Once the constraint of maximal local curvature is added, the percolation threshold and the limit of the L<sub>2</sub> phase can be explained by purely geometrical arguments (Barnes *et al* 1987b).

Another consequence of the high sensitivity of this packing parameter close to one (i.e. spontaneous curvature near to zero) is that any slight change in counter-ions has enormous effects on the phase diagrams (Chen *et al* 1987). The most striking case is the substitution of the bromide counter-ion by hydroxide, which makes the surfactant soluble in water. At low water content, didodecyldimethylammonium hydroxide spontaneously forms unilamellar vesicles in water (Ninham *et al* 1983b, Brady *et al* 1984, Kachar *et al* 1984a). At high water content, these vesicles shrink because of the high electrostatic interactions, turning back to micelles. It was commonly accepted that in dilute binary solutions, monomers are in equilibrium with spherical micelles; the picture that emerges slowly from studies on ternary microemulsions is that unilamellar vesicles can be in equilibrium with micelles. The origin of this behaviour in terms of free energy remains to be elucidated.

### 5.6. Non-ionic surfactant/oil/water ternary systems

Non-ionic surfactant polar headgroups are generally made of polyoxyethylene: these groups are hydrated with one to two water molecules per oxyethylene group. Steric constraints are less drastic than for ionic surfactants: area per polar head and chain length are able to adapt for different compositions. Therefore, the possibility of a large

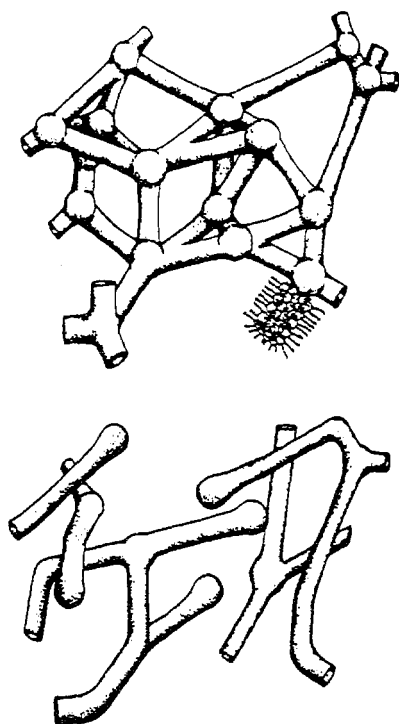


Figure 35. Schematic diagram of a typical microemulsion structure without cosurfactant, according to the DOC model. Top, connectivity  $Z = 4$ ; bottom, connectivity  $Z = 2$ .

stability domain of single-phase microemulsion exists. At low temperature, the non-ionic surfactant is soluble in water whereas at higher temperature the surfactant is soluble in oil: a certain temperature range around the phase inversion temperature allows then for three phase equilibria with middle phases without addition of cosurfactant. The phase diagrams of these systems are known (Kahlweit *et al* 1983, 1985).

General trends of the interfacial area per non-ionic headgroup versus volume fraction, which is the crucial measurement for structure investigation, are not yet known. A lower consolute point exists in binary solutions, where a simple model of interacting droplets holds to explain scattering spectra up to a concentration of 25% of surfactant.

### 5.7. Polymer-micelle mixtures

Most studies deal with water-soluble polymers in direct micelles. This is not micellar solubilisation but polymer adsorption at the micellar surface.

When the polymer solution (PEO) is dilute, the polymer is adsorbed around the SDS micelles. These keep the spontaneous size occurring in pure water (Cabane 1977). Opposite conclusions, i.e. smaller micelles, have been inferred since then for the same system by indirect methods (Gilyani and Wolfram 1981) and fluorescence probe studies (Zana *et al* 1985).

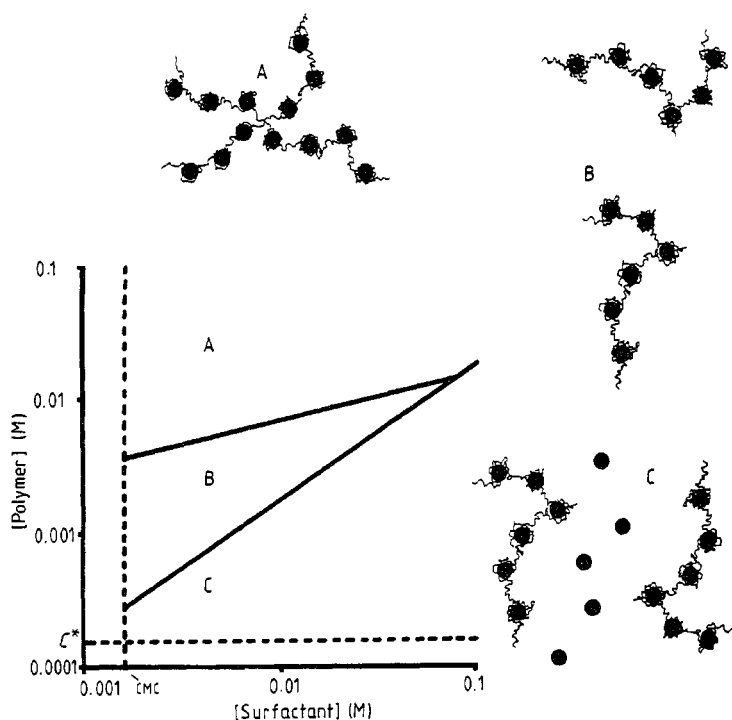
When the polymer is semidilute, two different structures in the polymer/micelle solution have been identified for the polyoxyethylene-SDS system (Cabane and Duplessix 1987): at high polymer-to-micelle ratio, the micellar structure is not perturbed

by the polymer. The presence of the polymer network does not perturb the set of interacting globular micelles.

At lower polymer-to-micelle ratio, one has necklaces of polymers decorated by SDS micelles: one observes a very peculiar scattering curve with two correlation peaks moving with different scaling laws towards concentration (Cabane and Duplessix 1985, 1987). The polymer strands are therefore seen as the origin of an attractive interaction between micelles: the system is a semidilute solution of independent necklaces (figure 36).

The complex formed between bovine serum albumin and lithium dodecylsulphate micelles has been studied by Chen and Teixeira (1986). At high surfactant concentration the protein is denatured, as are most proteins with ionic surfactants. From the linear behaviour of the logarithm of the intensity, a 'fractal' structure has been attributed to the protein-micelle aggregate. This is not the case when non-denaturing non-ionic surfactants are used: in the rhodopsin-detergent complex, the apolar part of the rhodopsin is in the core of the large non-ionic micelles (Osborne *et al* 1978).

Mixtures of polyelectrolytes and micelles have just started to be studied by an exploration of the phase diagram (Dubin *et al* 1985, Dubin and Davis 1985). Polyion-mixed micelle complexes have been detected. The main parameter controlling the phase diagram is the surface charge density of the micelles, which can be set by the anionic-to-non-ionic ratio in the surfactant mixture used. For the system SDS-cationic



**Figure 36.** Some possible structures of mixtures of a neutral water-soluble polymer (polyethyleneoxide) with SDS micelles. The polymer adsorbs at the micellar surface and makes bridges between micelles. The resulting necklace-like structures are of two types: A, polymer does not perturb the structure of the SDS solution; B, lower polymer concentration causes concentration of micelles in the polymer coils and micelle depletion in the polymer-free parts of solution; C, coexistence of structure B with excess micelles (Cabane and Duplessix 1987).

polyelectrolytes, addition of a stoichiometric quantity of SDS leads to precipitation. Further addition of surfactant allows redissolution of the complex, presumably by formation of an adsorbed SDS bilayer (Leung *et al* 1986). Phase diagrams and solubility data in mixed polymer surfactant systems have been reviewed by Goddard (1986).

Due to the diversity of the molecular forces present in these types of mixtures, a whole set of new structures will probably be discovered in the near future in these solutions.

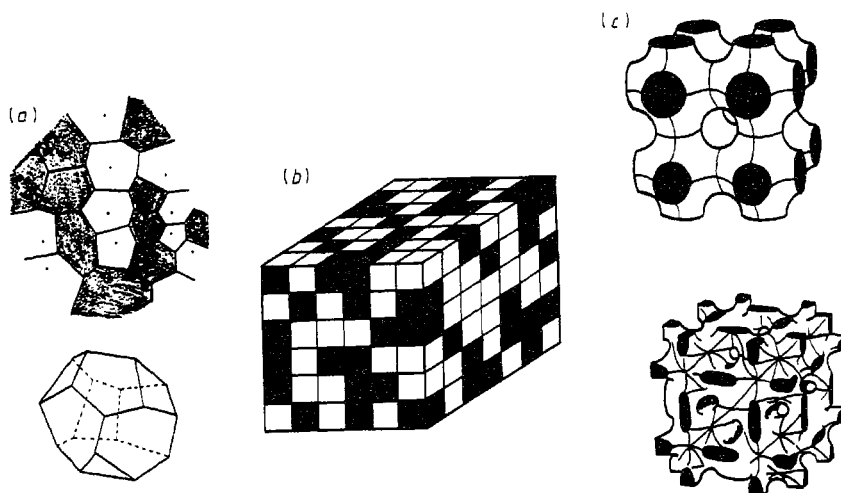
## 6. Quaternary systems

Isotropic solutions made of four chemical components are very complicated systems: the complete phase diagram needs to be represented in a tetrahedron and is often not available. Most systems are composed of a surfactant, a cosurfactant and two solvents: water and some kind of oil. The cosurfactant is usually a single-chain alcohol, although the interesting case of amine has been studied by Wormuth and Kaler (1987).

### 6.1. Current models of microemulsion microstructure

The terms 'bicontinuous', 'bicontiguous' or 'random' structure are used in most papers, but there is a lack of quantitative structural geometrical relations allowing a clear distinction between these intuitive concepts. Here, we give the main aspects of the most widely used structural models for which predictions can be tested experimentally (figure 37):

(i) The first intuition of a bicontinuous structure is due to Scriven (1976, 1977). Since the diffusion coefficients of water and oil are both high (Kaler *et al* 1983a, b, Clarkson *et al* 1985, Guéring and Lindman 1985, Lindman and Stilbs 1987), some continuity must exist between the water and oil domains. Since birefringence or Bragg



**Figure 37.** Schematic representations of bicontinuous microemulsion models: (a) Talmon and Prager (1978): random filling of Voronoi polyhedra with water or oil (top); unit water or oil cell (bottom). (b) De Gennes and Taupin (1982): random filling of cubes on a cubic lattice, the repetition distance equals the correlation length. (c) Scriven (1977): two examples of bicontinuous mesophase-like structures with minimal interfacial area.

peaks are not observed, the structure cannot be macroscopically ordered. Therefore, the microemulsion must be some kind of molten cubic phase. There was no experimental evidence of that idea at that time, nor was the model quantitative.

(ii) The first quantitative model of microemulsion is the Talmon and Prager model (Talmon and Prager 1978, 1982, Kaler and Prager 1982). The rules for constructing such a bicontinuous model are the following: first choose a set of random points (Poisson points); these points define a complete tessellation of space in adjacent cells (Voronoi tessellation). Each point in space belongs to the cell centred at the closest Poisson point. This is a very general method of generating a 'random lattice' whose geometrical properties have been studied extensively (Meijering 1953). Then, fill at random the cells with water or oil, according to the global volume fractions of water and oil. The surfactant film volume fraction is considered as negligible and surfactant film is distributed only along the faces common to two adjacent cells filled with different phases.

The great success of the Talmon-Prager model has been to generate an average size of the resulting oil and water domains which are of the order of magnitude of the observed size (Talmon and Prager 1982). It was the first available quantitative model which could be tested, since a clever method for deriving an analytical form of the scattering was worked out by Kaler and Prager (1982). But there are two major drawbacks: the conductivity of microemulsions should always exhibit percolation behaviour instead of the non-monotonic one observed in some systems and no peak should occur in the observed scattering. Unfortunately, except for few samples very close to a critical point, there is always a peak in the observed scattering. The Talmon-Prager model cannot represent the structure of a microemulsion as noticed by many authors (Zemb *et al* 1987, Vonk *et al* 1988, Teubner and Strey 1987).

The Talmon and Prager model has been improved by de Gennes and Taupin (1982) in the case where the spontaneous curvature is nearly zero: in that case, the characteristic length  $\xi_k$  is large (about 50 nm) and close to the size set by the low surfactant content. In that case, an approximate model of structure is the following: the random polyhedra are replaced (for the sake of simplicity in order to evaluate the phase diagram) by cubic cells, filled at random with water and oil, according to the macroscopic volume fraction set by composition. This is called the CRC (cubic random cell) model. The spontaneous average curvature  $\xi_k$  is then given by the film bending energy

$$\xi_k = a \exp\left(-\frac{4\pi K_0}{kT}\right)$$

where  $K_0$  is the film elasticity constant and  $a$  is a molecular size (the lower cut-off wavelength of the film undulations). The microemulsion is stable when the spontaneous length  $\xi_k$  is of the same order of magnitude as the size set by the volume fraction  $\varphi$  and the available area  $\Sigma$  (see below).

This curvature allows a natural scaling of the system, whose structure is approximated as follows: construct a regular lattice of cubes, of edge length  $\xi_k$ , filled at random with a probability given by the macroscopically imposed volumes of water and oil. As in the Talmon-Prager model, the interface is located at the edges of the cubic elementary cells (de Gennes and Taupin 1982). This model has essentially been built in order to allow calculation of phase diagrams, and this was made by Widom (1984a, b), who showed that the Winsor I-Winsor II-Winsor III behaviour can be obtained with CRC structure; it was further refined by Safran *et al* (1986).



Strictly speaking, the CRC model does not predict a peak in the observed scattering because a zero in the scattering of the individual cube wipes out the Bragg peak given by the cubic lattice. If any thermal disorder in the underlying perfect cubic lattice is introduced, however, a peak appears at the position related to the cube edge length  $a$ , given by

$$q = 2\pi/a.$$

The CRC model has been compared with the observed scattering by Auvray *et al* (1984a, b). Qualitatively, the behaviour is correct; a peak appears in the scattering and the variation of the peak position is approximately predicted by the CRC model. Also, bicontinuous structures can be stable (Schick and Wei-Heng 1986) and one can observe three phase equilibria (Widom 1984a, b). Both the CRC and Talmon-Prager models are compatible with the high self-diffusion coefficients of oil and water in the same sample in concentrated microemulsions (Lindman *et al* 1981, Guéring and Lindman 1985). Results of quasi-elastic light scattering too can be understood with these models (Chang and Kaler 1986), but the scattering peak position with the CRC model is systematically wrong by a factor of 1.5 to 3. Moreover, with the CRC model as with the Talmon-Prager model, the conductivity should always vary monotonically with composition. As shown by Auvray, a basic feature of the CRC model has to be modified to agree with the observed scattering: the cube edge length, instead of being fixed by the spontaneous curvature, should rather be set, as in ternary systems, by the available interface; the cube edge length is therefore dependent on the specific area covered by surfactant instead of the spontaneous curvature, and is given by

$$\xi_k = \frac{6\varphi(1-\varphi)}{\Sigma}.$$

The non-monotonic behaviour of the conductivity, already noticed by Baker *et al* (1984), is incompatible with the CRC or Talmon-Prager models, as is the observation of a long-range distance obtained in the Patterson function of the scattered intensity (De Geyer 1987). The conductivity shows a maximum in the middle of the  $L_2$  phase: this is a general feature which appears in three different systems where the surfactant is SDS, potassium oleate or AOT (Boned *et al* 1980, 1982). In some cases, the conductivity does not show a maximum but a percolation threshold which can be related to percolation of small spherical droplets (Laguës *et al* 1978).

The position of the scattering peak in Winsor III microemulsions is incompatible with the CRC model and could be explained in a rough approximation as a molten cubic phase, thus indicating a local complexity in terms of assembly of 'distorted minimal curvature' surfaces (De Geyer 1987). The same observation was also made earlier by Parodi (1984): here a locally lamellar structure is constructed. With these two models, the scattering curve can be reproduced assuming a very short correlation length of the local structure.

The DOC model of microemulsions has been introduced to explain the behaviour of ternary systems (see § 5). This model takes into account the fact that there is more interfacial area for a given structure than expected from other structural models. In the case where the volume fraction and the butanol partition can be calculated from dilution experiments, Abillon *et al* (1988) have shown that the peak position and the average chords in a quaternary system are compatible with the DOC model in the case of cylinders. The effect of the alcohol partition between interfacial film, water and oil is measured by a variation of the self-diffusion coefficient (Ceglie *et al* 1987). If butanol is used as cosurfactant, the molecular exchange is so rapid that all macroscopic

self-diffusion coefficients are high, suggesting that butanol destroys the organisation of interfacial film (Stilbs 1983).

In another quaternary system, Vonk *et al* (1988) have shown that the scattering could be calculated on the absolute scale for a large set of compositions, assuming only two parameters: thickness of local lamellar structure, and an orientational order parameter between the normals to planes. The distortion length found is on the order of 50 nm, and the direction persistence length on the order of 25 nm, as predicted by the DOC lamellar model (Zemb *et al* 1987, Barnes *et al* 1988a, b). A similar local lamellar structure has been proposed in the SDS/butanol/toluene/brine system by Bellocq *et al* (1979) and Bellocq and Fourche (1980).

Another model of microemulsions has been proposed by Berk (1987): a striking feature of neutron scattering of microemulsions is that they give a peak when oil or water is deuterated but the peak sometimes disappears when one observes the scattering of the film alone (both deuterated water and oil). This feature could be explained by exhibiting a random structure built by random interfering waves (Cahn 1965). This study showed that a bicontinuous structure can be constructed, avoiding the phase problem, by setting the volume fraction and the observed scattering. Two-dimensional analogues of microemulsions have been constructed in order to show that one can move continuously from water-in-oil to oil-in-water droplets while keeping the scattering strictly constant: the fact that the observed  $I(q)$  does not vary even when conductivity varies by orders of magnitude is demonstrated (Welberry and Zemb 1988).

A mechanical model of microemulsions came to the conclusion that within the same 'phase' in the phase diagram, different structures can coexist and transform continuously into each other: short cylinders towards spheres and local distorted lamellae (Neogi *et al* 1987). This model has not yet been compared with experiments due to lack of quantitative predictions.

The quaternary systems are mostly studied in the region of the phase diagram which is poor in surfactants, keeping realistic industrial applications such as detergency or oil recovery in mind. Since the complete phase diagram is a tetrahedron, it is usual to represent a slice of it, which is called pseudo-ternary phase diagram. Some ratio has to be fixed: the traditional way is to represent a projection of the phase diagram at constant surfactant-to-cosurfactant ratio. It is known that the water-to-surfactant ratio  $w$  is a good thermodynamic variable: the ratio  $w$  of water to surfactant is the same in two coexisting phases in a multiphasic region. Limits of a single-phase region often correspond to a constant  $w$  value; therefore, a representation at constant  $w$  is preferred (Cebula *et al* 1982). The same ratio  $w$  is a field variable in critical fluctuations (Roux and Bellocq 1984). Indeed, a given ratio  $w$  corresponds to a fixed average curvature (Friberg and Buraszenka 1978). This value is called the spontaneous radius of curvature. For ternary systems, the spontaneous curvatures  $H$  and  $K$  are given by

$$v/a_0l = 1 + \langle H \rangle l + \frac{1}{3} \langle K \rangle l^2$$

where  $l$  is the interfacial film thickness and  $H$  and  $K$  are the average mean and Gaussian curvatures. If the structure is a dispersion of large droplets, this constraint reduces to

$$v/a_0l \approx 1 + Rl$$

because the sphere radius, the average mean and Gaussian curvatures are the same ( $R = H = K$ ). This radius is then equal to the persistence length  $\xi_k$ . If  $H$  is different from  $K$ , a common approximation is to assume that  $\xi_k = H$ . In ternary systems,  $\xi_k$  is

therefore mainly set by molecular packing. When a cosurfactant is added to a single-chain surfactant, geometrical constraints are relaxed: the  $v/a_0l$  parameter is modified, since each cosurfactant molecule increases the effective volume  $v$  by the value of its own molecular volume. Interfacial chain length  $l$  is not modified but the area per surfactant headgroup increases by about  $0.2 \text{ nm}^2$  per alcohol molecule penetrating into the interfacial film.

The second important parameter to the microemulsion structure has different names in the literature. It can be intuitively understood as 'granularity' of the structure or 'cell size' or 'periodicity'. In the case where a peak is seen in the scattering at a value  $q^*$ , the cell size  $D^*$  is given by

$$D^* = 2\pi/q^*.$$

But a cell size  $D^*$  can also be defined without any peak in the scattering: the inverse interfacial area per unit volume  $1/S$  is a natural length in the system.  $1/S$  is, however, only an order of magnitude, the exact value of the cell size is given at fixed interfacial area and volume fraction by the peak position. If the structure of the microemulsion reduces to a dispersion of spheres,  $\xi_k$  is the radius of the spheres where  $D^*$  corresponds to the average distance between spheres: each sphere is at the centre of a Voronoi cell.

The third important parameter is the stiffness of the interface. The bending energy  $F$  of the interfacial film is set as (Helfrich 1978, De Gennes and Taupin 1982, Safran and Turkevitch 1983, Safran *et al* 1984)

$$Fb = \iint \left[ \frac{K_0}{2} \left( \frac{1}{R_1} + \frac{1}{R_2} - \frac{2}{\xi_k} \right)^2 + \frac{\bar{K}}{2} \left( \frac{1}{R_1} - \frac{1}{R_2} \right)^2 \right] ds.$$

A common simplifying assumption is that the saddle-splay term depending on  $\bar{K}$  is negligible compared with the bending energy depending upon  $K_0$ . This is only valid in favourable cases, but up to now no general theory which assumes that the two terms are equally important is available. The value of  $K_0$  can be evaluated by optical methods, assuming that the roughness of the interface between coexisting phases is related to  $K_0$  (Meunier 1985). The result of these studies shows that the value of  $K_0$  is much higher for ternary systems ( $K_0 > kT$ ) than for quaternary ones where  $K_0 > kT$  holds. A similar conclusion has been drawn from electric and magnetic birefringence measurements (Meyer *et al* 1982).

This allows us to introduce a reduced temperature  $\tau$  as

$$\tau = T/2\pi K_0.$$

When the reduced temperature  $\tau < 1$ , a zero-temperature model can be used: the spontaneous curvature is constant throughout the structure. Predicted microstructures are expected to be lamellar, cylindrical or spherical (Safran *et al* 1984). In that case, the DOC model developed for ternary systems applies. If  $\xi_k$  is lower than  $D^*$ , the microstructure can be approximated to connected cylinders (DOC cylinder model) (Ninham *et al* 1987). If  $\xi_k$  is larger than  $D^*$ , a sponge-like structure is expected (Cates *et al* 1988). The intermediate case is difficult: the best model could be a disordered dispersion of Voronoi cells, filled with either water or oil (Talmon and Prager 1982, Schick and Wei-Heng 1986, Andelmann *et al* 1987). Indeed, the flexibility of the film is very low and both distances  $\xi_k$  and  $D^*$  can be reduced to a single distance  $\xi_k$  and the original CRC model is used with a size-dependent bending constant. This model allows calculation of a phase diagram but is incompatible with the measured value of the average chord (Zemb *et al* 1987). A review comparing droplet models and the CRC model is available (Langevin 1986).

If  $\tau > 1$ , then thermal fluctuations are important: the interface has a bending radius which is no longer constant at every point and thermal fluctuations wipe out geometrical constraints. No geometrical model of microstructure holds any more. The dispersion has to be described in terms of cells whose size is fixed by geometrical constraints. Some arbitrary energy terms have to be associated with local contiguity of water and oil domains (Andelmann *et al* 1987, Welberry and Zemb 1988, Milner *et al* 1988).

The phenomenological expression derived by Teubner and Strey (1987) for ternary systems can also be used here. In ternary systems, the persistence length does not vary with concentration whereas addition of butanol allows a large variation of  $\xi_k$  with concentration.

At high reduced temperature, if the persistence length  $\xi_k$  is large and the volume fractions are low, a dispersion of lamellae, without long-range order is predicted in the region between the microemulsion and the lamellar phase (Cates *et al* 1988). This phase has indeed been observed (Porte *et al* 1986) and is of crucial importance: its recent discovery creates a continuous conceptual path between micelles, microemulsions and vesicles. Geometrical rules for construction of a random dispersion of lamellae are the following:

(i) Divide space into a cubic lattice (for calculation of a phase diagram) or into a Voronoi tessellation (for structural investigations). The lattice size is set by interfacial area and volumes.

(ii) Fill the cells with two 'species' of oil, species 1 and 2. These species are of course really identical. The species are distributed at random within the lattice. The only parameter is the 'pseudo-volume fraction'  $\psi$  of species 1.

(iii) If two adjacent cells are filled with different 'species' of oil, the interface between them is a thin layer of water covered by two surfactant films. The cells are filled with the major solvent.

In the following, we refer to these construction rules as the random lamellar model. The resulting structure is a disordered 'sponge' of water in oil. If  $\psi$  is about 0.5, the structure is close to a folded and compressed sheet of paper. The interfacial film is continuous, it has no edge. For low  $\psi$ , the structure turns naturally towards dilute vesicles (Cates *et al* 1988).

## 6.2. Experimental evidence

A review devoted to experimental methods of studying microemulsions is available (Kahlweit *et al* 1987). Some observations on the structure of microemulsions can be directly measured by scattering without use of any model. These points are direct evidence obtained with x-ray or light scattering, and are more powerful than indirect measurement obtained by spectroscopic methods. These are:

(i) The direct measurement of the average radius of curvature by Auvray *et al* (1984a, b) in the system SDS/H<sub>2</sub>O/butanol/oil. When the oil and water have equal volume fractions, the average radius of curvature is zero. There is no information about the Gaussian curvature of the interface.

(ii) The direct comparison of the average chord through the structure towards the position of the scattering peak: the CRC model and all models filling random cells predict these two quantities to be equal. Locally lamellar, hard sphere and DOC models predict the observed ratio of 2 between these two quantities.

Up to now, there is no quaternary system whose microstructure is understood well enough to derive the scattering from composition alone. The problem is that volume fraction and specific interface  $S$  cannot be precisely determined. If one uses a model of ellipsoids (as in Caponetti *et al* (1986)), any scattering curve can be fitted to interacting monodisperse ellipsoids. The good agreement between fitted scattering curves and experiment is therefore not a definite proof of a structural model.

Quantitative determination of the microstructure in quaternary systems is very difficult because there are four components which are distributed in three pseudo-phases: the water phase, which contains a negligible proportion of surfactant, the oil phase and the surfactant film, where most of the surfactant molecules are located. The main problem is the distribution of the alcohol between these three pseudo-phases (Biais *et al* 1981, 1986). In the pseudo-phase framework, one can always remove one of the three pseudo-phases, for instance the interfacial film. Therefore, it should always be possible to find a surfactant-free ternary solution with the same vapour pressure as the complete four-component microemulsion. This hypothesis holds as a first approximation in all systems studied so far.

Another approach consists (for some favourable cases, i.e. near the water or the oil corner of the phase diagram) in diluting the microemulsion at constant droplet size with an appropriate mixture of oil and surfactant. A correct trajectory in the multicomponent phase diagram has thus to be defined where the diluent chemical composition is the same as that of the continuous phase. The procedure consists in a dilution along (close to) a diphasic domain boundary in such a way that the overall composition of additions remains constant (Schulman *et al* 1959, Graciaa *et al* 1976, 1977, 1978). It is supported by the observation that the water-to-surfactant ratio is constant along a demixing line in water-in-oil microemulsion phase diagrams, this ratio governing the size if the area per headgroup is constant (Roux 1984, Roux and Bellocoq 1984). The particle size is expected to reach its maximum value along such a pathway, further growth causing instability and phase separation. Under such a dilution procedure, the chemical potentials of all the components remain constant. The constant chemical composition of droplets has been checked by experiments (Laguës *et al* 1978), but interactions may influence droplet size, even at constant composition (Cazabat 1983). Another problem is that the proximity of critical points has to be avoided. This dilution procedure has allowed light scattering studies of the interaction between droplets (Cazabat 1983). At high dilution, near the water corner or near the oil corner, hard sphere interaction gives a good approximation of the scattering (Cebula *et al* 1982). In this latter paper, interpretations concerning concentrated systems are, however, questionable, since the interparticle scattering was oversimplified.

Some systems can be considered as pseudo-ternary systems, where brine replaces pure water in the composition of the sample. Adding salt is a very handy way of testing the importance of electrostatics in microemulsions since the two main effects of salt addition are to screen out the electrostatic interactions and to reduce the area per headgroup, thus varying the spontaneous curvature and the packing parameter  $v/a_0l_c$  introduced to roughly predict the phase behaviour. This has been predicted in the first zero-order geometrical unified theory of micelles, microemulsions and vesicles (Mitchell and Ninham 1981), where the theory of self-aggregation is extended from micelles to vesicles and microemulsions. The main observed trends with temperature variation, oil type variation or salt concentration are explained with the effective packing parameter  $v/a_0l_c$ . Predictions have been verified in terms of extension of the stability region of single-phase microemulsions:  $v/a_0l_c$  has to be increased close to

unity with a flexible interface in order to allow a large single-phase microemulsion (Lang *et al* 1984). This observation is called 'chain length compatibility' (Bansal *et al* 1980, Leung *et al* 1986). Since the  $v/a_0l_c$  parameter of common single-chain surfactants is lower than one, addition of small quantities of quaternary ammonium surfactants increases the spontaneous radius of curvature in SDS microemulsions (Sauterey *et al* 1978).

The salinity variation is an easy way of varying the geometrical packing parameter  $v/a_0l_c$  and hence the behaviour of a pseudo-quaternary system (since brine is used instead of pure water). At low salt content, one has equilibrium between microemulsion and excess oil (Winsor I), then a microemulsion in equilibrium with excess water and oil (Winsor III) and finally, at salinities of the order of  $50\text{ g l}^{-1}$ , excess water in equilibrium with a microemulsion (Winsor II). Usually, the term 'middle phase' refers to the microemulsion part of a Winsor III system. This 'Winsor' behaviour occurs in most systems, but not always: two microemulsion phases in equilibrium are sometimes observed (Roux *et al* 1983b). If the microstructure remains as spherical droplets, this phase behaviour can be understood in terms of van der Waals attraction between droplets (Huh 1984), but the same behaviour can occur with any other microstructure.

The structure of quaternary microemulsions seems to be very close to perfectly monodisperse spheres of oil in water near the oil corner (roughly more than 95% of oil) or near the water corner of the phase diagram. When the picture of closed droplets holds, the microemulsion is called water-in-oil ( $L_2$ ) or oil-in-water ( $L_1$ ) microemulsion, the minor compound is supposed to be located 'inside'. A detailed review of the structure and the thermodynamics of microemulsions in these 'dilute' cases is available (Bellocq *et al* 1984). For water-in-oil ( $L_2$ ) microemulsions, the dilution procedure at the boundary of a demixing curve was used in a large number of light, x-ray and neutron scattering studies about the structure in the dilute water-in-oil regime: the emerging picture is that the spherical droplets interact through an attractive potential which is due to the chain overlapping (Cazabat and Langevin 1981, Bellocq *et al* 1984, Huang *et al* 1984, Dichristina *et al* 1985). Evidence has been obtained for liquid gas transitions and critical points due to this attraction (Fourche *et al* 1982, Cazabat *et al* 1982a). The relationship between critical points and percolation thresholds has been studied (Cazabat *et al* 1984a). The effect of critical concentration fluctuations of attractive droplets near a coexistence curve allows the calculation of the observed scattering curve over four decades in intensity for the CTAB/octane/butanol/brine system (Magid *et al* 1984b).

The problem of microemulsion microstructure is much more complicated in the centre of the phase diagram, where any conductivity or molecular diffusion measurement suggests a 'bicontinuous' structure (Lindman *et al* 1980a, b, Larché *et al* 1980). This latter term is not defined in most papers dealing with structure; 'bicontinuous' means that there exists a continuous path completely included in the two solvents and extending over macroscopic distances. Bicontinuity at lower water content is easily proved by simple conductimetric measurements; for  $L_1$  microemulsions, more sophisticated techniques like pulsed field gradient spin-echo NMR are needed. Construction of a quantitative model of the relative distribution of water and oil is the central problem of the 'structure of microemulsions'. This description should be ordered on a microscopic scale and disordered on a macroscopic scale.

A critical discussion about the concentration range where a spherical droplet model breaks down is due to Kaler *et al* (1983a, b) and, as an alternative, the Talmon-Prager model using randomly filled Voronoi cells is proposed. In this work, however, the

correlation peak in the scattering was missed because of the linear collimation used in the x-ray camera. Many and various results clearly show that the simple CRC or droplet model cannot be reconciled with experimental measurements: these are lines of isoconductivity and isoviscosity in the  $L_2$  microemulsion phase diagram (Rushforth *et al* 1986), similar and apparently paradoxical conductivity behaviours (Georges *et al* 1987) could also only be interpreted through a sphere-to-cylinder or sphere-to-lamella transition (Singh *et al* 1983). Also, in the case of oil-in-water microemulsions, fluorescence studies could not be reconciled with spherical droplets of constant radius (Kano *et al* 1984) and molar volumes measurements lead to the same conclusion (Roux *et al* 1981). None of these observations can be reconciled with the occurrence of dispersed droplets, but they are also in opposition to CRC models or Talmon-Prager models: the only structural model allowing complex conductivity behaviour is the DOC model.

The experimental problem is that the scattering curves can give 'reasonable fits' by any model of interacting oblate ellipsoids, even in the case where there should be some continuity in water domains as indicated by electrical conductivity (as in Caponetti *et al* (1986)). The only way to discriminate between models is to measure precisely the volume of the three pseudo-phases and also take into account the value of the total interfacial area: the plot of the maximum of the scattering against volume fraction is the most critical test of any structural model because of topological constraints. This type of plot has been extensively used to show that the model of spherical droplets has to break down at high concentration (Auvray *et al* 1984a, b, De Geyer 1987, Abillon *et al* 1988, Zemb *et al* 1987).

Microstructure is intimately related to the phase diagram: a dispersion of droplets can always be diluted by adding a mixture which has the same composition as the continuous phase. But connected lamellae or cylinders cannot be diluted without microstructural change: therefore, for this type of structure, the monophasic domain is restricted to the centre of the phase diagram and is not connected to the edges (Friberg and Buraszenka 1978). This crucial observation has often been overlooked.

Geometrical packing constraints forbid a structure in droplets of water or oil in the reverse solvent for purely geometrical reasons in a large area of the phase diagram (Biais *et al* 1981, Israelachvili 1985). In regions of high surfactant content or near a two-phase boundary, the structure is unknown and there is no simple model of it, since the conductivity varies in some complex way: one observes increases and decreases of the conductivity by varying water content (Rushforth *et al* 1986). The first observation that conductivity does not increase as water is added was due to Schulman and Riley (1948) who called these samples 'anomalous' microemulsions. This conductivity drop has been studied in detail in the system potassium oleate/hexanol/hexadecane/brine (Shah *et al* 1977); as in ternary systems, this conductivity drop by adding water is only consistent with the DOC model if one assumes a maximum bending radius for the interface (Barnes *et al* 1988b).

This problem has been addressed by Clausee *et al* (1981, 1985, 1987a, b) and Heil *et al* (1982). These authors propose a general classification of all microemulsions into two broad classes: the type S class where the  $L_1$  (oil-in-water) is disjoint from the  $L_2$  phase in the phase diagram and the so-called U class where it is possible to follow a continuous path from the oil corner to the water corner of the phase diagram. A very tiny chemical modification of the cosurfactant is needed to switch from a U-type phase diagram to an S-type phase diagram. There is no quantitative theory of the origin of that effect, except that it has been known for a very long time that short-chain alcohols, which distribute more equally between oil and water, favour U-type structures (Ekwall

1969, Stilbs *et al* 1983b). A borderline case between U- and S-type microemulsions has been studied by Bassereau *et al* (1985).

More microscopic structural properties are no different in quaternary microemulsions than in micelles. Microwave absorption in the GHz range shows that the 'hydration water', i.e. the number of water molecules directly bound to ionic headgroups and adsorbed counter-ions, is also of the order of 10 molecules per ionic headgroup (Epstein *et al* 1983). NMR relaxation studies lead to similar order parameter profiles in microemulsions to those observed in binary micelles (Söderman and Walderhaug 1986).

## 7. Conclusion

### 7.1. Chemical reactivity in micellar solutions

Micellar solutions have been recognised to be powerful solvents for mixing reagents that cannot be dissolved in the same solvent (Duynstee and Grunwald 1959, Menger *et al* 1975). With such a solvent, two reagents, one being soluble in water and the other one being soluble in apolar solvents, can be put together in a microdiphasic solution with a very large interfacial area between two media; this improves considerably reaction rates as compared with phase transfer reactions. Apart from this obvious but interesting use, it is expected that reactions occurring at an interface could be oriented by the host structure. When several chemical reaction paths are allowed (as always in chemistry), the interface structure may accelerate or inhibit some of them and, as a consequence, induce selectivity which cannot be expected from a bulk solvent. Some reactions (but only few) that do not occur in bulk solvents were shown to occur at an interface.

Very intensive work has been done on these ideas, and it should be recognised that results are worse than expectations: induced selectivities do exist but are not large enough to be of practical interest. The origin of this partial failure lies in the fast dynamics of the structures. For compartmentalisation and orientation of reagents to be efficient, it should remain so throughout the reaction time, so that large structure-induced selectivities are expected only when reaction rates are faster than micellar dynamics (especially exchange processes, see § 4.1.9). On this basis successful results were obtained with fast reactions, mainly photochemical ones, and structures with slow dynamics (bilayers and vesicles).

Microheterogeneous media are, moreover, simple models for biostructures and attempts are currently being made to mimic biological reactions (Tsuchida *et al* 1984). This part of the review addresses mainly general concepts; several reviews exist and the books of Fendler and Fendler (1975) and Fendler (1982) contain exhaustive bibliographies.

**7.1.1. The chemist's point of view.** A chemist involved in synthesis expects reactions with good yields and preparative scale procedures, low solvent quantities and products which are easily recoverable and pure. This is not achieved by micellar solvents which solubilise low quantities, contain surfactants which are hard to remove, and which foam on vacuum evaporation. As the reaction proceeds, chemical composition changes and structure modifications may occur. Few preparative scale reactions involving micellar solutions are described. For use of large quantities of reagents, microemulsions are more suitable, one reagent replacing oil (Sakellariou-Fargues *et al* 1982, Perez *et*



*al* 1987), provided they remain monophasic all along the reaction course. Reactions are generally better carried out in diphasic media under phase transfer conditions. Phase transfer catalysts which are often added are amphiphilic compounds, so that some micelle-like structures may be the origin of the catalytic efficiency.

Most works on reactivity are studies of mechanisms, they are performed under conditions such that interpretation is easier: low reagent concentrations (for minimal structure modification) and dilute solutions. In-line analysis is generally performed, so that no product recovery is attempted. Rate modifications are often observed with subsequent selectivity changes. They are estimated with reference to bulk solvents or solutions below the CMC, which is a problem as both aqueous and hydrocarbon solvents are present in micellar solutions. Given a chemical mechanism, three main quantities influence reaction rates: the concentrations of reagents at the reaction site, the height of the energy barrier for reaction to occur (activation energy) and the diffusion rate of reagents. When the activation energy is low, reactions are diffusion controlled (photochemical or acid-base ones), while most organic reactions are slow and controlled by the activation energy.

A typical local concentration change by micelle intervention comes from the electrostatic potential around the micelles when ionic reagents are present. Reactions of micelle-solubilised hydrophobic reagents with reactive ions are accelerated when the ion and micelle charges are opposite because of electrostatic attraction (local high concentrations); when charges are of the same sign, reaction is slowed down (Duynstee and Grunwald 1959). Quantitative explanations are possible with Poisson-Boltzmann calculations of local concentrations (Chevalier and Chachaty 1985, Bunton and Moffatt 1985). In such cases, the reaction mechanism remains the same as in bulk solvents. Cases of failure of electrostatics-based interpretation are then of greater interest since new mechanisms are involved (Gonsalves *et al* 1985, Bunton and Cuenca 1987). An effort to separate trivial concentration effects and true structure-induced mechanism modifications is on hand (Romsted 1985).

Apart from electrostatics, local medium (solvent) effects are called forth but further analysis is prevented by the poor basic knowledge of solvent effects on chemical reactions and because interface physical properties cannot be related to any equivalent bulk solvent. The solvent effect is often a lowering of activation energy, but nothing is stated about the structural origin of this lowering. Generally speaking, it is difficult to understand chemistry in heterogeneous media without basic information about the homogeneous homologue reaction, especially in water (Rideout and Breslow 1980, Lubineau 1986).

Structure-induced changes in diffusion of species are important when reagents are confined in small structures (micelles or microemulsion droplets): fast reaction occurs between species solubilised in the same micelle whereas diffusion across the solvent or through collision of micelles for solubilise exchange is necessary for further reaction (Fletcher *et al* 1982, Brochette *et al* 1987). Confinement in small droplets makes the encounter probability of molecules larger and accelerates diffusion controlled reactions (Grätzel 1987). A dimensionality effect is expected when reactions occur at an interface (two dimensional) for which diffusion is very fast and encounter probabilities high (Hatlee *et al* 1980, Astumian and Chock 1985).

**7.1.2. The physicist's point of view.** The main question of interest is: what is the relation between reactivity and structure in microheterogeneous media? The role of the physicist is to rationalise the chemist's interpretation (based on macroscopic thermodynamic

quantities) on structural grounds (Zana 1986). The rich polymorphism of structures (spatial as well as dynamic) should provide a means for controlling reactivity at will upon external medium changes (Friberg *et al* 1973, Ramesh and Labes 1986). More than pictorial descriptions of 'empty' (without reactive species) and time-averaged solution structures, structural influences on diffusion and intermolecular forces on different time and spatial scales seem to be the basic concepts that chemists need for choosing appropriate reaction media. Of course, here also, structural information is not the only information needed to understand experimental results; the phase boundaries are also set by delicate energetic considerations (see as an example Guérin and Bellocq (1988) or as reviews Roux and Coulon (1986) or Kahlweit *et al* (1988)).

*7.1.3. Some examples.* The influence of structure on competition between two reaction paths is well illustrated by fluorescence quantum yield enhancement of molecules dissolved in micelles (Humphry-Baker *et al* 1980). This is a monomolecular reaction. A molecule in its singlet excited state recovers its ground state via photon re-emission (fluorescence) or non-radiative transitions, energy is then transferred to the molecule and its surroundings through heat (vibrational energy). The fluorescence quantum yield is related to the relative rates of these two processes. In a micellar solution, quantum yield enhancement is observed, which corresponds to a reduction of the rate of non-radiative transitions to the ground state. Micellar influence is to squeeze the fluorescent molecule in such a way that its vibrational abilities are reduced, re-emission of a fluorescence photon is then preferred. This picture is consistent with the dependence of quantum yield on viscosity in bulk solvents (Mialocq 1982).

The stabilisation of radical molecules by micelles is a good example of reaction inhibition by electrostatic effects on dimolecular reactions (Chevalier *et al* 1984, Lerebours *et al* 1985, Pileni *et al* 1985b). As a dicationic molecule bearing hydrophobic chains (dialkylviologen) is reduced in water by an electron donor, it gets monocharged and becomes able to enter a cationic micelle. If the electron donor is cationic too, the recombination of the two parent radicals is prevented by an electrostatic barrier. The lifetime of the radical is then increased from 2.5 ms to some 150 ms, and the back reaction becomes first order instead of second order without micelles, which suggests that the rate-controlling back reaction step is the micelle-to-water exchange of the trapped radical. Radical stabilisation by such charge separation is a key for the use of light for energy production by photochemical means (Grätzel 1980, 1981).

Classical organic reactions are generally slower than photochemical ones, the influence of loose aggregates such as micelles or microemulsions is thus small. They are, moreover, mainly controlled by activation energy barriers; diffusion or compartmentation of species have little importance. For such reactions, vesicles are preferred because of their slower dynamics. A fine illustration of the favourable intervention of confinement on an interface is the spontaneous polymerisation of esters of alkylglycine as vesicles which does not occur in bulk solvents (Folda *et al* 1982). Spontaneous polyamide formation could also be performed in reverse micelles (Armstrong *et al* 1978). Such reactions are of interest since a biodegradable polypeptide of controlled morphology is built up.

## *7.2. Relation between solubilising power and structure*

The influence of solubilisates on structure was discussed in §§ 5 and 6. This section deals with the influence of structure on solubilisation. The practical limitation of the

use of micellar solvents for chemical reactivity control is due in part to the poor solubilising power of micelles. Solubilisation properties are, moreover, of importance in detergency and related applications. We restrict ourselves here to water-insoluble solubilisates, addressing only true micellar solubilisation, for which micelles are necessary for effective solubilisation (solubilisation only occurs when surfactant concentration is above the CMC, i.e. the solute does not promote micelle formation (Stearns *et al* 1947)). In case of non-negligible solubility in water, as with short-chain alcohols, a partition between micelles and water occurs which obscures the interpretation. Water-insoluble hydrophobic molecules may solubilise either in the micellar core or in the interfacial region (Abuin *et al* 1984).

For in-core solubilisation, the solubility follows linearly the surfactant concentration, so that solubilising power can be defined as the amount of solubilate per surfactant molecule. The general result is then a lower solubilising power of micellar cores as compared with bulk alkanes (Elworthy *et al* 1968). If swollen micelles are modelled as hydrocarbon droplets separated from water by a curved surface of tension, a Laplace pressure  $P$  occurs inside the droplet because of curvature  $R$  ( $P = \gamma/R$ ,  $\gamma$  being the interfacial tension), which reduces the solubility of hydrocarbons (Wishnia 1963). This explanation works very well (Mukerjee 1979, 1980) but is phenomenological: it does not inform us about the physical origin of micellar solubilisation.

The curvature radius is indeed that of the swollen micelle, not an intrinsic property of the surfactant or solute molecule. When solute is incorporated into micelles, it can fill the centre of the micellar core, so that surfactant molecules get more freedom in conformation (surfactant methyl groups do not have to be present at the micelle centre), the micellar radius can thus exceed the length of a fully extended chain  $l_c$ , and micelles can grow without deformation. In other words, a solute-free micelle is a frustrated system, solute introduction releases the frustration of packing at constant internal density and a swollen micelle should be more stable. The packing criterion  $a_0 l_c/v$  no longer applies. Swollen micelles are indeed larger than solute-free ones (Almgren and Swarup 1982).

Solubilisation is then limited by the curvature of the swollen aggregate: when the area per headgroup has reached its 'natural' value, surfactant coverage is optimum and no more solubilisation is possible. The relation between solubilising ability and structure is then simple: it is governed by the optimal curvature (or area per headgroup  $a_0$ ) of the surfactant film at a hydrocarbon-water interface; the lower this curvature, the larger the solubilisation. Experiments show this trend: long-chain or double-chain surfactants solubilise more hydrocarbon (even if the solubilising power is defined as the solubilisation per surfactant methylene) (McBain and Johnson 1944, Heller and Klevens 1946, Kolthoff and Stricks 1948, Mukerjee 1979, Shinoda and Sagitani 1983), large micelles (with low  $a_0$ ) are more efficient than spherical ones (Brown and Nauman 1967, Robins and Thomas 1968, Ozeki and Ikeda 1985, Imae *et al* 1986, Abe *et al* 1987), salt addition in ionic micellar solutions which decreases  $a_0$ , increases the solubilising power (McBain and Johnson 1944, McBain and Green 1946, Venable and Nauman 1964), the solubilisation ability of non-ionic alkylpolyoxyethylene ethers follows the hydrophobicity (related to  $a_0$ ) of their headgroups (Nakagawa 1966).

For solubilisation to increase further, a fourth component is added, which solubilises at the interface: short-chain alcohols then give way to microemulsions. Very large solubilising powers are obtained with microemulsions in which the cosurfactant is able to adjust  $a_0$  to the aggregate curvature (see § 6). For molecules which solubilise at the interface (long-chain alcohols, aromatic hydrocarbons (Eriksson and Gillberg 1966,

Johansson *et al* 1976, Samori and Mattivi 1986), hydrophobic dyes (Ganesh *et al* 1982, 1984)), larger solubilisations are then generally obtained. The curvature reduction is easily shown by a transition to elongated micelles (Henriksson *et al* 1977b, Eriksson *et al* 1982) or lamellar mesophases (Robins and Thomas 1968, Ekwall *et al* 1969b) as solute is added. Interesting synergism effects of benzene-cyclohexane or methyl methacrylate-hexane solute mixtures were obtained in this way (Cáceres and Lissi 1984, Nagarajan and Ruckenstein 1984, Nagarajan *et al* 1984, Chaiko *et al* 1984).

### 7.3. Some controversial structures

We have somewhat arbitrarily eliminated all structures presenting a crystalline long-range order as lyotropic liquid crystals. Cubic phases are usually very stiff and show a large number of crystalline reflection orders and can therefore be unambiguously detected by diffraction experiments. The lamellar and hexagonal phases have an optical anisotropy which is easily observed between crossed polarisers. The class of lyotropic nematic phases is borderline and they could be classified in either of the two categories: microemulsions or liquid crystals. It has been noticed that there should be structural relations between lyotropic liquid crystals and microemulsions. When their chemical compositions are close, the first Bragg reflection of a liquid crystal occurs at the same angle as the maximum of the broad peak characteristic of microemulsions (Fontell *et al* 1983). However, one should be aware that a systematic relation between the ordered structure of liquid crystalline phases and the liquid one of adjacent isotropic solutions does not exist (see § 4.1.3).

The transition between micelles or microemulsions (dilute 'gas' or concentrated 'liquid' state) and the more ordered phases (solid at least in one direction) occurs through a well defined phase transition. Second-order phase transitions can be expected from phase diagrams, but have not yet been shown in surfactant systems. Very slow phase separations are difficult to distinguish from new types of liquid-liquid equilibria with very different structures, as opposed to critical fluctuations. In the very dilute microemulsions made of toluene, brine, butanol and less than 0.1% of SDS, the low interfacial area available induces very large characteristic sizes in surfactant aggregates. The typical length in those systems is 100 nm, as revealed by the yellow to red colours of the samples. If the basic structure were spheres, one would have only very dilute droplets. This work shows that not only lamellar phases, but also microemulsions can be swelled by dilution. Moreover, very thick interfaces, in the millimetre range, of unknown structure appear in that system (Gambi *et al* 1987a, b).

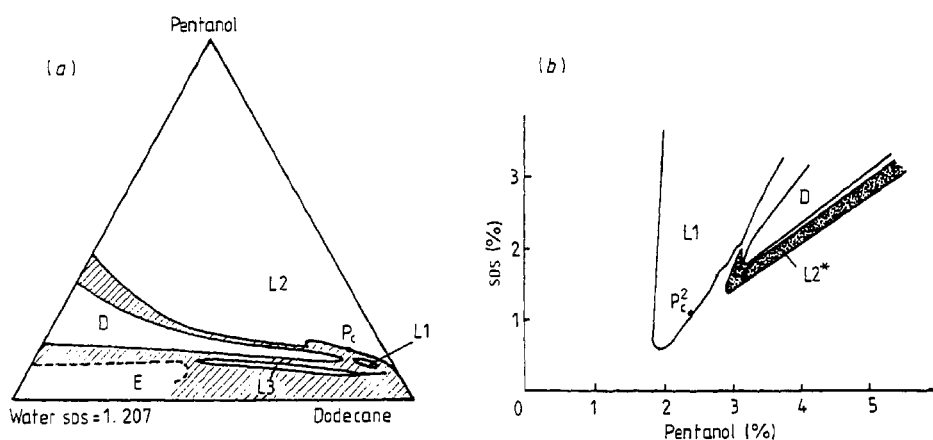
The term liquid crystalline microemulsion or birefringent microemulsion, sometimes used in the past, should no longer be used, because this nomenclature hides the fact that these phases are identical to the classical lamellar phases when they are swollen by large amounts of oil or water.

The emerging picture of the very large number of structural studies of optically isotropic systems is that simple models of interacting spheres hold surprisingly well in the dilute region. Spherical droplets are very loose, fluctuating, with fast molecular exchange processes. Interactions are mainly excluded volume effects and either a long-range screened electrostatic potential for direct ionic micelles, or a short-range attractive one for non-ionic and/or reverse systems. In the concentrated regime, the DOC model is currently the only proposed structure compatible with the peak position in scattering patterns, but alternative descriptive models of bicontinuous materials could probably work.

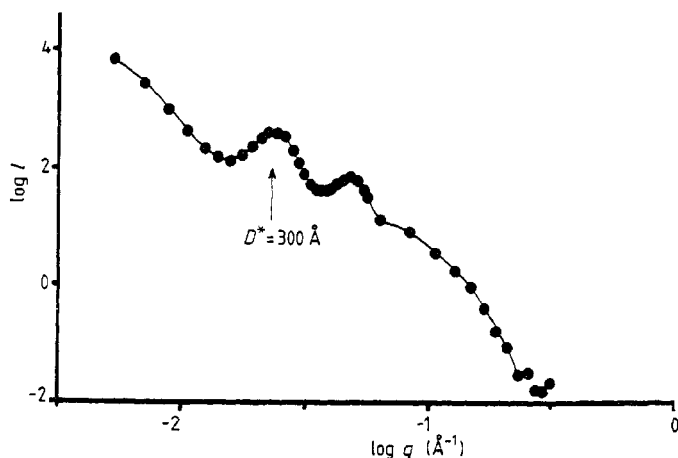
The main point is that one phase, defined as a monophasic area in a phase diagram, does not correspond in surfactant systems to one single type of structure: cylinders can gradually transform into coalescing spheres or lamellae of unilamellar vesicles can, at some temperature, emerge from a solution of micelles (Gershfeld *et al* 1986) without leaving this monophasic area. It is now well established that micelles are in equilibrium with surfactant as a molecular dispersion in the solvent. It is not yet firmly established whether micelles of a pure compound can also dynamically coexist with unilamellar vesicles or if first reports about it are due to impurities (Brady *et al* 1984). However, the usual dispersions of bilayers called 'vesicles' are prepared by sonication and correspond to a kinetic stability: the corresponding point in a phase diagram belongs to a lamellar phase. The scattering spectra are close to spectra obtained in microemulsions and can be understood using similar methods as for micellar systems (Muddle *et al* 1983).

In the near future, the structure of the 'anomalous' isotropic phases called  $L_3$  in phase diagrams (figure 23) has also to be elucidated, together with the mechanism of the viscosity 'accidents' occurring close to lamellar phases in ordinary direct micellar solutions (Bellocq and Roux 1987) (figure 38). Also some strange scattering spectra of large structures have been observed but the phase diagram of this system is not known (Tabony *et al* 1987). Some new types of foams, disordered cylinders with large periodicity (100 nm), predicted by the DOC model, remain to be quantitatively studied.

This continuity in structural change within one 'phase' of the diagram has been identified as a key point for progress in surfactant structures by Shinoda (1985), Ninham and Evans (1986) and Evans and Ninham (1986). For instance, it is now widely accepted that there is no difference in nature between emulsions and microemulsions. A solution is called an emulsion when it is opaque, due to a very strong scattering of light: the characteristic size of oil and water domains is at least one order of magnitude higher than in microemulsions. Emulsions which cannot be broken appear frequently in the vicinity of a microemulsion domain in phase diagrams (Fontell *et al* 1986). In the ternary system nonylphenol-9-polyoxyethylene/water/cyclohexanone, spon-



**Figure 38.** Localisation of 'anomalous' disordered phases in microemulsion phase diagrams: (a) in the SDS/pentanol/water/dodecane system.  $L_2$  is the microemulsion, D and E are lamellar and hexagonal mesophases,  $L_1$  and  $L_3$  are flow birefringent phases of unknown structure (Bellocq and Roux 1987). (b) In the water-rich corner of the SDS/pentanol/brine ( $20 \text{ g l}^{-1} \text{ NaCl}$ )/dodecane (1%) system, near the direct micellar phase ( $L_1$ ) and the lamellar mesophase (D), a flow birefringent phase ( $L_2^*$ ) is found (Guérin and Bellocq 1988).



**Figure 39.** Small-angle neutron scattering spectrum of a spontaneous ternary emulsion (DDAB/water/dodecane): two reflection orders are superimposed to a very strong zero-angle scattering. This type of scattering is not observed with micelles, nor with a dispersion of lamellar microcrystallites.

taneous emulsion and microemulsion coexist within the same 'single-phase' region of the phase diagram. The opaque emulsion region is then formally distinguished from the microemulsion by partial molecular volumes. No theory of microemulsion exists which can predict the appearance of a disordered microemulsion state near the liquid crystalline phase. Therefore, it is not known whether the kinetic stability of emulsions involves the same or a different set of molecular interactions. Typically small-angle scattering spectra of emulsions are very different from those of microemulsions (figure 39). Whether or not structures are retained upon cooling is still studied: the 'gel-microemulsion' transition involves complete modification of x-ray scattering (Nimtz *et al* 1986). At very high cooling rates, electron microscopy has been used: droplet-like structures have been observed. Optical diffractograms of the transmission electron micrograph pictures are similar to common small-angle scattering data (Dubochet *et al* 1984).

The key to the understanding of the occurrence of different microstructures in a single macroscopic 'phase' in a composition diagram is the relation between molecular properties and aggregation phenomena. Indeed, the elementary 'particle' in colloidal physics is a flexible molecule or an aggregate. Up to now, the interfacial curvature energy, the electrostatic energy averaged over few molecular distances and the water-to-oil transfer energies have received most attention in theoretical work. The lowest possible approximation level is given by the molecular shape parameter  $a_0 l_c / v$ . A more refined general framework has recently been published (Overbeek *et al* 1987) and the predictions remain to be tested. As was expressed by Ninham and Evans (1986), the relation between geometrical constraints due to steric hindrance in molecular packing and the statistical mechanics of disordered interfaces in heterogeneous systems is the major challenge to a theory offering some predictive power of microemulsion stability.

## References

- Abe A, Imae T and Ikeda S 1987 *Colloid Polym. Sci.* **265** 637  
 Abillon O, Binks B P, Otero C, Langevin D and Ober R 1988 *J. Phys. Chem.* **92** 4411

- Abragam A 1961 *The Principles of Nuclear Magnetism* (Oxford: Clarendon Press)
- Abraham M H 1979 *J. Am. Chem. Soc.* **101** 5477
- Abu-Hamdiyyah M and Rahman I A 1987 *J. Phys. Chem.* **91** 1530
- Abuin E B and Lissi E A 1983 *J. Colloid Interface Sci.* **95** 198
- Abuin E B, Valenzuela E and Lissi E A 1984 *J. Colloid Interface Sci.* **101** 401
- Ackerson B J 1978 *J. Chem. Phys.* **69** 684
- Ahlnäs T, Söderman O, Hjelm C and Lindman B 1983 *J. Phys. Chem.* **87** 822
- Ahlnäs T, Söderman O, Walderhaug H and Lindman B 1984 *Surfactants in Solution* vol 1 ed K L Mittal and B Lindman (New York: Plenum) p 107
- Åkesson T and Jönsson B 1985 *J. Phys. Chem.* **89** 2401
- Alexandre M, Fouchet C and Rigny P 1973 *J. Chim. Phys.* **70** 1073; *J. Physique* **34** C8-15
- Alexandrowicz Z 1962 *J. Polym. Sci.* **56** 97, 115
- Alexandrowicz Z and Katchalsky A 1963 *J. Polym. Sci. A* **1** 3231
- Allen M, Evans D F, Mitchell D J and Ninham B W 1987 *J. Phys. Chem.* **91** 2320
- Almgren M, Aniansson E A G and Holmåker K 1977a *Chem. Phys.* **19** 1
- Almgren M, Aniansson E A G, Wall S N and Holmåker K 1977b *Micellization, Solubilization, and Microemulsions* vol 1 ed K L Mittal (New York: Plenum) p 329
- Almgren M, Löfroth J E and Rydholm R 1979 *Chem. Phys. Lett.* **63** 265
- Almgren M, Stilbs P, Alsins J, Linse P and Kamenka N 1985 *J. Phys. Chem.* **89** 2666
- Almgren M and Swarup S 1982 *J. Phys. Chem.* **86** 4212
- Altenberger A R, Dahler J S and Tirrel M V 1985 *Macromolecules* **18** 2752
- Amorim da Costa A M and Santos E B H 1983 *Colloid Polym. Sci.* **261** 58
- Anacker E W and Ghose H M 1968 *J. Am. Chem. Soc.* **90** 3161
- Andelmann D, Cates M E, Roux D and Safran S A 1987 *J. Chem. Phys.* **12** 7229
- Angel L R, Evans D F and Ninham B W 1983 *J. Phys. Chem.* **87** 538
- Angel M, Hoffmann H, Löbl M, Reizlein K, Thurn H and Wunderlich I 1984 *Prog. Colloid Polym. Sci.* **69** 12
- Aniansson E A G, Wall S N, Almgren M, Hoffmann H, Kielmann I, Ulbricht W, Zana R, Lang J and Tondre C 1976 *J. Phys. Chem.* **80** 905
- Aoudia M, Rodgers M A J and Wade W H 1984a *J. Colloid Interface Sci.* **101** 472
- 1984b *J. Phys. Chem.* **88** 5008
- Appell J and Porte G 1981 *J. Colloid Interface Sci.* **81** 85
- 1983 *J. Phys. Lett.* **44** 689
- Appell J, Porte G and Poggi Y 1982 *J. Colloid Interface Sci.* **87** 492
- Armstrong D W, Seguin R, McNeal C J, McFarlane R D and Fendler J H 1978 *J. Am. Chem. Soc.* **100** 4605
- Asakawa T, Johten K, Miyagishi S and Nishida M 1985 *Langmuir* **1** 347
- Astumian R D and Chock P B 1985 *J. Phys. Chem.* **89** 3477
- Auvray L, Cotton J P, Ober R and Taupin C 1984a *J. Physique* **45** 913
- 1984b *J. Phys. Chem.* **88** 4586
- 1985 *Physics of Complex and Supramolecular Fluids* ed S A Safran and N A Clark (New York: Wiley) p 449
- Backus J K and Scheraga H A 1951 *J. Colloid Sci.* **6** 508
- Badiali J P 1985 *Mol. Phys.* **55** 939
- Baker R C, Florence A T, Ottewill R H and Tadros T F 1984 *J. Colloid Interface Sci.* **100** 332
- Bansal V K, Shah D O and O'Connell J P 1980 *J. Colloid Interface Sci.* **75** 462
- Barnes I S, Hyde S T, Ninham B W, Derian P J, Drifford M, Warr G G and Zemb T 1988b *Prog. Colloid Polym. Sci.* **76** 90
- Barnes I S, Hyde S T, Ninham B W, Derian P J, Drifford M and Zemb T 1988a *J. Phys. Chem.* **92** 2286
- Barnes I S and Zemb T 1988 *J. Appl. Crystallogr.* **21** 373
- Bassereau P, Marignan J and Appell J 1985 *J. Phys. Lett.* **46** L609
- Bassereau P, Marignan J and Porte G 1987 *J. Physique* **48** 673
- Bayer O, Hoffmann H, Ulbricht W and Thurn H 1986 *Adv. Colloid Interface Sci.* **26** 177
- Becher P 1960 *J. Phys. Chem.* **64** 1221
- Belaïd S and Chachaty C 1984 *Surfactants in Solution* vol 1 ed K L Mittal and B Lindman (New York: Plenum) p 501
- Bell G M and Dunning A J 1970 *Trans. Faraday Soc.* **66** 500
- Belle J and Bothorel P 1974 *Biochem. Biophys. Res. Commun.* **58** 433
- Belloq A M, Biais J, Bothorel P, Clin B, Fourche G, Lalanne P, Lemaire B, Lemanceau B and Roux D 1984 *Adv. Colloid Interface Sci.* **20** 167
- Belloq A M, Biais J, Clin B, Lalanne P and Lemanceau B 1979 *J. Colloid Interface Sci.* **70** 524

- Belloq A M, Bourbon D and Lemanceau B 1981 *J. Colloid Interface Sci.* **79** 419
- Belloq A M and Fourche G 1980 *J. Colloid Interface Sci.* **78** 275
- Belloq A M and Roux D 1987 *Microemulsions* ed S E Friberg and P Bothorel (Boca Raton, FL: CRC) p 34
- Belloni L 1985 *Chem. Phys.* **99** 43
- 1986a *Phys. Rev. Lett.* **57** 2026
- 1986b *J. Chem. Phys.* **85** 519
- Belloni L, Drifford M and Turq P 1984 *Chem. Phys.* **83** 147
- Bendedouch D and Chen S H 1983 *J. Phys. Chem.* **87** 1653
- 1984 *J. Phys. Chem.* **88** 648
- Bendedouch D, Chen S H and Koehler W C 1983a *J. Phys. Chem.* **87** 153
- 1983b *J. Phys. Chem.* **87** 2621
- Benjamin L 1966 *J. Phys. Chem.* **70** 3790
- Ben-Shaul A and Gelbart W M 1982 *J. Phys. Chem.* **86** 316
- 1985 *Ann. Rev. Phys. Chem.* **36** 179
- Ben-Shaul A, Rorman D H, Hartland G V and Gelbart W M 1986 *J. Phys. Chem.* **90** 5277
- Ben-Shaul A, Szeifer I and Gelbart W M 1984 *Proc. Natl. Acad. Sci. USA* **81** 4601
- 1985 *J. Chem. Phys.* **83** 3597
- Benton W J and Miller C A 1983 *J. Phys. Chem.* **87** 4981
- Beresford-Smith B and Chan D Y C 1982 *Chem. Phys. Lett.* **92** 474
- Berk N F 1987 *Phys. Rev. Lett.* **58** 2718
- Berr S S, Caponetti E, Johnson J S Jr, Jones R R M and Magid L J 1986a *J. Phys. Chem.* **90** 5766
- Berr S S, Coleman M J, Jones R R M and Johnson J S Jr 1986b *J. Phys. Chem.* **90** 6492
- Biais J, Bodet J F, Clin B, Lalanne P and Roux D 1986 *J. Phys. Chem.* **90** 5835
- Biais J, Bothorel P, Clin B and Lalanne P 1981 *J. Colloid Interface Sci.* **80** 136
- Biedermann W and Datyner A 1981 *J. Colloid Interface Sci.* **82** 276
- Blankshtein D, Thurston G M and Benedek G B 1985 *Phys. Rev. Lett.* **54** 955
- Blokhuis A M, Hoiland H and Backlund S 1986 *J. Colloid Interface Sci.* **114** 9
- Bloor D M, Gormally J and Wyn-Jones E 1984 *J. Chem. Soc. Faraday Trans. I* **80** 1915
- Blum F D, Pickup S, Ninham B W, Chen S J and Evans D F 1985 *J. Phys. Chem.* **89** 711
- Boden N, Corne S A and Jolley K W 1987a *J. Phys. Chem.* **91** 4092
- Boden N, Jackson P H, McMullen K and Holmes M C 1979 *Chem. Phys. Lett.* **65** 476
- Boden N, Jones S A and Sixl F 1987b *J. Phys. Chem.* **91** 137
- Boned C, Clausse M, Lagourette B, Peyrelasse J, McClean V and Sheppard R J 1980 *J. Phys. Chem.* **84** 1520
- Boned C, Peyrelasse J, Heil J, Zradba A and Clausse M 1982 *J. Colloid Interface Sci.* **88** 603
- Bothorel P 1968 *J. Colloid Interface Sci.* **27** 529
- 1974 *J. Chim. Phys.* **71** 1133
- Bothorel P, Belle J and Lemaire B 1974 *Chem. Phys. Lipids* **12** 96
- Bothorel P, Such C and Clément C 1972 *J. Chim. Phys.* **69** 1453
- Botré C, Crescenzi V L and Mele A 1959 *J. Phys. Chem.* **63** 650
- Bourrel M, Bernard D and Graciaa A 1984 *Tenside Deterg.* **21** 311
- Bourrel M and Schechter R S 1988 *Microemulsions and Related Systems: Formulation, Solvency, and Physical Properties* (Surfactant Science series vol 30) (New York: Dekker)
- Brady J E, Evans D F, Kachar B and Ninham B W 1984 *J. Am. Chem. Soc.* **106** 4279
- Brady J E, Evans D F, Warr G G, Grieser F and Ninham B W 1986 *J. Phys. Chem.* **90** 1853
- Bratko D, Jönsson B and Wennerström H 1986 *Chem. Phys. Lett.* **128** 449
- Bratko D and Vlachy V 1982 *Chem. Phys. Lett.* **90** 434
- Brévard C and Granger P 1981 *Handbook of High Resolution Multinuclear NMR* (New York: Wiley)
- Brochette P, Zemb T, Mathis P and Pileni M P 1987 *J. Phys. Chem.* **91** 1444
- Brooker M H 1984 *J. Chem. Soc. Faraday Trans. I* **80** 73
- Brown D and Clarke J H R 1988 *J. Phys. Chem.* **92** 2881
- Brown J B and Nauman R V 1967 *J. Colloid Interface Sci.* **23** 302
- Brown M S, Grant D M, Horton W J, Mayne C L and Evans G T 1985 *J. Am. Chem. Soc.* **107** 6698
- Brown W, Johnsen R, Stilbs P and Lindman B 1983 *J. Phys. Chem.* **87** 4548
- Brown W and Rymden R 1987 *J. Phys. Chem.* **91** 3565
- Brunetti S, Roux D, Belloq A M, Fourche G and Bothorel P 1983 *J. Phys. Chem.* **87** 1028
- Bunton C A and Cuenca A 1987 *J. Org. Chem.* **52** 901
- Bunton C A and Moffatt J R 1985 *J. Phys. Chem.* **89** 4166
- Burns R A Jr, Roberts M F, Dluhy R and Mendelsohn R 1982 *J. Am. Chem. Soc.* **104** 430
- Cabane B 1977 *J. Phys. Chem.* **81** 1639



- 1984 *Colloïdes et interfaces* ed A M Cazabat and M Veyssié (Paris: Les Editions de Physique) p 101
- 1987 *Surfactant Solutions. New Methods of Investigation* ed R Zana (Surfactant science series vol 22) (New York: Dekker) p 57
- Cabane B and Duplessix R 1985 *Colloids Surf.* **13** 19
- 1987 *J. Physique* **48** 651
- Cabane B, Duplessix R and Zemb T 1984 *Surfactants in Solution* vol 1 ed K L Mittal and B Lindman (New York: Plenum) p 373
- 1985 *J. Physique* **46** 2161
- Cabani S, Gianni P, Mollica V and Lepori L 1981 *J. Solution Chem.* **10** 563
- Cabos C 1983 *J. Phys. Lett.* **44** 997
- Cabos C and Delord P 1979 *J. Appl. Crystallogr.* **12** 502
- 1980 *J. Phys. Lett.* **41** 455
- Cabos C, Delord P and Martin J C 1979 *J. Phys. Lett.* **40** 407
- Cabos C and Marignan J 1985 *J. Phys. Lett.* **46** 267
- Câceres T and Lissi E A 1984 *J. Colloid Interface Sci.* **97** 298
- Cahn J W 1965 *J. Chem. Phys.* **42** 93
- Cameron D G, Umemura J, Wong P T T and Mantsch H H 1982 *Colloids Surfaces* **4** 131
- Candau S J, Hirsch E and Zana R 1982 *J. Colloid Interface Sci.* **88** 428
- 1984 *J. Physique* **45** 1263
- 1985 *J. Colloid Interface Sci.* **105** 521
- Candau S J and Zana R 1981 *J. Colloid Interface Sci.* **84** 206
- Cantor R S and Dill K A 1984 *Macromolecules* **17** 384
- Cantù L, Corti M, Degiorgio V, Minero C and Piazza R 1985 *J. Colloid Interface Sci.* **105** 628
- Caponetti E, Magid L J, Hayter J B and Johnson J S Jr 1986 *Langmuir* **2** 722
- Caron G and Desnoyers J E 1987 *J. Colloid Interface Sci.* **119** 141
- Cates M E, Roux D, Andelmann D, Milner S T and Safran S A 1988 *Europhys. Lett.* **5** 733
- Cazabat A M 1983 *J. Phys. Lett.* **44** 593
- Cazabat A M, Chatenay D, Guéring P and Langevin D 1984b *Reverse Micelles* ed P L Luisi and B E Straub (New York: Plenum) p 121
- Cazabat A M, Chatenay D, Guéring P, Langevin D, Meunier J, Sorba O, Lang J, Zana R and Paillette M 1984a *Surfactants in Solution* vol 3 ed K L Mittal and B Lindman (New York: Plenum) p 1737
- Cazabat A M, Chatenay D, Langevin D and Meunier J 1983 *Faraday Discuss. Chem. Soc.* **76** 291
- Cazabat A M and Langevin D 1981 *J. Chem. Phys.* **74** 3148
- Cazabat A M, Langevin D, Meunier J and Pouchelon A 1982a *Adv. Colloid Interface Sci.* **16** 175
- 1982b. *J. Phys. Lett.* **4** 89
- Cazabat A M and Veyssié M ed 1984 *Colloïdes et Interfaces* (Paris: Les Editions de Physique)
- Cebula D J, Myers D Y and Ottewill R H 1982 *Colloid Polym. Sci.* **260** 96
- Ceglie A, Das K P and Lindman B 1987 *J. Colloid Interface Sci.* **115** 115
- Chaiko M A, Nagarajan R and Ruckenstein E 1984 *J. Colloid Interface Sci.* **99** 168
- Chakhovskoy N 1956 *Bull. Soc. Chim. Belg.* **65** 474
- Chang J N, Billman J F, Licklider R A and Kaler E W 1987 *Statistical Thermodynamics of Micellar and Microemulsion Systems* ed S H Chen (Berlin: Springer)
- Chang J N and Kaler E W 1986 *Langmuir* **2** 184
- Chao Y S, Sheu E Y and Chen S H 1985 *J. Phys. Chem.* **89** 4395
- Charvolin J 1983 *J. Chim. Phys.* **80** 15
- Charvolin J and Hendriks Y 1985 *Nuclear Magnetic Resonance of Liquid Crystals* ed J W Emsley (Dordrecht: Reidel) p 449
- Charvolin J, Hendriks Y and Rawiso M 1984 *Surfactants in Solution* vol 1 ed K L Mittal and B Lindman (New York: Plenum) p 59
- Charvolin J, Levelut A M and Samulski E T 1979 *J. Phys. Lett.* **40** 587
- Charvolin J and Rigny P 1971a *J. Magn. Reson.* **4** 40
- 1971b *Mol. Cryst. Liquid Cryst.* **15** 211
- 1973 *J. Chem. Phys.* **58** 3999
- Chatenay D, Urbach W, Nicot C, Vacher M and Waks M 1987 *J. Phys. Chem.* **91** 2198
- Chen S H 1986a *Physica B* **137** 183
- 1986b *Ann. Rev. Phys. Chem.* **37** 351
- Chen S H and Kotlarichyk M 1985 *Physics of Amphiphiles* ed M Corti and V De giorgio (Amsterdam: North Holland) p 281
- Chen S H and Teixeira J 1986 *Phys. Rev. Lett.* **57** 2583

- Chen S J, Evans D F and Ninham B W 1984 *J. Phys. Chem.* **88** 1631  
— 1987 *J. Phys. Chem.* **91** 1823  
Chen S J, Evans D F, Ninham B W, Mitchell D J, Blum F D and Pickup S 1986 *J. Phys. Chem.* **90** 842  
Chevalier S, Lerebours B and Pileni M P 1984 *J. Photochem.* **27** 301  
Chevalier Y, Belloni L, Hayter J B and Zemb T 1985 *J. Physique* **46** 749  
Chevalier Y and Chachaty C 1984 *Colloid Polym. Sci.* **262** 489  
— 1985 *J. Am. Chem. Soc.* **107** 1102  
Clark N A and Ackerson B J 1980 *Phys. Rev. Lett.* **44** 1005  
Clarkson M T, Beaglehole D and Callaghan P T 1985 *Phys. Rev. Lett.* **54** 1722  
Clausse M and Heil J 1983 *Lett. Nuovo Cimento* **36** 369  
Clausse M, Nicolas-Morgantini L, Zradba A and Touraud J 1987a *Microemulsion Systems* (Surfactant Science Series vol 24) ed H L Rosano and M Clausse (New York: Dekker)  
Clausse M, Peyrelasse J, Heil J, Boned C and Lagourette B 1981 *Nature* **293** 636  
Clausse M, Zradba A, Heil J, Rouvière J and Sohounhloué K 1987b *Microemulsion Systems* (Surfactant Science Series vol 24) ed H L Rosano and M Clausse (New York: Dekker)  
Clausse M, Zradba A and Nicolas-Morgantini L 1985 *Colloid Polym. Sci.* **263** 767  
Clemett C J 1970 *J. Chem. Soc. A* 2251  
Clifford J and Pethica B A 1964 *Trans. Faraday Soc.* **60** 216  
Corkill J M, Gemmell K W, Goodman J F and Walker T 1970 *Trans. Faraday Soc.* **66** 1817  
Corkill J M, Goodman J F and Walker T 1967 *Trans. Faraday Soc.* **63** 768  
Corkill J M, Goodman J F, Walker T and Wyer J 1969 *Proc. R. Soc. A* **312** 243  
Corti M and Degiorgio V 1981 *J. Phys. Chem.* **85** 711  
Corti M, Minero C and Degiorgio V 1984 *J. Phys. Chem.* **88** 309  
Courchene W L 1964 *J. Phys. Chem.* **68** 1870  
Cummins P G, Hayter J B, Penfold J and Staples E 1987b *Chem. Phys. Lett.* **138** 436  
Cummins P G, Staples E, Hayter J B and Penfold J 1987a *J. Chem. Soc. Faraday Trans. I* **83** 2773  
Cutler S G, Meares P and Hall D G 1978 *J. Chem. Soc. Faraday Trans. I* **74** 1758  
Danielsson I, Friman R and Sjöblom J 1982 *J. Colloid Interface Sci.* **85** 442  
Danielsson I and Lindman B 1981 *Colloids Surfaces* **3** 391  
Davies D G and Bury C R 1930 *J. Chem. Soc.* 2263  
Davis J H, Jeffrey K R and Bloom M 1978 *J. Magn. Reson.* **29** 191  
Davis H T and Scriven L E 1982 *Adv. Chem. Phys.* **49** ed I Prigogine and S A Rice (New York: Wiley) p 357  
De Gennes P G and Taupin C 1982 *J. Phys. Chem.* **86** 2294  
De Geyer A 1987 *PhD Thesis* University Paris VI  
De Geyer A and Tabony J 1985 *Chem. Phys. Lett.* **113** 83  
— 1986 *Chem. Phys. Lett.* **124** 357  
Degiorgio V and Corti M (eds) 1985 *Physics of Amphiphiles: Micelles, Vesicles and Microemulsions* (Amsterdam: North Holland)  
Degiorgio V and Piazza R 1987 *Prog. Colloid Polym. Sci.* **73** 76  
De Lisi R, Genova C, Testa R and Turco Liveri V 1984 *J. Solution Chem.* **13** 121  
De Lisi R, Milioto S and Turco Liveri V 1987 *J. Colloid Interface Sci.* **117** 64  
Delord P and Larché F C 1984 *J. Colloid Interface Sci.* **98** 277  
Delville A 1980 *Chem. Phys. Lett.* **69** 386  
— 1984 *Chem. Phys. Lett.* **105** 83  
Delville A, Herwats L and Laszlo P 1984 *Nouv. J. Chim.* **8** 557  
Dennis E A 1986 *Adv. Colloid Interface Sci.* **26** 155  
Desnoyers J E and Arel M 1967 *Can. J. Chem.* **45** 359  
Desnoyers J E, De Lisi R, Ostiguy C and Perron G 1979 *Solution Chemistry of Surfactants* vol 2 ed K L Mittal (New York: Plenum) p 221  
Desnoyers J E, De Lisi R and Perron G 1980 *Pure Appl. Chem.* **54** 433  
Desnoyers J E, Roberts D, De Lisi R and Perron G 1982 *Solution Behavior of Surfactants* vol 1 ed K L Mittal and E J Fendler (New York: Plenum) p 343  
Dichristina T, Roux D, Bellocq A M and Bothorel P 1985 *J. Phys. Chem.* **89** 1433  
Dickinson E 1979 *Chem. Phys. Lett.* **66** 500  
Dill K A 1984 *Surfactants in Solution* vol 1 ed K L Mittal and B Lindman (New York: Plenum) p 307  
Dill K A and Cantor R S 1984 *Macromolecules* **17** 380  
Dill K A and Flory P J 1980 *Proc. Natl. Acad. Sci. USA* **77** 3115  
— 1981 *Proc. Natl. Acad. Sci. USA* **78** 676

- Di Meglio J M, Dvolaitzky M and Taupin C 1987 *Microemulsion Systems* (Surfactant Science Series vol 24) ed H L Rosano and M Clausse (New York: Dekker)
- Donchi K F, Robert G P, Ternai N R and Derrick P J 1980 *Aust. J. Chem.* **33** 2199
- Doolittle A K 1951 *J. Appl. Phys.* **22** 1471
- Dormoy Y, Hirsch E, Candau S J and Zana R 1987 *Prog. Colloid Polym Sci.* **73** 80
- Dreyfus B 1985 *Macromolecules* **18** 284
- Drifford M, Belloni L, Dalbiez J P and Chattopadhyay A K 1985 *J. Colloid Interface Sci.* **105** 587
- Drifford M, Zemb T, Hayoun M and Jehanno A 1984 *Surfactants in Solution* vol 1 ed K L Mittal and B Lindman (New York: Plenum) p 487
- Dubin P L and Davis D 1985 *Colloids Surfaces* **13** 113
- Dubin P L, Rigsbee D R and McQuigg D W 1985 *J. Colloid Interface Sci.* **105** 509
- Dubochet J, Adrian M, Texeira J, Alba C, Kadiyala R K, McFarlane D R and Angell C A 1984 *J. Phys. Chem.* **88** 6727
- Dufourc E J and Smith I C P 1986 *Chem. Phys. Lipids* **41** 123
- Duynstee E F J and Grunwald E 1959 *J. Am. Chem. Soc.* **81** 4540, 4542
- Dvolaitzky M, Lagües M, Le Pesant J P, Ober R, Sauterey C and Taupin C 1980 *J. Phys. Chem.* **84** 1532
- Edholm O 1982 *Chem. Phys.* **65** 259
- Eicke H F 1979 *J. Colloid Interface Sci.* **68** 440
- 1980 *Top. Curr. Chem.* **87** 85
- Eicke H F and Christen H 1974 *J. Colloid Interface Sci.* **46** 417
- 1978 *Helv. Chim. Acta* **61** 2258
- Eicke H F, Hilfiker R and Thomas H 1986 *Chem. Phys. Lett.* **125** 295
- Eicke H F, Hopmann R F W and Christen H 1975 *Ber. Bunsenges. Phys. Chem.* **79** 667
- Eicke H F and Kvita P 1984 *Reverse Micelles* ed P L Luisi and B E Straub (New York: Plenum) p 21
- Eicke H F and Rehak J 1976 *Helv. Chim. Acta* **59** 2883
- Eisenberg A 1970 *Macromolecules* **3** 147
- Ekwall P 1969 *J. Colloid Interface Sci.* **29** 16
- 1975 *Adv. Liquid Cryst.* **1** 1 ed G H Brown (New York: Academic)
- Ekwall P, Eikrem H and Mandell L 1963 *Acta Chem. Scand.* **17** 111
- Ekwall P and Holmberg P 1965 *Acta Chem. Scand.* **19** 455
- Ekwall P, Mandell L and Fontell K 1969a *Mol. Cryst. Liq. Cryst.* **8** 157
- 1969b *J. Colloid Interface Sci.* **29** 638
- 1970 *J. Colloid Interface Sci.* **33** 215
- 1986 *Colloid Polym. Sci.* **264** 542
- Ekwall P, Mandell L and Solyom P 1971 *J. Colloid Interface Sci.* **35** 519
- Elkoubi D, Turq P and Hansen J P 1977 *Chem. Phys. Lett.* **52** 493
- Elvingson C and Wall S 1986 *J. Phys. Chem.* **90** 5250
- Elworthy P H 1959 *J. Chem. Soc.* 813, 1951
- Elworthy P H, Florence A T and McFarlane C B 1968 *Solubilization by Surface-Active Agents* (New York: Chapman and Hall)
- Emerson M F and Holtzer A 1967 *J. Phys. Chem.* **71** 1898
- Emert J, Behrens C and Goldenberg M 1979 *J. Am. Chem. Soc.* **101** 771
- Encinas M V and Lissi E A 1986 *Chem. Phys. Lett.* **132** 545
- Epstein B R, Foster K R and Mackay R A 1983 *J. Colloid Interface Sci.* **95** 218
- Eriksson F, Eriksson J C and Stenius P 1979 *Solution Chemistry of Surfactants* vol 1 ed K L Mittal (New York: Plenum) p 297
- 1981 *Colloids Surfaces* **3** 339
- Eriksson J C and Gillberg G 1966 *Acta Chem. Scand.* **20** 2019
- Eriksson J C, Henriksson U, Klason T and Ödberg L 1982 *Solution Behavior of Surfactants* vol 2 ed K L Mittal and E J Fendler (New York: Plenum) p 907
- Eriksson P O and Lindblom G 1980 *Springer Ser. Chem. Phys.* **11** 290
- Eriksson P O, Lindblom G and Arvidson G 1987 *J. Phys. Chem.* **91** 846
- Evans D F, Brady J, Kachar B and Ninham B W 1985 *J. Solution Chem.* **14** 141
- Evans D F, Mitchell D J and Ninham B W 1984 *J. Phys. Chem.* **88** 6344
- 1986b *J. Phys. Chem.* **90** 2817
- Evans D F and Ninham B W 1983 *J. Phys. Chem.* **87** 5025
- 1986 *J. Phys. Chem.* **90** 226
- Evans D F, Sen R and Warr G G 1986a *J. Phys. Chem.* **90** 5500
- Evans H C 1956 *J. Chem. Soc.* 579

- Faure A, Ahlanäs T, Tistchenko A M and Chachaty C 1987 *J. Phys. Chem.* **91** 1827
- Faure A, Tistchenko A M, Zemb T and Chachaty C 1985 *J. Phys. Chem.* **89** 3373
- Fendler J H 1982 *Membrane Mimetic Chemistry* (New York: Wiley)
- Fendler J H and Fendler E J 1975 *Catalysis in Micellar and Macromolecular Systems* (New York: Academic)
- Flamberg A and Pecora R 1984 *J. Phys. Chem.* **88** 3026
- Fletcher P D I, Perrins N M, Robinson B H and Toprakcioglu C 1984 *Reverse Micelles* ed P L Luisi and B E Straub (New York: Plenum) p 69
- Fletcher P D I, Robinson B H, Bermejo-Barrera F, Oakenfull D G, Dore J C and Steytler D C 1982 *Microemulsions* ed I D Robb (New York: Plenum) p 221
- Flory P J 1969 *Statistical Mechanics of Chain Molecules* (New York: Wiley)
- Folda T, Gros L and Ringsdorf H 1982 *Makromol. Chem. Rapid Commun.* **3** 167
- Fontell K, Ceglie A, Lindman B and Ninham B W 1986 *Acta Chem. Scand. A* **40** 247
- Fontell K, Hernquist L, Larsson K and Sjöblom J 1983 *J. Colloid Interface Sci.* **93** 453
- Forrest B J, Fujiwara F Y and Reeves L W 1980 *J. Phys. Chem.* **84** 662
- Forrest B J and Reeves L W 1981 *Chem. Rev.* **81** 1
- Fourche G, Bellocq A M and Brunetti S 1982 *J. Colloid Interface Sci.* **88** 302
- Fourche G and Bothorel P 1969 *J. Chim. Phys.* **66** 54
- Fowkes F M 1962 *J. Phys. Chem.* **66** 1843
- Franks F and Smith H T 1968 *Trans. Faraday Soc.* **64** 2962
- Friberg S E 1982 *Colloids Surfaces* **4** 201
- Friberg S E and Buraszenka I 1978 *Prog. Colloid Polym. Sci.* **63** 1
- Friberg S E and Liang P 1986 *Colloid Polym. Sci.* **264** 449
- Friberg S, Rydhag O and Lindblom G 1973 *J. Phys. Chem.* **77** 1280
- Friedman M E and Scheraga H A 1965 *J. Phys. Chem.* **69** 3795
- Fromherz P 1981a *Chem. Phys. Lett.* **77** 460
- 1981b *Ber. Bunsenges. Phys. Chem.* **85** 891
- 1984 *Surfactants in Solution* vol 1 ed K L Mittal and B Lindman (New York: Plenum) p 321
- Fujiwara F Y and Reeves L W 1980 *J. Phys. Chem.* **84** 653
- Fujiwara F Y, Reeves L W, Suzuki M and Vanin J A 1979 *Solution Chemistry of Surfactants* vol 1 ed K L Mittal (New York: Plenum) p 63
- Funasaki N, Hada S and Neya S 1984 *J. Phys. Chem.* **88** 1243
- 1986 *J. Phys. Chem.* **90** 5469
- Fuson M M and Prestegard J H 1983 *J. Am. Chem. Soc.* **105** 168
- Gaboriaud R, Charbit G and Dorion F 1984 *J. Colloid Interface Sci.* **98** 583
- Gambi C M C, Léger L and Taupin C 1987a *Europhys. Lett.* **3** 213
- 1987b *J. Phys. Chem.* **91** 4536
- Ganesh K N, Mitra P and Balasubramanian D 1982 *J. Phys. Chem.* **86** 4291
- 1984 *Surfactants in Solution* vol 1 ed K L Mittal and B Lindman (New York: Plenum) p 565
- Ganz A M and Boegfer B E 1986 *J. Colloid Interface Sci.* **109** 504
- Gelbart W M, Ben-Shaul A, McMullen W E and Masters A 1984 *J. Phys. Chem.* **88** 861
- Gelbart W M, McMullen W E and Ben-Shaul A 1985 *J. Physique* **46** 1137
- Georges J, Chen J W and Arnaud N 1987 *Colloid Polym. Sci.* **265** 45
- Gershfeld N L, Stevens Jr W F and Nossal R J 1986 *Faraday Discuss. Chem. Soc.* **81** 19
- Gilányi T and Wolfram E 1981 *Colloids Surfaces* **3** 181
- Goddard E D 1986 *Colloids Surfaces* **19** 255, 301
- Goldstein R E 1986 *J. Chem. Phys.* **84** 3367
- Gomati R, Appell J, Bassereau P, Marignan J and Porte G 1987 *J. Phys. Chem.* **91** 24
- Gonsalves M, Probst S, Rezende M C, Nome F, Zucco C and Zanette D 1985 *J. Phys. Chem.* **89** 1127
- Graciaa A, Lachaise J, Chabrat P, Letamendia L, Rouch J and Vaucamps 1978 *J. Phys. Lett.* **39** 235
- Graciaa A, Lachaise J, Martinez A, Bourrel M and Chambu C 1976 *C.R. Acad. Sci., Paris B* **272** 547
- Graciaa A, Lachaise J, Martinez A and Rousset A 1977 *C.R. Acad. Sci., Paris B* **285** 295
- Grätzel M 1980 *Ber. Bunsenges. Phys. Chem.* **84** 981
- 1981 *Acc. Chem. Res.* **14** 376
- 1987 *Tetrahedron* **43** 1679
- Gravsholt S 1976 *J. Colloid Interface Sci.* **57** 575
- Grest G S, Webman J, Safran S A and Bug A 1985 *Phys. Rev. A* **32** 506
- Grimson M J 1987 *Mol. Phys.* **60** 179
- Gruen D W R 1981 *J. Colloid Interface Sci.* **84** 281
- 1985a *J. Phys. Chem.* **89** 146, 153

- 1985b *Progr. Colloid Polym. Sci.* **70** 6
- Gruen D W R and De Lacey E H B 1984 *Surfactants in Solution* vol 1 ed K L Mittal and B Lindman (New York: Plenum) p 279
- Guérin F and Bellocq A M 1988 *J. Phys. Chem.* **92** 2250
- Guéring P and Cazabat A M 1983 *J. Phys. Lett.* **44** 601
- Guéring P and Lindman B 1985 *Langmuir* **1** 464
- Guéring P, Nilsson P G and Lindman B 1985 *J. Colloid Interface Sci.* **105** 41
- Guldbrand L, Jönsson B, Wennerström H and Linse P 1984 *J. Chem. Phys.* **80** 2221
- Gunnarsson G and Gustavsson H 1982 *J. Chem. Soc. Faraday Trans. I* **78** 2901
- Gunnarsson G, Jönsson B and Wennerström H 1980 *J. Phys. Chem.* **84** 3114
- Gustavsson H and Lindman B 1975 *J. Am. Chem. Soc.* **97** 3923
- Güveli D E 1984 *J. Colloid Interface Sci.* **101** 344
- Haan S W and Pratt L R 1981 *Chem. Phys. Lett.* **79** 436 (erratum 1981 **81** 386)
- Haering G, Luisi P L and Meussdorffer F 1985 *Biochim. Biophys. Acta* **127** 9
- Haile J M and O'Connell J P 1984 *J. Phys. Chem.* **88** 6363
- Hall D G and Pethica B A 1966 *Nonionic Surfactants* (Surfactant Science Series vol 1) ed M J Schick (New York: Dekker) p 516
- Halle B, Landgren M and Jönsson B 1988 *J. Physique* **49** 1235
- Halle B and Wennerström H 1981 *J. Chem. Phys.* **75** 1928
- Hansen J P and McDonald I R 1976 *Theory of Simple Liquids* (New York: Academic)
- Hartland G V, Greiser F and White L R 1987 *J. Chem. Soc. Faraday Trans. I* **83** 591
- Hartley G S 1936 *Aqueous Solutions of Paraffin Chain Salts* (Paris: Hermann)
- Hashimoto S, Thomas J K, Evans D F, Mukherjee S and Ninham B W 1983 *J. Colloid Interface Sci.* **95** 594
- Hatlee M D, Kozak J J, Rothenberger G, Infelta P P and Grätzel M 1980 *J. Phys. Chem.* **84** 1508
- Hayashi S and Ikeda S 1980 *J. Phys. Chem.* **84** 744
- Hayter J B 1981 *Ber. Bunsenges. Phys. Chem.* **85** 887
- 1985 *Physics of Amphiphiles: Micelles, Vesicles and Microemulsions* ed V Degiorgio and M Corti (Amsterdam: North Holland) p 59
- 1987 *Physics of Complex and Supramolecular Fluids* ed S A Safran and N A Clark (New York: Wiley) p 21
- Hayter J B, Hayoun M and Zemb T 1984 *Colloid Polym. Sci.* **262** 798
- Hayter J B and Penfold J 1981a *J. Chem. Soc. Faraday Trans. I* **77** 1851
- 1981b *Mol. Phys.* **42** 109
- 1983 *Colloid Polym. Sci.* **261** 1022
- 1984 *J. Phys. Chem.* **88** 4589
- Hayter J B and Zemb T 1982 *Chem. Phys. Lett.* **93** 91
- Hayter J B and Zulauf M 1982 *Colloid Polym. Sci.* **260** 1023
- Heil J, Clausse M, Peyrelasse J and Boned C 1982 *Colloid Polym. Sci.* **260** 93
- Helfrich W 1978 *Z. Naturf.* **33a** 305
- Heller W and Klevens H B 1946 *J. Chem. Phys.* **14** 567
- Hendriks Y and Charvolin J 1981 *J. Physique* **42** 1427
- Hendriks Y, Charvolin J and Rawiso M 1984 *J. Colloid Interface Sci.* **100** 597
- Hendriks Y, Charvolin J, Rawiso M, Liébert L and Holmes M C 1983 *J. Phys. Chem.* **87** 3991
- Henriksson U, Klason T, Ödberg L and Eriksson J C 1977b *Chem. Phys. Lett.* **52** 554
- Henriksson U, Ödberg L, Eriksson J C and Westman L 1977a *J. Phys. Chem.* **81** 76
- Herrmann K W, Brushmiller J G and Courchene W L 1966 *J. Phys. Chem.* **70** 2909
- Hey M J, McTaggart J W and Rochester C H 1985 *J. Chem. Soc. Faraday Trans. I* **81** 207
- Hirsch E, Candau S and Zana R 1984b *J. Colloid Interface Sci.* **97** 318
- Hirsch E, Debeauvais F, Candau F, Lang J and Zana R 1984a *J. Physique* **45** 257
- Hoffmann H 1978 *Ber. Bunsenges. Phys. Chem.* **82** 988
- 1984 *Ber. Bunsenges. Phys. Chem.* **88** 1078
- Hoffmann H, Kalus J and Schwander B 1987 *Ber. Bunsenges. Phys. Chem.* **91** 99
- Hoffmann H, Kalus J, Thurn H and Ibel K 1983 *Ber. Bunsenges. Phys. Chem.* **87** 1120
- Hoffmann H, Kielman H S, Pavlovic D, Platz G and Ulbricht W 1981d *J. Colloid Interface Sci.* **80** 237
- Hoffmann H, Platz G, Rehage H and Schorr W 1981c *Ber. Bunsenges. Phys. Chem.* **85** 877
- Hoffmann H, Platz G, Rehage H, Schorr W and Ulbricht W 1981b *Ber. Bunsenges. Phys. Chem.* **85** 255
- Hoffmann H, Platz G and Ulbricht W 1981a *J. Phys. Chem.* **85** 1418
- Hoffmann H, Platz G and Ulbricht W 1986 *Ber. Bunsenges. Phys. Chem.* **90** 877
- Hoffmann H and Schorr W 1985 *Ber. Bunsenges. Phys. Chem.* **89** 538
- Hoffmann H and Ulbricht W 1987 *Tenside Deterg.* **24** 23

- Hoiland H 1976 *J. Solution Chem.* **5** 773
- Hoiland H, Blokhuis A M, Kvammen O J and Backlund S 1985 *J. Colloid Interface Sci.* **107** 576
- Howe A M, Toprakcioglu C, Dore J C and Robinson B H J 1986 *J. Chem. Soc. Faraday Trans. I* **82** 2411
- Hua X T and Rosen M J 1982 *J. Colloid Interface Sci.* **90** 212
- Huang J S and Kim M W 1981 *Phys. Rev. Lett.* **47** 1462
- Huang J S and Kotlarchyk M 1986 *Phys. Rev. Lett.* **57** 2587
- Huang J S, Milner S T, Farago B and Richter D 1987 *Phys. Rev. Lett.* **59** 2600
- Huang J S, Safran S A, Kim M W, Grest G S, Kotlarchyk M and Quirke N 1984 *Phys. Rev. Lett.* **53** 592
- Hubbard P S 1969 *J. Chem. Phys.* **51** 1647
- 1970 *J. Chem. Phys.* **53** 985
- Huggins M L 1941 *J. Am. Chem. Soc.* **63** 116
- Huh C 1984 *J. Colloid Interface Sci.* **97** 210
- Humphry-Baker R, Grätzel M and Steiger R 1980 *J. Am. Chem. Soc.* **102** 847
- Ikeda S, Tsunoda M and Maeda H 1979 *J. Colloid Interface Sci.* **70** 448
- Imae T, Abe A, Taguchi Y and Ikeda S 1986 *J. Colloid Interface Sci.* **109** 567
- Imae T, Kamiya R and Ikeda S 1985 *J. Colloid Interface Sci.* **108** 215
- Imae T and Ikeda S 1984a *J. Colloid Interface Sci.* **98** 363
- 1984b *Colloid Polym. Sci.* **262** 497
- 1985 *Colloid Polym. Sci.* **263** 756
- 1986 *J. Phys. Chem.* **90** 5216
- Israelachvili J N 1984 *Physics of Amphiphiles* ed M Corti and V Degiorgio (Amsterdam: Elsevier)
- 1985 *Chem. Scr.* **25** 7
- Israelachvili J N, Marcelja S and Horn R G 1980 *Q. Rev. Biophys.* **13** 121
- Israelachvili J N, Mitchell D J and Ninham B W 1976 *J. Chem. Soc. Faraday Trans. II* **72** 1525
- Jahn W and Strey R 1988 *J. Phys. Chem.* **92** 2294
- Jansson M, Puyong L and Stilbs P 1987 *J. Phys. Chem.* **91** 5279
- Jansson M and Stilbs P 1985 *J. Phys. Chem.* **89** 4868
- 1987 *J. Phys. Chem.* **91** 113
- Johansson L B Å, Lindblom G and Nordén B 1976 *Chem. Phys. Lett.* **39** 128
- Jönsson B, Linse P, Åkesson T and Wennerström H 1984 *Surfactants in Solution* vol 3 ed K L Mittal and B Lindman (New York: Plenum) p 2033
- Jönsson B and Wennerström H 1987 *J. Phys. Chem.* **91** 338
- Jönsson B, Wennerström H, Nilsson P G and Linse P 1986 *Colloid Polym. Sci.* **264** 77
- Jorgensen W L, Madura J D and Swenson C J 1984 *J. Am. Chem. Soc.* **106** 6638
- Jouffroy J, Levinson P and De Gennes P G 1982 *J. Physique* **43** 1241
- Kachar B, Evans D F and Ninham B W 1984a *J. Colloid Interface Sci.* **99** 593
- 1984b *J. Colloid Interface Sci.* **100** 28
- Kadam K L 1986 *Enzyme Microbiol. Technol.* **8** 266
- Kahlweit M, Lessner E and Strey R 1983 *J. Phys. Chem.* **87** 5032
- Kahlweit M, Strey R, Firman P, Haase D, Jen J and Schomäcker R 1988 *Langmuir* **4** 499
- Kahlweit M, Strey R and Haase D 1985 *J. Phys. Chem.* **89** 163
- Kahlweit M, Strey R, Haase D, Kuneida H, Schmeling T, Faulhaber B, Borkovec M, Eicke H, F, Busse G, Eggers F, Funck T, Richmann H, Magid L, Söderman O, Stilbs P, Winkler J, Dittrich A and Jahn W 1987 *J. Colloid Interface Sci.* **118** 436
- Kaler E W, Bennett K E, Davis H T and Scriven L E 1983a *J. Chem. Phys.* **79** 5673
- Kaler E W, Davis H T and Scriven L E 1983b *J. Chem. Phys.* **79** 5685
- Kaler E W and Prager S 1982 *J. Colloid Interface Sci.* **86** 359
- Kalus J, Hoffmann H, Reizlein K, Ulbricht W and Ibel K 1982 *Ber. Bunsenges. Phys. Chem.* **86** 37
- Kalyanasundaram K, Gratzel M and Thomas J K 1975 *J. Am. Chem. Soc.* **97** 3916
- Kalyanasundaram K and Thomas J K 1976 *J. Phys. Chem.* **80** 1462
- 1977 *J. Am. Chem. Soc.* **99** 2039
- Kamenka N, Fabre H, Chorro M and Lindman B 1977 *J. Chim. Phys.* **74** 510
- Kamenka N, Lindman B and Brun B 1974 *Colloid Polym. Sci.* **252** 144
- Kamenka N, Puyal M C, Brun B, Haouche G and Lindman B 1984 *Surfactants in Solution* vol 1 ed K L Mittal and B Lindman (New York: Plenum) p 359
- Kamogawa K, Tajima K, Hayakawa K and Kitagawa T 1984 *J. Phys. Chem.* **88** 2494
- Kano K, Yamaguchi T and Ogawa T 1984 *J. Phys. Chem.* **88** 793
- Katchalsky A, Alexandrowicz Z and Kedem O 1966 *Chemical Physics of Ionic Solutions* ed B E Conway and R G Barradas (New York: Wiley)
- Kato T, Anzai S I and Seimiya T 1987 *J. Phys. Chem.* **91** 4655

- Kato T and Seimiya T 1986 *J. Phys. Chem.* **90** 3159
- Kawai T, Umemura J and Takenaka T 1984 *Colloid Polym. Sci.* **262** 61
- Keh E and Valeur B 1981 *J. Colloid Interface Sci.* **79** 465
- Kertes A S, Tsimering L and Garti N 1985 *Colloid Polym. Sci.* **263** 67
- Kimizuka H and Satake I 1962 *Bull. Chem. Soc. Japan* **35** 251
- Kitahara A 1965 *J. Phys. Chem.* **69** 2788
- Kitahara A, Kobayashi T and Tachibana T 1962 *J. Phys. Chem.* **66** 363
- Kjellander R 1982 *J. Chem. Soc. Faraday Trans. II* **78** 2025
- Kjellander R and Marcelja S 1984 *Chem. Phys. Lett.* **112** 49
- 1985a *J. Chem. Phys.* **82** 2122
- 1985b *Chem. Scr.* **25** 112
- 1986a *Chem. Phys. Lett.* **127** 402
- 1986b *J. Phys. Chem.* **90** 1230
- Kolthoff I M and Stricks W 1948 *J. Phys. Colloid Chem.* **52** 915
- Kotlarchyk M, Chen S H and Huang J S 1982 *J. Phys. Chem.* **86** 3273
- 1983 *Phys. Rev. A* **28** 508
- Kotlarchyk M, Chen S H, Huang J S and Kim M W 1984a *Phys. Rev. Lett.* **53** 941
- 1984b *Phys. Rev. A* **29** 2054
- Kotlarchyk M, Huang J S and Chen S H 1985 *J. Phys. Chem.* **89** 4382
- Kotlarchyk M, Stephens R B and Huang J S 1988 *J. Phys. Chem.* **92** 1533
- Kumar V V, Kumar C and Raghunathan P 1984 *J. Colloid Interface Sci.* **99** 315
- Kunieda H and Shinoda K 1979 *J. Colloid Interface Sci.* **70** 477
- Kushner L M and Hubbard W D 1954 *J. Phys. Chem.* **58** 1163
- Labourette B, Peyrelasse J, Boned C and Clausse M 1979 *Nature* **281** 60
- Laguës M 1979 *J. Phys. Lett.* **40** 331
- Laguës M, Ober R and Taupin C 1978 *J. Phys. Lett.* **39** 487
- Laguës M and Sauterey C 1980 *J. Phys. Chem.* **84** 3503
- Lang J, Rueff R, Dinh-Cao M and Zana R 1984 *J. Colloid Interface Sci.* **101** 184
- Lang J, Tondre C, Zana R, Bauer R, Hoffmann H and Ulbricht W 1975 *J. Phys. Chem.* **79** 276
- Lang J and Zana R 1986 *J. Phys. Chem.* **90** 5258
- Lang J C 1984 *Surfactants in Solution* vol 1 ed K L Mittal and B Lindman (New York: Plenum) p 35
- Lang J C and Morgan R D 1980 *J. Chem. Phys.* **73** 5849
- Langevin D 1986 *Phys. Scr.* **13** 252
- Larché F and Delord P 1985 *Fluid Phase Equil.* **20** 257
- Larché F, Rouvière F, Delord P, Brun B and Dussossoy J L 1980 *J. Phys. Lett.* **41** L 437
- Laurent M and Samori B 1987 *J. Am. Chem. Soc.* **109** 5109
- Lawson K D and Flaatt T J 1967 *J. Am. Chem. Soc.* **89** 5489
- Lemaire B, Bothorel P and Roux D 1983 *J. Phys. Chem.* **87** 1023
- Lerebours B, Chevalier Y and Piléni M P 1985 *Chem. Phys. Lett.* **117** 89
- Lessner E, Teubner M and Kahlweit M 1981a *J. Phys. Chem.* **85** 1529
- 1981b *J. Phys. Chem.* **85** 3167
- Leung R, Shah D O and O'Connell J P 1986 *J. Colloid Interface Sci.* **111** 286
- Lianos P and Lang J 1983 *J. Colloid Interface Sci.* **96** 222
- Lianos P and Zana R 1983 *J. Phys. Chem.* **87** 1289
- Lichtenberg D 1985 *Biochim. Biophys. Acta* **821** 470
- Lim K C, Fincher Jr C R and Heeger A J 1983 *Phys. Rev. Lett.* **50** 1934
- Lindblom G, Larsson K, Johansson L, Fontell K and Forsen S 1979 *J. Am. Chem. Soc.* **101** 5465
- Lindblom G and Wennerström H 1977 *Biophys. Chem.* **6** 167
- Lindman B, Ahlén T, Söderman O, Walderhaug H, Rapacki K and Stilbs P 1983 *Faraday Discuss. Chem. Soc.* **76** 317
- Lindman B and Brun B 1973 *J. Colloid Interface Sci.* **42** 388
- Lindman B, Kamenka N and Brun B 1976 *J. Colloid Interface Sci.* **56** 328
- Lindman B, Kamenka N, Kathopoulos T M, Brun B and Nilsson P G 1980a *J. Phys. Chem.* **84** 2485
- Lindman B, Kamenka N, Puyal M C, Brun B and Jönsson B 1984a *J. Phys. Chem.* **88** 53
- Lindman B, Lindblom G, Wennerström H and Gustavsson H 1977 *Micellization, Solubilization, and Microemulsions* vol 1 ed K L Mittal (New York: Plenum) p 195
- Lindman B, Puyal M C, Kamenka N, Brun B and Gunnarsson G 1982 *J. Phys. Chem.* **86** 1702
- Lindman B, Puyal M C, Kamenka N, Rymdén R and Stilbs P 1984b *J. Phys. Chem.* **88** 5048
- Lindman B and Stilbs P 1987 *Microemulsions* ed S E Friberg and P Bothorel (Boca Raton, FL: CRC)

- Lindman B, Stilbs P and Moseley M E 1981 *J. Colloid Interface Sci.* **83** 569
- Lindman B and Wennerström T 1980 *Top. Curr. Chem.* **87** 1
- Lindman B, Wennerström H, Gustavsson H, Kamenka N and Brun B 1980b *Pure Appl. Chem.* **52** 1307
- Lindström B, Khan A, Söderman O, Kamenka N and Lindman B 1985 *J. Phys. Chem.* **89** 5313
- Linse P 1986 *J. Phys. Chem.* **90** 6821
- Linse P, Gunnarsson G and Jönsson B 1982 *J. Phys. Chem.* **86** 413
- Linse P and Jönsson B 1983 *J. Chem. Phys.* **78** 3167
- Lippert J L and Petitcolas W L 1972 *Biochim. Biophys. Acta* **282** 8
- Little R C 1970 *J. Phys. Chem.* **74** 1817
- Little R C and Singleterry C R 1964 *J. Phys. Chem.* **68** 3453
- Ljunggren S and Eriksson J C 1984 *J. Chem. Soc. Faraday Trans. II* **80** 489
- Llor A and Rigny P 1986 *J. Am. Chem. Soc.* **108** 7533
- Löbl M, Thurn H and Hoffmann H 1984 *Ber. Bunsenges. Phys. Chem.* **88** 1102
- Löfroth J E and Almgren M 1984 *Surfactants in Solution* vol 1 ed K L Mittal and B Lindman (New York: Plenum) p 627
- Lubineau A 1986 *J. Org. Chem.* **51** 2142
- Luisi P L, Giomini M, Pileni M P and Robinson B H 1988 *Biophys. Biochim. Acta* **947** 209
- Luisi P L and Straub B E 1984 (ed) *Reverse Micelles* (New York: Plenum)
- Lum Wan J A, Warr G G, White L R and Grieser F 1987 *Colloid Polym. Sci.* **265** 528
- Luzzati V, Tardieu A, Mateu I and Stuhmann H B 1976 *J. Mol. Biol.* **101** 115
- Magid L J 1986 *Colloids Surfaces* **19** 129
- Magid L J, Daus K A, Butler P D and Quincy R B 1983 *J. Phys. Chem.* **87** 5472
- Magid L J, Triolo R and Johnsson J R Jr 1984a *J. Chem. Phys.* **81** 5161
- 1984b *J. Phys. Chem.* **88** 5730
- Maitra A and Eicke H F 1981 *J. Phys. Chem.* **85** 2687
- Maitra A, Vasta G and Eicke H F 1983 *J. Colloid Interface Sci.* **93** 383
- Malliaris A, Lang J, Sturm J and Zana R 1987 *J. Phys. Chem.* **91** 1475
- Marcus R A 1955 *J. Chem. Phys.* **23** 1057
- Marignan J, Bassereau P and Delord P 1986 *J. Phys. Chem.* **90** 645
- Marignan J, Delichère A and Larché F C 1983 *J. Phys. Lett.* **44** 609
- Marra J 1986 *J. Phys. Chem.* **90** 2145
- Marsden S S and McBain J W 1948 *J. Phys. Chem.* **52** 110
- Martens F M and Verhoeven J W 1981 *J. Phys. Chem.* **85** 1773
- Martin C A and Magid L J 1981 *J. Phys. Chem.* **85** 3938
- Mazer N A, Benedek G B and Carey M C 1976 *J. Phys. Chem.* **80** 1075
- Mazer N A, Carey M C and Benedek G B 1977 *Micellization, Solubilization, and Microemulsions* vol 1 ed K L Mittal (New York: Plenum) p 359
- Mazer N A and Olofsson G 1982 *J. Phys. Chem.* **86** 4584
- McAuliffe C 1966 *J. Phys. Chem.* **70** 1267
- McBain J W 1913 *Trans. Faraday Soc.* **9** 99
- McBain J W and Johnson K E 1944 *J. Am. Chem. Soc.* **66** 9
- McBain M E L and Hutchinson E 1955 *Solubilisation and Related Phenomena* (New York: Academic)
- McMullen W E, Ben-Shaul A and Gelbart W M 1984a *J. Colloid Interface Sci.* **98** 523
- McMullen W E, Gelbart W M and Ben-Shaul A 1984b *J. Phys. Chem.* **88** 6649
- Meijering J L 1953 *Phillips Res. Rep.* **8** 270
- Mély B, Charvolin J and Keller P 1975 *Chem. Phys. Lipids* **15** 161
- Menger F M 1979 *Acc. Chem. Res.* **12** 111
- 1984 *Surfactants in Solution* vol 1 ed K L Mittal and B Lindman (New York: Plenum) p 347
- Menger F M and Bonicamp J M 1981b *J. Am. Chem. Soc.* **103** 2140
- Menger F M and Boyer B J 1980 *J. Am. Chem. Soc.* **102** 5936
- Menger F M and Carnahan D W 1986 *J. Am. Chem. Soc.* **108** 1297
- Menger F M and Chow J F 1983 *J. Am. Chem. Soc.* **105** 5501
- Menger F M and Doll D W 1984 *J. Am. Chem. Soc.* **106** 1109
- Menger F M, Dulany M A, Carnahan D W and Lee L H 1987 *J. Am. Chem. Soc.* **109** 6899
- Menger F M and Jerkunica J M 1977 *Tetrahedron Lett.* **52** 4569
- Menger F M, Jerkunica J M and Johnston J C 1978 *J. Am. Chem. Soc.* **100** 4676
- Menger F M, Rhee J U and Rhee H K 1975 *J. Org. Chem.* **40** 3803
- Menger F M, Yoshinaga H, Venkatasubban K S and Das A R 1981a *J. Org. Chem.* **46** 415
- Méring P and Tchoubar D 1968 *J. Appl. Crystallogr.* **1** 153



- Meunier J 1985 *J. Phys. Lett.* **46** 1005
- Meyer C T, Poggi Y and Maret G 1982 *J. Physique* **43** 827
- Mialocq J C 1982 *Chem. Phys.* **73** 107
- Miller D D, Bellare J R, Evans D F, Talmon Y and Ninham B W 1987a *J. Phys. Chem.* **91** 674
- Miller D D, Evans D F, Warr G G, Bellare J R and Ninham B W 1987b *J. Colloid Interface Sci.* **116** 598
- Millero F J 1971 *Chem. Rev.* **71** 147
- Milner S T, Safran S A, Andelmann D, Cates M E and Roux D 1988 *J. Physique* **49** 1065
- Minch M J and Shah S S 1979 *J. Org. Chem.* **44** 3252
- Missel P J, Mazer N A, Benedek G B and Carey M C 1983 *J. Phys. Chem.* **87** 1264
- Missel P J, Mazer N A, Benedek G B, Young C Y and Carey M C 1980 *J. Phys. Chem.* **84** 1044
- Missel P J, Mazer N A, Carey M C and Benedek G B 1982 *Solution Behavior of Surfactants* vol 1 ed K L Mittal and E J Fendler (New York: Plenum) p 373
- Mitchell D J and Ninham B W 1981 *J. Chem. Soc. Faraday Trans. II* **77** 601
- Mitchell D J, Tiddy G J T, Waring L, Bostock T and McDonald M P 1983 *J. Chem. Soc. Faraday Trans. I* **79** 975
- Mitra P, Ganesh K N and Balasubramian D 1984 *J. Phys. Chem.* **88** 318
- Mittal K L (ed) 1977 *Micellization, Solubilization, and Microemulsions* vols 1 and 2 (New York: Plenum)
- (ed) 1979 *Solution Chemistry of Surfactants* vols 1 and 2 (New York: Plenum)
- Mittal K L and Bothorel P (eds) 1986 *Surfactants in Solution* vols 4–6 (New York: Plenum)
- Mittal K L and Fendler E J (eds) 1982 *Solution Behavior of Surfactants* vols 1 and 2 (New York: Plenum)
- Mittal K L and Lindman B (eds) 1984 *Surfactants in Solution* vols 1–3 (New York: Plenum)
- Mizushima S I and Simanouti T 1949 *J. Am. Chem. Soc.* **71** 1320
- Muddle A G, Higgins J S, Cummons P G, Staples E J and Lyle I G 1983 *Faraday Discuss. Chem. Soc.* **76** 77
- Mukerjee P 1964 *J. Colloid Sci.* **19** 722
- 1971 *J. Phys. Chem.* **60** 1528
- 1979 *Solution Chemistry of Surfactants* vol 1 ed K L Mittal (New York: Plenum) p 153
- 1980 *Pure Appl. Chem.* **52** 1317
- Mukerjee P, Cardinal J R and Desai N R 1977 *Micellization, Solubilization, and Microemulsions* vol 1 ed K L Mittal (New York: Plenum) p 241
- Mukerjee P and Mysels K J 1971 Critical micelle concentrations of aqueous surfactant systems *Nat. Stand. Ref. Data Ser.* NSRDS-NBS36 (Washington, DC: NBS)
- Mukerjee P, Mysels K J and Kapauan P 1967 *J. Phys. Chem.* **71** 4166
- Müller N, Wolff T and von Bünau G 1984 *J. Photochem.* **24** 37
- Muller V M and Derjaguin B V 1983 *Colloids Surfaces* **6** 205
- Musbally G M, Perron G and Desnoyers J E 1974 *J. Colloid Interface Sci.* **48** 494
- Mysels K J and Prinsen L H 1959 *J. Phys. Chem.* **63** 1696
- Nagarajan R 1985 *Langmuir* **1** 331
- 1986 *Adv. Colloid Interface Sci.* **26** 205
- Nagarajan R, Chaiko M A and Ruckenstein E 1984 *J. Phys. Chem.* **88** 2916
- Nagarajan R and Ruckenstein E 1984 *Surfactants in Solution* vol 2 ed K L Mittal and B Lindman (New York: Plenum) p 923
- Nagarajan R, Shah K M and Hammond S 1982 *Colloids Surfaces* **4** 147
- Nägele G, Klein R and Medina-Noyola M 1985 *J. Chem. Phys.* **83** 2560
- Nakagawa T 1966 *Nonionic Surfactants* (Surfactant Science Series vol 1) ed M J Schick (New York: Dekker) p 558
- Nash M E, Jennings B R and Tiddy G J T 1987 *J. Colloid Interface Sci.* **120** 542
- Neeson P G, Jennings B R and Tiddy G J T 1983 *Chem. Phys. Lett.* **95** 533
- Neogi P, Kim M and Friberg S E 1987 *J. Phys. Chem.* **91** 605
- Néry H, Söderman O, Canet D, Walderhaug H and Lindman B 1986 *J. Phys. Chem.* **90** 5802
- Nguyen C M, Rathman J F and Scamehorn J F 1986 *J. Colloid Interface Sci.* **112** 438
- Nilsson P G and Lindman B 1983 *J. Phys. Chem.* **87** 4756
- 1984 *J. Phys. Chem.* **88** 4764
- Nilsson P G, Wennerström H and Lindman B 1983 *J. Phys. Chem.* **87** 1377
- 1985 *Chem. Scr.* **25** 67
- Nimtz G, Binggeli B, Borgnen L, Marquardt P, Schafer U and Zorn R 1986 *Europhys. Lett.* **2** 103
- Ninham B W, Barnes I S, Hyde S T, Dérian P J and Zemb T 1987 *Europhys. Lett.* **4** 561
- Ninham B W, Chen S J and Evans D F 1984 *J. Phys. Chem.* **88** 5855
- Ninham B W, Evans D F and Wei G J 1983c *J. Phys. Chem.* **87** 5020
- Ninham B W and Evans D F 1986 *Faraday Discuss. Chem. Soc.* **81** 1

- Ninham B W, Hashimoto S and Thomas J K 1983a *J. Colloid Interface Sci.* **95** 594  
Ninham B W, Talmon Y and Evans D F 1983b *Science* **221** 1048  
Oakenfull D 1980 *J. Chem. Soc. Faraday Trans. I* **76** 1875  
Okabayashi H and Abe M 1980 *J. Phys. Chem.* **84** 999  
Okabayashi H and Matsushita K 1984 *Surfactants in Solution* vol 1 ed K L Mittal and B Lindman (New York: Plenum) p 517  
Okabayashi H, Okuyama M and Kitagawa T 1975 *Bull. Chem. Soc. Japan* **48** 2264  
Okabayashi H, Okuyama M, Kitagawa T and Miyazawa T 1974 *Bull. Chem. Soc. Japan* **47** 1075  
Okabayashi H, Yoshida T, Ikeda T, Matsuura H and Kitagawa T 1982 *J. Am. Chem. Soc.* **104** 5399  
Olsson U, Söderman O and Guéring P 1986 *J. Phys. Chem.* **90** 5223  
Osborne H B, Sardet C, Michel-Villaz M and Chabre M 1978 *J. Mol. Biol.* **123** 177  
Overbeek J T G, Verhoeckx G J, De Bruyn P L and Lekkerkerker H N W 1987 *J. Colloid Interface Sci.* **119** 422  
Owenson B and Pratt L R 1984 *J. Phys. Chem.* **88** 2905  
Ozeki S and Ikeda S 1985 *J. Phys. Chem.* **89** 5088  
Paddy M R, Dahlquist F W, Davis J H and Bloom M 1981 *Biochemistry* **20** 3152  
Parodi O 1984 *Colloïdes et Interfaces* ed A M Cazabat and M Veyssié (Paris: Les Editions de Physique)  
Pashley R M and Israelachvili J N 1984 *J. Colloid Interface Sci.* **101** 511  
Pashley R M, McGuiggan P M and Ninham B W 1986 *J. Phys. Chem.* **90** 5841  
Pecora R 1984 *Pure Appl. Chem.* **56** 1391  
Percus J K and Yevick G J 1958 *Phys. Rev.* **110** 1  
Perez E, Alandis N, Laval J P, Rico I and Lattes A 1987 *Tetrahedron Lett.* **28** 2343  
Pethica B A 1955 *Trans. Faraday Soc.* **51** 1402  
Peyrelasse J, Boned C and Marin G 1986 *Colloid Polym. Sci.* **264** 143  
Peyrelasse J, Boned C and Moha-Ouchane M 1988 *Proc. 2nd World Surfactant Congr.* vol 3 (Paris: CESIO) p 157  
Pileni M P, Brochette P, Hickel B and Lerebours B 1984 *J. Colloid Interface Sci.* **98** 549  
Pileni M P, Hickel B, Ferradini C and Pucheault J 1982 *Chem. Phys. Lett.* **92** 308  
Pileni M P, Lerebours B, Brochette P and Chevalier Y 1985b *J. Photochem.* **28** 273  
Pileni M P, Zemb T, Brochette P, Hickel B and Milhaud J 1986 *Surfactants in Solutions* vol 5 ed K L Mittal and P Bothorel (New York: Plenum) p 685  
Pileni M P, Zemb T and Petit C 1985a *Chem. Phys. Lett.* **118** 414  
Pincus P A and Safran S A 1987 *J. Chem. Phys.* **86** 2294  
Podo F, Ray A and Némethy G 1973 *J. Am. Chem. Soc.* **95** 6164  
Porte G 1983 *J. Phys. Chem.* **87** 3541  
Porte G and Appell J 1981 *J. Phys. Chem.* **85** 2511  
— 1984 *Surfactants in Solution* vol 2 ed K L Mittal and B Lindman (New York: Plenum) p 805  
Porte G, Appell J and Poggi Y 1980 *J. Phys. Chem.* **84** 3105  
Porte G, Gomati R, El Haitamy O, Appell J and Marignan J 1986 *J. Phys. Chem.* **90** 5746  
Porte G, Marignan J, Bassereau P and May R 1988 *J. Physique* **49** 511  
Porte G, Poggi Y, Appell J and Maret G 1984 *J. Phys. Chem.* **88** 5713  
Pownall H J and Smith L C 1973 *J. Am. Chem. Soc.* **95** 3136  
Pratt L R, Owenson B and Sun Z 1986 *Adv. Colloid Interface Sci.* **26** 69  
Prestegard J H and Grant D M 1978 *J. Am. Chem. Soc.* **100** 4664  
Puig J E, Hemker D L, Gupta A, Davis H T and Scriven L E 1987 *J. Phys. Chem.* **91** 1137  
Pusey P N 1978 *J. Phys. A: Math. Gen.* **11** 119  
Quirion F and Magid L J 1986 *J. Phys. Chem.* **90** 5193  
Radley K, Reeves L W and Tracey A S 1976 *J. Phys. Chem.* **80** 174  
Ramesh V and Labes M M 1986 *J. Am. Chem. Soc.* **108** 4643  
Rathman J F and Scamehorn J F 1987 *Langmuir* **3** 372  
Ravey J C 1983 *J. Colloid Interface Sci.* **94** 289  
Ravey J C and Buzier M 1984 *Reverse Micelles* ed P L Luisi and B E Straub (New York: Plenum) p 195  
Ravey J C, Buzier M and Dupont G 1986 *Microemulsions* (Surfactant Science Series vol 24) ed M Clausse and H Rosano (New York: Dekker) p 163  
Ravey J C, Buzier M and Picot C 1984 *J. Colloid Interface Sci.* **97** 9  
Reatto L and Tau M 1984 *Chem. Phys. Lett.* **108** 292  
— 1987 *Europhys. Lett.* **3** 527  
Rehage H, Wunderlich I and Hoffmann H 1986 *Prog. Colloid Polym. Sci.* **72** 51  
Reiss-Husson F and Luzzati V 1964 *J. Phys. Chem.* **68** 3905

- Rideout D C and Breslow R 1980 *J. Am. Chem. Soc.* **102** 7816
- Roberts R T 1973 *Nature* **242** 348
- Roberts R T and Chachaty C 1973 *Chem. Phys. Lett.* **22** 348
- Robins D C and Thomas I L 1968 *J. Colloid Interface Sci.* **26** 422
- Robinson B H, Toprakcioglu C, Dore J C and Chieux P 1984 *J. Chem. Soc. Faraday Trans. I* **80** 13
- Rogers F J 1980 *J. Chem. Phys.* **73** 6272
- Romsted L S 1985 *J. Phys. Chem.* **89** 5107, 5113
- Rosenblatt C and Zolty N 1985 *J. Phys. Lett.* **46** 1191
- Rosenholm J B, Larsson K and Dinh-Nguyen N 1977 *Colloid Polym. Sci.* **255** 1098
- Rosenholm J B, Stenius P and Danielsson I 1976 *J. Colloid Interface Sci.* **57** 551
- Rosevear F B 1968 *J. Soc. Cosmet. Chem.* **19** 581
- Rouvière J, Couret J M, Lindheimer M, Dejardin J L and Marrony R 1979a *J. Chim. Phys.* **76** 289
- Rouvière J, Couret J M, Lindheimer A, Lindheimer M and Brun B 1979b *J. Chim. Phys.* **76** 297
- Roux A H, Roux G, Grolier J P and Viallard A 1981 *J. Colloid Interface Sci.* **84** 250
- Roux D 1984 *PhD Thesis*, Bordeaux I University, France
- Roux D and Bellocq A M 1984 *Phys. Rev. Lett.* **52** 1895
- Roux D, Bellocq A M and Bothorel P 1984a *Prog. Colloid Polym. Sci.* **69** 1
- 1984b *Surfactants in Solutions* vol 3 ed K L Mittal and B Lindman (New York: Plenum) p 1843
- Roux D, Bellocq A M and Leblanc M S 1983 *Chem. Phys. Lett.* **94** 156
- Roux D and Coulon C 1986 *J. Physique* **47** 1257
- Rubingh D N 1979 *Solution Chemistry of Surfactants* vol 1 ed K L Mittal (New York: Plenum) p 337
- Rushforth D S, Sanchez-Rubio M, Santos-Vidals L M, Worsmuth K R, Kaler E W, Cuevas R and Puig J E 1986 *J. Phys. Chem.* **90** 6668
- Russell J C and Whitten D G 1982 *J. Am. Chem. Soc.* **104** 5937
- Russell J C, Wild U R and Whitten D G 1986 *J. Phys. Chem.* **90** 1319
- Safinya C R, Roux D, Smith G S, Sinha S K, Dimon P, Clark N A and Bellocq A M 1986 *Phys. Rev. Lett.* **57** 2718
- Safran S A 1983 *J. Chem. Phys.* **78** 2073
- Safran S A, Roux D, Cates M E and Andelman D 1986 *Phys. Rev. Lett.* **57** 491
- Safran S A and Turkevitch L A 1983 *Phys. Rev. Lett.* **50** 1930
- Safran S A, Turkevich L A and Pincus P 1984 *J. Phys. Lett.* **45** 69
- Sagitani H and Friberg S E 1983 *Colloid Polym. Sci.* **26** 862
- Sakellariou-Fargues R, Maurette M T, Oliveros E, Rivière M and Lattes A 1982 *J. Photochem.* **18** 101
- Samori B and Mattivi F 1986 *J. Am. Chem. Soc.* **108** 1679
- Sasaki T, Hattori M, Sasaki J and Nukina K 1975 *Bull. Chem. Soc. Japan* **48** 1397
- Satake I, Iwamatsu I, Hosokawa S and Matuura R 1963 *Bull. Chem. Soc. Japan* **36** 204
- Satake I, Tahara T and Matuura R 1969 *Bull. Chem. Soc. Japan* **42** 319
- Sauterey C, Taupin C and Dvolaitsky M 1978 *C.R. Acad. Sci., Paris C* **287** 77
- Schatzberg P 1963 *J. Phys. Chem.* **67** 776
- Scheraga H A and Backus J K 1951 *J. Am. Chem. Soc.* **73** 5108
- Schick M and Wei-Heng S. 1986 *Phys. Rev. B* **34** 1797
- Schick M J, Atlas S M and Eirich F R 1962 *J. Phys. Chem.* **66** 1326
- Schorr W and Hoffmann H 1981 *J. Phys. Chem.* **85** 3160
- Schott H 1967 *J. Colloid Interface Sci.* **24** 193
- Schulman J H and Riley D P 1948 *J. Colloid Sci.* **3** 383
- Schulman J H, Stoeckenius W and Prince L M 1959 *J. Phys. Chem.* **63** 1677
- Scriven L E 1976 *Nature* **263** 123
- 1977 *Micellization, Solubilization, and Microemulsions* vol 2 ed K L Mittal (New York: Plenum) p 877
- Sepulveda L and Cortès J 1985 *J. Phys. Chem.* **89** 5322
- Shah D O, Bansal V K and Hsieh W C 1977 *Improved Oil Recovery and Polymer Flooding* (New York: Academic)
- Sheu E Y, Chen S H and Huang J S 1987 *J. Phys. Chem.* **91** 1535
- Sheu E Y, Goklen K E, Hatton T A and Chen S H 1986 *Biotechnol. Prog.* **2** 175
- Sheu E Y, Wu C F, Chen S H and Blum L 1985 *Phys. Rev. A* **32** 3807
- Shih L B and Williams R W 1986 *J. Phys. Chem.* **90** 1615
- Shinitzky M, Dianoux A C, Gitler C and Weber G 1971 *Biochemistry* **10** 2106
- Shinoda K 1985 *J. Phys. Chem.* **29** 2429
- Shinoda K and Sagitani H 1983 *J. Phys. Chem.* **87** 2018
- Shinoda K and Soda T 1963 *J. Phys. Chem.* **67** 2072

- Singer L A 1982 *Solution Behavior of Surfactants* vol 1 ed K L Mittal and E J Fendler (New York: Plenum) p 73
- Singh H N, Swarup S, Singh R P and Saleem S M 1983 *Ber. Bunsenges. Phys. Chem.* **87** 1115
- Sirianni A F, Cowie J M G and Puddington I E 1962 *Can. J. Chem.* **40** 957
- Sjoblom E and Friberg S E 1978 *J. Colloid Interface Sci.* **67** 16
- Skoulios A 1978 *Ann. Phys., Paris* **3** 421
- Smith D H 1979 *J. Colloid Interface Sci.* **68** 70
- Smith E R and Rowlinson J S 1980 *J. Chem. Soc. Faraday Trans. II* **76** 1468
- Smith G A, Christian S D, Tucker E E and Scamehorn J F 1987 *Langmuir* **3** 598
- Smith G D, Donelan C E and Barden R E 1977 *J. Colloid Interface Sci.* **60** 488
- Snyder R G 1967 *J. Chem. Phys.* **47** 1316
- Snyder R G, Strauss H L and Elliger C A 1982 *J. Phys. Chem.* **86** 5145
- Söderman O, Henriksson U and Olsson U 1987 *J. Phys. Chem.* **91** 116
- Söderman O and Walderhaug H 1986 *Langmuir* **2** 57
- Söderman O, Walderhaug H, Henriksson U and Stilbs P 1985b *J. Phys. Chem.* **89** 3693
- Söderman O, Walderhaug H and Lindman B 1985a *J. Phys. Chem.* **89** 1795
- Staples E J and Tiddy G J T 1978 *J. Chem. Soc. Faraday Trans. I* **74** 2530
- Stearns R S, Oppenheimer H, Simon E and Harkins W D 1947 *J. Chem. Phys.* **15** 496
- Stébé M J, Serratrice G and Delpuech J J 1985 *J. Phys. Chem.* **89** 2837
- Stecker M M and Benedek G B 1984 *J. Phys. Chem.* **88** 6519
- Stenius P and Ekwall P 1967 *Acta Chem. Scand.* **21** 1643
- Stilbs P 1982a *J. Colloid Interface Sci.* **87** 385
- 1982b *J. Colloid Interface Sci.* **89** 547
- 1983 *J. Colloid Interface Sci.* **94** 463
- 1987 *Prog. NMR Spectrosc.* **19** 1
- Stilbs P and Lindman B 1981 *J. Phys. Chem.* **85** 2587
- Stilbs P, Rapacki K and Lindman B 1983b *J. Colloid Interface Sci.* **95** 583
- Stilbs P, Walderhaug H and Lindman B 1983a *J. Phys. Chem.* **87** 4762
- Sudhölter E J R and Engberts J B F N 1982 *J. Phys. Chem.* **86** 263
- Svensson B and Jönsson B 1984 *Chem. Phys. Lett.* **108** 580
- Szeifer I, Ben-Shaul A and Gelbart W M 1985 *J. Chem. Phys.* **83** 3612
- Tabony J 1984 *Mol. Phys.* **51** 975
- 1985 *Chem. Phys. Lett.* **113** 75
- Tabony J, de Geyer A and Braganza L F 1987 *Nature* **327** 321
- Tabony J, Llor A and Drifford M 1983 *Colloid Polym. Sci.* **261** 938
- Talmon Y 1986 *Colloids Surfaces* **19** 237
- Talmon Y, Evans D F and Ninham B W 1983 *Science* **221** 1047
- Talmon Y and Prager S 1978 *J. Chem. Phys.* **69** 2984
- 1982 *J. Chem. Phys.* **76** 1535
- Tanford C 1972 *J. Phys. Chem.* **76** 3020
- 1973 *The Hydrophobic Effect: Formation of Micelles and Biological Membranes* (New York: Wiley)
- 1974a *Proc. Natl. Acad. Sci. USA* **71** 1811
- 1974b *J. Phys. Chem.* **78** 2469
- 1977 *Micellization, Solubilization, and Microemulsions* vol 1 ed K L Mittal (New York: Plenum) p 119
- Tarazona P and Vicente L 1985 *Mol. Phys.* **56** 557
- Tardieu A, Laporte D and Delaye M 1987 *J. Physique* **48** 1207
- Tartar H V 1955 *J. Phys. Chem.* **59** 1195
- Tau M and Reatto L 1985 *J. Chem. Phys.* **83** 1921
- Tavernier S M F, Vonk C G and Gijbels R 1981 *J. Colloid Interface Sci.* **81** 341
- Teubner M and Strey R 1987 *J. Chem. Phys.* **87** 3195
- Thurn H, Kalus J and Hoffmann H 1984 *J. Chem. Phys.* **80** 3440
- Tocanne J F, Ververgaert P H J Th, Verkleij A J and Van Deenen L L M 1974 *Chem. Phys. Lipids* **12** 201
- Toprakcioglu C, Dore J C, Robinson B H and Howe A M 1984 *J. Chem. Soc. Faraday Trans. I* **80** 413
- Trauble H and Eibl H 1974 *Proc. Natl. Acad. Sci. USA* **71** 214
- Treiner C, Boquet J F and Pommier C 1986 *J. Phys. Chem.* **90** 3052
- Treiner C and Mannebach M H 1987 *J. Colloid Interface Sci.* **118** 243
- Triolo R, Caponetti E and Graziano V 1985 *J. Phys. Chem.* **89** 5743
- Triolo R, Hayter J B, Magid L J and Johnson J S Jr 1983 *J. Chem. Phys.* **79** 1977
- Triolo R, Magid L J, Johnson J S Jr and Child H R 1982 *J. Phys. Chem.* **86** 3689

- Tsuchida E, Nishide H, Yuasa M, Hasegawa E and Matsushita Y I 1984 *J. Chem. Soc. Dalton Trans.* 1147
- Turro N J, Aikawa M and Yekta A 1979 *J. Am. Chem. Soc.* **101** 774
- Turro N J and Okubo T 1981 *J. Am. Chem. Soc.* **103** 7224
- Ueno M and Kishimoto H 1983 *J. Phys. Chem.* **87** 850
- Ulmius J, Lindman B, Lindblom G and Drakenberg T 1978 *J. Colloid Interface Sci.* **65** 88
- Ulmius J, Wennerström H, Johansson L B A, Lindblom G and Gravsholt S 1979 *J. Phys. Chem.* **83** 2232
- Umemura J, Cameron D G and Mantsch H H 1980 *J. Phys. Chem.* **84** 2272
- Umemura J, Mantsch H H and Cameron D G 1981 *J. Colloid Interface Sci.* **83** 558
- Valeur B and Keh E 1979 *J. Phys. Chem.* **83** 3305
- Van der Ploeg P and Berendsen H J C 1983 *Mol. Phys.* **49** 233
- van de Sande W and Persoons A 1985 *J. Phys. Chem.* **89** 404
- van Gunsteren W F, Berendsen H J C and Rullmann J A C 1981 *Mol. Phys.* **44** 69
- Venable R L and Nauman R V 1964 *J. Phys. Chem.* **68** 3498
- Verkleij A J, De Kruffy B, Vervegaert P H J Th, Tocanne J F and Van Deenen L L M 1974 *Biochim. Biophys. Acta* **339** 432
- Vikingsstad E 1979 *J. Colloid Interface Sci.* **72** 68
- Vikingsstad E, Skauge A and Hoiland H 1978 *J. Colloid Interface Sci.* **66** 240
- Vold R L and Vold R R 1978 *Prog. Nucl. Magn. Reson. Spectrosc.* **12** 79
- Vonk C G, Billman J F and Kaler E W 1988 *J. Chem. Phys.* **88** 3970
- Walderhaug H, Söderman O and Stilbs P 1984 *J. Phys. Chem.* **88** 1655
- Wall S and Elvingston C 1985 *J. Phys. Chem.* **89** 2695
- Wall S N and Aniansson E A G 1980 *J. Phys. Chem.* **84** 727
- Warr G G, Sen R, Evans D F and Trend J E 1988b *J. Phys. Chem.* **92** 774
- Warr G G, Zemb T, Barnes I S and Ninham B W 1988a *Proc. 2nd World Surfactant Congr.* vol 2 (Paris: CESIO) p 361
- Welberry T R and Zemb T 1988 *J. Colloid Interface Sci.* **123** 413
- Wennerström H 1979 *J. Colloid Interface Sci.* **68** 589
- Wennerström H, Jönsson B and Linse P 1982 *J. Chem. Phys.* **76** 4665
- Wennerström H, Lindblom G and Lindman B 1974 *Chem. Scr.* **6** 97
- Wennerström H and Lindman B 1979a *Phys. Rep.* **52** 1
- 1979b *J. Phys. Chem.* **83** 2931
- Wennerström H, Lindman B, Söderman O, Drakenberg T and Rosenholm J B 1979 *J. Am. Chem. Soc.* **101** 6860
- Werbelow L G and Grant D M 1977 *Adv. Magn. Reson.* **9** 189
- Westra S W T and Leyte J C 1979 *Ber. Bunsenges. Phys. Chem.* **83** 672, 678
- Widom B 1984a *J. Chem. Phys.* **81** 1030
- 1984b *J. Phys. Chem.* **88** 6508
- Winsor P A 1948 *Trans. Faraday Soc.* **44** 376
- Wishnia A 1963 *J. Phys. Chem.* **67** 2079
- Wolff T, Suck T A, Emming C S and von Büнау G 1987 *Prog. Colloid Polym. Sci.* **73** 18
- Wolff T and von Büнау G 1984a *J. Colloid Interface Sci.* **99** 299
- 1984b *Ber. Bunsenges. Phys. Chem.* **88** 1098
- 1986 *J. Photochem.* **35** 239
- Wong M and Thomas J K 1977 *Micellization, Solubilization, and Microemulsions* vol 2 ed K L Mittal (New York: Plenum) p 647
- Wong M, Thomas J K and Gratzel M 1976 *J. Am. Chem. Soc.* **98** 2391
- Wong M, Thomas J K and Nowak T 1977 *J. Am. Chem. Soc.* **99** 4730
- Woods M C, Haile J M and O'Connell J P 1986 *J. Phys. Chem.* **90** 1875
- Wormuth K R and Kaler E W 1987 *J. Phys. Chem.* **91** 611
- Yarusso D J and Cooper S L 1983 *Macromolecules* **16** 1871
- Yiv S, Zana R, Ulbricht W and Hoffmann H 1981 *J. Colloid Interface Sci.* **80** 224
- Zachariasse K A 1978 *Chem. Phys. Lett.* **57** 429
- Zachariasse K A, Kozankiewicz B and Kuhnle W 1984 *Surfactants in Solution* vol 1 ed K L Mittal and B Lindman (New York: Plenum) p 565
- Zachariasse K A, Phuc N V and Kozankiewicz B 1981 *J. Phys. Chem.* **85** 2676
- Zampieri G G, Jaecle H and Luisi P L 1986 *J. Phys. Chem.* **90** 1849
- Zana R 1980 *J. Colloid Interface Sci.* **78** 330
- 1986 *J. Chim. Phys.* **83** 603
- (ed) 1987 *Surfactant Solutions, New Methods of Investigation* (New York: Dekker)

- Zana R, Lang J, Sorba O, Cazabat A M and Langevin D 1982 *J. Phys. Lett.* **43** 829
- Zana R, Lang J, Yiv S H, Djavanbakt A and Abad C 1977 *Micellization, Solubilization, and Microemulsions* vol 1 ed K L Mittal (New York: Plenum) p 291
- Zana R, Lianos P and Lang J 1985 *J. Phys. Chem.* **89** 41
- Zana R and Mackay R A 1986 *Langmuir* **2** 109
- Zana R, Picot C and Duplessix R 1983 *J. Colloid Interface Sci.* **93** 43
- Zana R and Weill C 1985 *J. Phys. Lett.* **46** 953
- Zana R, Yiv S, Strazielle C and Lianos P 1981 *J. Colloid Interface Sci.* **80** 208
- Zemb T, Drifford M, Hayoun M and Jehanno A 1983 *J. Phys. Chem.* **87** 4524
- Zemb T, Hyde S T, Derian P J, Barnes I S and Ninham B W 1987 *J. Phys. Chem.* **91** 3814
- Zulauf M and Eicke H F 1979 *J. Phys. Chem.* **83** 480
- Zulauf M and Rosenbusch J P 1983 *J. Phys. Chem.* **87** 856
- Zulauf M, Weckström K, Hayter J B, Degiorgio V and Corti M 1985 *J. Phys. Chem.* **89** 3411